

Emmy Noether

Current English Translation Drafts

Reader note

This current working reader combines the available English translation pieces for Emmy Noether currently in hand: *Zur Invariantentheorie der Formen von n Variablen* (1910) and *Der Endlichkeitssatz der Invarianten endlicher Gruppen* (1916), the current draft and continuation on integral rational representation of invariants, the current draft and continuation on an absolute irreducibility criterion for representations, further current drafts covering formal variational calculus, differential invariants, polynomial ideals and resultants, elimination theory, general ideal theory, short reports on abstract ideal theory, Hilbert numbers, group characters, the opening of abstract ideal theory in algebraic number and function fields, and the latest delivered continuation on hypercomplex quantities and representation theory. The translations are intended as continuation bases for checking and refinement, not as proofed editions.

On the invariant theory of forms in n variables

In the projective invariant theory of forms in n variables, the principal problem has been solved: the finiteness of the system of forms has been proved. A number of further questions, however, have not been treated, namely those that concern methods for investigating the relations among forms. I should like briefly to communicate the results that I have obtained with respect to such questions, but for the sake of clarity I shall first sketch the questions in the ternary case, where they are already known.

I think first of the two fundamental theorems of the symbolic method: the theorem that all invariant formations can be represented symbolically, and the second theorem that all relations among invariants are obtained by successive application of a small number of symbolic identities; in other words, that the symbolic method gives all relations without leaving the domain of invariants.

Built upon these are the processes for generating forms: the symbolic contraction process, the non-symbolic differentiation processes, the theory of expansions into series, and so on. In this connection I should also mention a theorem proved in my dissertation: all contractions can be generated by successive application of the two possible *cogredient* contractions; by this I mean, for a symbolic product s, t, u, v , the two contractions (stu) and (vx) . Upon this theorem one can build the theory of reducentia and the reduction of systems of forms, analogously to the binary case, as I showed in my dissertation.

When we pass to the n -ary case, already the first theorem of the symbolic method holds only in a qualified sense. As is well known, for quaternary forms there already appear, besides point and plane coordinates x and u , line coordinates p_ρ , namely the minors obtained from two rows of point coordinates. Analogously, in the n -ary case we have rows of variables p_ρ whose individual elements are the minors taken from ρ rows x , and symbolic rows S_ρ whose elements behave like minors taken from ρ rows s , the latter being cogredient to the u .

The symbolic representation by determinants is known to hold only when the symbolic rows S_ρ are resolved into the rows s and these are distributed among different determinants;

then only those aggregates of determinants have a real meaning that depend on the S_ρ . Which determinants depend only on the S_ρ can, to be sure, be recognized by differential equations; but in that way one obtains no survey of the totality of invariant formations, nor of the processes by which they are generated.

At this point the theorem on corresponding matrices becomes decisive. To every matrix product

$$(v^{(1)} \dots v^{(\rho)} \mid y^{(1)} \dots y^{(\rho)}),$$

where the rows of the second matrix are contragredient to those of the first, there corresponds a determinant; and conversely every determinant can be transformed into a matrix product with a prescribed number ρ of rows. Hence determinant and matrix product appear as completely equivalent, and the first theorem of the symbolic method takes the following form:

All invariant formations can be represented symbolically by matrix products.

From two symbolic rows S_ρ, T_τ one can now, with the help of rows of variables, form very easily a matrix product that contains these symbolic rows explicitly and therefore represents an invariant formation of the type $(S_\rho \mid p_\rho)(T_\tau \mid p_\tau)$, namely

$$(G_{n-\rho-\lambda} S_\rho T_\tau \mid p_{\rho+\lambda} p_{\tau-\lambda}), \quad \lambda = 1, 2, \dots, \tau \text{ or } n - \rho. \quad (1)$$

More important is the converse statement: all invariant formations can be represented explicitly by matrix products, so that the above process is the genuine extension of the binary and ternary contraction process. To prove this converse it is necessary to transfer the second theorem of the symbolic method to the n -ary case, namely to determine the totality of the independent indecomposable identities in which all rows S_ρ, p_ρ are treated as indecomposable. These identities arise from the known identities containing only rows x and u by replacing the product theorem for determinants by a system of product theorems for matrices and then combining several matrix products into a single one by means of these same product theorems. In this way one arrives at two dual formulas, of which I cite only one:

$$\sum_{i=1}^k (S_{\rho_1} \dots S_{\rho_k} \mid p_\rho x^{(i)})(S_\rho \mid x^{(i)}) = (S_{\rho_1} \dots S_{\rho_k} S_\rho \mid p_\rho). \quad (2)$$

The only specialization contained in these formulas is that a row x appears. For completely general symbolic and variable rows one reaches a *decomposition identity*, that is, an identity in which a row p_ρ appears as though resolved into the individual rows x . Clebsch already used the simplest special case of the decomposition identity in his “fundamental problem of invariant theory” in order to derive the elementary covariants occurring in series expansions.

With the aid of these identities, which mediate the passage from the non-explicit determinant expressions to matrix products, one can in fact prove that matrix representation yields the desired explicit symbolic representation.

The same formalism then gives the differentiation processes — for a row of variables gives rise to a row of symbols by replacing its entries with differential symbols — and by means of the decomposition identity it becomes possible to transfer Mertens’s proof almost literally to the n -ary case.

The finiteness theorem for the invariants of finite groups

Let the finite group G consist of the h linear transformations (with nonvanishing determinant)

$$A_1, \dots, A_h,$$

where A_k is represented by the linear transformation

$$x_\nu^{(k)} = \sum_{\mu=1}^n a_{\nu\mu}^{(k)} x_\mu \quad (\nu = 1, \dots, n),$$

or, more briefly, by

$$(x^{(k)}) = A_k(x).$$

Thus the group G carries the row (x) with elements $x_1, \dots, x_n$ into the rows $(x^{(k)})$ with elements $x_1^{(k)}, \dots, x_n^{(k)}$. Since the identity must occur among $A_1, \dots, A_h$, the row (x) itself also occurs among the rows $(x^{(k)})$.

By an *integral rational (absolute) invariant* of the group I mean an integral rational function of $x_1, \dots, x_n$ that remains identically unchanged under application of $A_1, \dots, A_h$; thus for such an invariant $f(x)$ one has

$$f(x) = f(x^{(1)}) = f(x^{(2)}) = \dots = f(x^{(h)}). \quad (1)$$

1. A full invariant system from the Galois resolvent

Formula (1) expresses that $f(x)$ is an integral rational symmetric function of the rows $(x^{(1)}), \dots, (x^{(h)})$ — in fact, since each summand $f(x)$ contains only the single row (x) , this is the simplest case, usually called the *uniform* case. By the known theorem on symmetric functions of rows, $f(x)$ can therefore be expressed integrally and rationally through the elementary symmetric functions of these rows, that is, through the coefficients $G_{\alpha_1 \dots \alpha_n}(x)$ of the “Galois resolvent”

$$\Phi(z, u) = \prod_{k=1}^h (z - (u_1 x_1^{(k)} + u_2 x_2^{(k)} + \dots + u_n x_n^{(k)})) = z^h + \sum G_{\alpha_1 \dots \alpha_n}(x) u_1^{\alpha_1} \dots u_n^{\alpha_n} z^{\alpha_0}, \quad (2)$$

where the $G_{\alpha_1 \dots \alpha_n}(x)$ are invariants of degree $\alpha_1 + \dots + \alpha_n$ in the variables x .

Thus we have proved:

Theorem. *The coefficients $G_{\alpha_1 \dots \alpha_n}(x)$ of the Galois resolvent form a complete system of invariants of the group, in such a way that every invariant can be expressed integrally and rationally in terms of these finitely many invariants.*

2. A second complete system

Starting from (1) one can also make the following still more elementary observation, which at the same time leads to a second complete system.

Suppose

$$f(x) = a + b x_\nu + \dots + c x_\nu^m + \dots,$$

where $a, b, \dots, c$ are constants. Then from (1) one has

$$h \cdot f(x) = h \cdot a + b \sum_{k=1}^h x_\nu^{(k)} + \dots + c \sum_{k=1}^h (x_\nu^{(k)})^m + \dots.$$

Hence every invariant can be assembled integrally and linearly from the special quantities

$$s_m = \sum_{k=1}^h (x_\nu^{(k)})^m \quad (m = 0, 1, 2, \dots),$$

so it is enough to prove the theorem for these special functions.

Up to a numerical factor, s_m is the coefficient of $u_1^{\alpha_1} \cdots u_n^{\alpha_n}$ in the expression

$$h \cdot s_m = \sum_{k=1}^h (u_1 x_1^{(k)} + \cdots + u_n x_n^{(k)})^m,$$

where $u = u_1 + \cdots + u_n$ is written only as a convenient abbreviation, and this expression represents the u -power sum of the h linear forms.

Now it is well known that the infinitely many power sums S_m are integral rational functions of $S_1, S_2, \dots, S_h$; their coefficients are determined by the invariants

$$S_m = \sum_{k=1}^h (x_1^{(k)} + \cdots + x_n^{(k)})^m \quad (m \leq h).$$

Thus we obtain a second complete system:

Theorem. *A complete invariant system of the group is given by all invariants $J_{\alpha_1 \dots \alpha_n}$ for which*

$$\alpha_1 + \cdots + \alpha_n \leq h,$$

where h is the order of the group.

The connection with part 1 is given by the remark that the power sums $S_1, \dots, S_h$ may be replaced by the elementary symmetric functions of $x_1^{(k)}, \dots, x_n^{(k)}$, which leads back to the complete system there given by the $G_{\alpha_1 \dots \alpha_n}(x)$.

Both results show that all invariants can be expressed integrally and rationally through those whose degree in the x does not exceed the order h of the group.

3. Rational representation

From what has just been proved one obtains consequences for rational representation. Every rational absolute invariant can — as is seen immediately by multiplying numerator and denominator by the conjugates of the denominator with respect to the group — be represented as the quotient of two integral rational absolute invariants, not necessarily relatively prime. It follows that every rational absolute invariant can be expressed rationally through the coefficients $G_{\alpha_1 \dots \alpha_n}(x)$ of the Galois resolvent $\Phi(z, u)$.

This theorem can also be proved in various easy ways without passing through the finiteness theorem; it is already formulated in Weber II, §58.

4. Final remark

Finally, I should mention that the results derived here are already implicitly contained in my paper *Körper und Systeme rationaler Funktionen*: namely, the complete system given in part 1 appears there in Theorems VI and VII, while an argument for rational representation independent of the present finiteness theorem appears there in Theorem III. Both the proof and the result become, however, considerably simpler in the special situation treated here than in the general theory.

I was led to apply those general investigations to the invariants of finite groups by conversations with E. Fischer, to whom the proof given in part 2 is due in substance as well. In the general case the complete system arising there is not rationally definable, and the remark in part 3 likewise goes back to him.

Erlangen, May 1915.

On integral rational representation of the invariants of a system of arbitrarily many basic forms

At the end of his contribution to the Schwarz Festschrift, *On the invariants of a system of arbitrarily many basic forms*, Hilbert expressed the conjecture that these invariants can be represented integrally and rationally by the finitely many invariants of the system (J, PJ) , where J denotes a complete system of invariants for the linearly independent basic forms and PJ denotes the invariants obtained from J by polar processes.

In what follows I show, in Part I, that this conjecture is indeed correct, and in fact as a simple consequence of the reduction theorem: every form in arbitrarily many rows of n variables can be represented as a sum of polars of forms that involve only n fixed rows.

In Part II I give a more conceptual proof of the reduction theorem by interpreting the question as a problem of equivalence of linear families of forms. I owe this point of view, and likewise an essential part of the reasoning employed, to Fischer's paper *On algebraic systems of modules* and to unpublished propositions of E. Fischer, whose use he kindly permitted me. Finally, in Part III I indicate how analogous considerations also lead to the most general series expansions.

I. The invariant-theoretic consequence of the reduction theorem

Let Θ be a form homogeneous in each of the $N > n$ rows

$$A_1, A_2, \dots, A_N,$$

where in general the row A_i consists of the n elements

$$a_1^{(i)}, \dots, a_n^{(i)}.$$

The coefficients of Θ may be indeterminates or quantities belonging to a given field of rationality. Let P denote an integral rational function—with rational numerical coefficients—of the polar operators

$$P_{ih} = a_1^{(i)} \frac{\partial}{\partial a_1^{(h)}} + a_2^{(i)} \frac{\partial}{\partial a_2^{(h)}} + \dots + a_n^{(i)} \frac{\partial}{\partial a_n^{(h)}},$$

where by a product such as $P_{ih}P_{jk}\Theta$ one understands the action of P_{ih} on $P_{jk}\Theta$.

Then the reduction theorem referred to above is expressed by the identity

$$\Theta = \sum P\Theta_0, \tag{1}$$

where the forms Θ_0 involve only the rows

$$A_1, A_2, \dots, A_n$$

and are obtained from Θ by polar processes P .

Now consider the basic system (F) consisting of the N forms of the same order and the same number of variables,

$$F_1, F_2, \dots, F_N,$$

whose coefficients are taken respectively to be the N rows

$$A_1, A_2, \dots, A_N.$$

Let (J) denote a complete system of invariants of

$$F_1, \dots, F_n,$$

that is, a finite set of invariants such that every invariant of

$$F_1, \dots, F_n$$

can be expressed integrally and rationally through the invariants (J) . Hilbert's conjecture to be proved is then the following: for the simultaneous invariants S of

$$F_1, \dots, F_N$$

a complete system is given by the invariants (J, PJ) .

Applying the reduction theorem (1) to a simultaneous invariant S —which can itself be regarded as a form in the N rows A_i —one sees at once that every simultaneous invariant is representable as a sum of polars of invariants that involve only the n rows

$$A_1, \dots, A_n.$$

But such forms are invariants of

$$F_1, \dots, F_n,$$

and hence can be expressed integrally and rationally through the invariants (J) . On the other hand, by the very definition of the polar processes, every form obtained from an invariant of

$$F_1, \dots, F_n$$

by a polar process is again a simultaneous invariant of

$$F_1, \dots, F_N.$$

Since the polar process preserves integral rational expressibility, it follows that every simultaneous invariant S of

$$F_1, \dots, F_N$$

can be represented integrally and rationally through the invariants (J, PJ) .

Thus we have proved:

Theorem. Every simultaneous invariant of a system of arbitrarily many basic forms is integrally and rationally expressible through the finite family (J, PJ) , where J is a complete invariant system for n linearly independent basic forms and PJ denotes all invariants obtained from J by polar operations.

This establishes Hilbert's conjecture.

On integral rational representation of the invariants of a system of arbitrarily many basic forms

II. Linear families and the proof of the reduction theorem

For the proof of the reduction theorem, we first insert some preliminaries about linear families of forms. By a *linear family of forms* over K we mean a system of forms of the same dimension so complete that, together with θ_1 and θ_2 , every combination

$$c_1\theta_1 + c_2\theta_2$$

again belongs to the system, where c_1, c_2 lie in a prescribed field K , and K is also the coefficient field of the θ . Among these forms there is always a finite number — say ϱ — that are linearly independent over K , whereas every $\varrho + 1$ forms satisfy a linear relation with coefficients in K ; the family has rank ϱ with respect to K .

We shall use the following evident fact: if L and T are two linear families over K of the same rank ϱ , and if T is a subfamily of L , then L and T are identical (equivalent).

With this understood, we return to the reduction theorem. The identity proved in Section I,

$$\theta = \sum P \cdot Z, \tag{1}$$

remains valid in analogous form if the same polar process P is applied to both sides. Thus the reduction theorem simultaneously gives a representation of *all* polars of θ by sums of the special polars PZ . On the other hand, these special polars certainly belong to the family generated by θ . The problem is therefore to prove the equivalence of suitable linear families.

For simplicity let $\theta(x, y, z)$ be of dimensions α, β, γ in the three binary rows $x_1, x_2, y_1, y_2, z_1, z_2$; the coefficients of θ are at first regarded as indeterminates a (or, if one prefers, rational numbers). We then consider the following three linear families.

1. The family L . This consists of all forms $f(x, y, z)$ obtained from θ by polar processes and having, in each of the rows x, y, z , the same dimensions as θ . In particular, L contains θ itself. If one views the $f(x, y, z)$ as forms in x, y, z and the indeterminates a , then the coefficient field is the field R of rational numbers; K must likewise be taken to be R , since only for rational c_1, c_2 does $c_1P_1 + c_2P_2$ remain a polar process. Let the rank of L over R be ϱ .

2. The family G . This consists of all forms $g(\Lambda; x, y)$ defined by

$$g(\Lambda; x, y) = f(x, y, \Lambda_1x + \Lambda_2y),$$

or, written through polar processes,

$$g(\Lambda; x, y) = \sum_{i=0}^{\gamma} \binom{\gamma}{i} \Lambda_1^{\gamma-i} \Lambda_2^i f(x, y, z)^{(i)},$$

where

$$f(x, y, z)^{(i)} = \frac{1}{i!(\gamma-i)!} \left(y \frac{\partial}{\partial z} \right)^{\gamma-i} \left(x \frac{\partial}{\partial z'} \right)^i f(x, y, z),$$

and $f(x, y, z)^{(i)}$ arises successively from $f(x, y, z)$ by the process $(yz') \frac{\partial^2}{\partial z \partial z'}$. The coefficients $g_i(x, y)$ occurring in g are therefore forms Z of the identity (1). Let G have rank δ over R .

3. The family T . This is obtained from G by the inverse polar construction by which G arose from L . Its forms $h(x, y, z)$ are defined by

$$h(x, y, z) = \sum_{i=0}^{\gamma} \binom{\gamma}{i} z_1^{\gamma-i} z_2^i g_i(x, y),$$

with

$$g_i(x, y) = \binom{\gamma}{i} f(x, y, z)^{(i)}.$$

The individual h_i , and therefore also h , are thus forms of the type PZ and sums of such forms. Let the rank of T be r .

As the construction shows, every $h(x, y, z)$ of T is obtained from θ by polar processes and has, in the rows x, y, z , the same dimensions as θ . Hence T is a subfamily of L . To conclude equivalence it remains only to prove that the ranks are equal.

Lemma a). From

$$f(x, y, \Lambda_1 x + \Lambda_2 y) = 0$$

(identically in Λ) it follows necessarily that

$$f(x, y, z) = 0.$$

This follows immediately from the identically valid relation among three binary rows

$$(xy)^2 z + (yz)^2 x + (zx)^2 y = 0,$$

which yields

$$f(x, y, (xy)^2 z + (yz)^2 x + (zx)^2 y) = 0.$$

Since $(xy)^2 = (x_1 y_2 - x_2 y_1)^2 \neq 0$, the substitution $\Lambda_1 = (yz)^2$, $\Lambda_2 = (zx)^2$ in the identity $f(x, y, \Lambda_1 x + \Lambda_2 y) = 0$ gives

$$f(x, y, z) (xy)^2 = 0,$$

hence $f(x, y, z) = 0$.

Thus there is a one-to-one correspondence between the forms of G and those of L ; moreover every linear relation in G induces, and is induced by, a linear relation in L . Therefore

$$\varrho = \delta.$$

Lemma b). From

$$h(x, y, z) = 0$$

it follows necessarily that

$$g(\Lambda; x, y) = 0.$$

Indeed, $h(x, y, z) = 0$ means that each of the terms

$$h_i(x, y) z_1^{\gamma-i} z_2^i$$

vanishes, and therefore so does every linear combination of these terms. In particular,

$$g(\Lambda; \xi, \eta) = \sum_{i=0}^{\gamma} \binom{\gamma}{i} \Lambda_1^{\gamma-i} \Lambda_2^i h_i(\xi, \eta) = 0$$

for all ξ, η , and consequently $g(\Lambda; x, y) = 0$. This proves

$$\delta = r.$$

Since $T \subseteq L$ and $\varrho = \delta = r$, the two families are equivalent. Therefore every form in L , and in particular θ itself, can be represented as a linear combination of linearly independent forms $h^{(\nu)}(x, y, z)$:

$$\theta = \sum c_\nu h^{(\nu)}(x, y, z) = \sum P \cdot Z.$$

Because this identity was derived for indeterminate coefficients a , it remains valid after replacing those indeterminates by elements of an arbitrary rationality field. Hence the reduction theorem is proved in the case of three binary rows.

The general proof, for N rows each of n variables, proceeds in completely parallel fashion. One again considers three linear families L , G , and T : the family L of all forms obtained from $\theta(A)$ by polar processes; the family G obtained from the forms of L by substituting

$$A_{n1} = \Lambda_1 A_{11} + \cdots + \Lambda_n A_{1n}, \quad \dots, \quad A_{nn} = \Lambda_1 A_{n1} + \cdots + \Lambda_n A_{n,n-1};$$

and the family T consisting of sums of forms PZ . The same two lemmas remain valid — in particular because among any $n + 1$ rows there is always an identically valid linear relation — and therefore L and T have the same rank. This yields once again the identity

$$\theta = \sum P \cdot Z,$$

first for indeterminate coefficients, hence also for coefficients lying in an arbitrary rationality field. Thus the reduction theorem is established in full generality.

III. General series expansions

We now indicate briefly how analogous considerations also produce the most general series expansions (for $N < n$). Let Z be a separately homogeneous form in the n rows $A_1, \dots, A_n$, each consisting of n variables, and let Δ denote the determinant of these rows. Then we first prove the identity corresponding to (1):

$$Z = \sum P \cdot H \pmod{\Delta}, \quad (2)$$

where the forms H contain only the rows $A_1, \dots, A_{n-1}$ and are derived from Z by processes P .

The proof consists in converting the relations derived in II into congruences modulo Δ . For simplicity let us again consider the case of three ternary rows x, y, z . The three linear families L, G, T are defined as before; write ϱ, δ, r for their ordinary ranks, and ϱ^*, r^* for the ranks of L and T modulo Δ .

One obtains, in place of Lemma a), the following.

Lemma a'). From

$$f(x, y, \Lambda_1 x + \Lambda_2 y) = 0$$

it follows necessarily that

$$f(x, y, z) = 0 \pmod{\Delta}.$$

Indeed, for any four ternary rows there is an identically valid relation

$$\Delta_x x - \Delta_y y + \Delta_z z = \Delta_t t,$$

and therefore, modulo Δ , always a linear relation among three rows,

$$\Lambda_x x - \Lambda_y y + \Lambda_z z \equiv 0 \pmod{\Delta}.$$

From this one deduces, exactly as before, that

$$f(x, y, \Lambda_1 x + \Lambda_2 y) \equiv 0 \pmod{\Delta} \Rightarrow f(x, y, z) \equiv 0 \pmod{\Delta}.$$

Hence $\varrho^* = \delta$.

Lemma b) from Section II remains valid without change, so $\delta = r^*$. To show in addition that $r = r^*$, Noether introduces another lemma.

Lemma c). From

$$h(x, y, z) \equiv 0 \pmod{\Delta}$$

it follows necessarily that

$$h(x, y, z) = 0.$$

The reason is that $h(x, y, z)$, being a polar form, vanishes under the familiar Ω -process. Thus one has a differential equation of the form

$$\Omega h = \sum_i \frac{\partial^2 h}{\partial x_i \partial y_i} = 0.$$

If now

$$h(x, y, z) = \Delta(x, y, z) k(x, y, z),$$

then after differentiating one obtains

$$\Omega h = (xy) k + \dots,$$

and from this it follows that h itself must vanish. Consequently

$$\varrho^* = \delta = r = r^*,$$

and because T is a subfamily of L , identity (2) is proved.

Applying identity (2) to the multiplier of Δ then leads to an expansion

$$Z = \varphi + \Delta \varphi_1 + \Delta^2 \varphi_2 + \dots + \Delta^r \varphi_r,$$

where each φ_i vanishes under the Ω -process. Repeated application of Ω and the elementary identity

$$\Omega(\Delta \psi) = \psi + \Delta \Omega(\psi)$$

then yields

$$\varphi_i = \sum P \cdot H_i,$$

and therefore the general expansion

$$Z = \sum P \cdot H + \Delta \sum P \cdot H_1 + \Delta^2 \sum P \cdot H_2 + \dots + \Delta^r \sum P \cdot H_r. \quad (3)$$

This is the most general series expansion of a form in n rows of n variables in terms of polars of forms involving only $n - 1$ rows. The corresponding expansion for forms in N rows of n variables ($N < n$) is then obtained, as Noether remarks, by a simple substitution argument.

She concludes that, together with the reductions and their combinations, these expansions exhaust the known expansion formulas for forms in arbitrarily many rows of n cogredient variables.

Erlangen, 5 January 1915.

An algebraic criterion for absolute irreducibility

A polynomial with coefficients in an arbitrarily given field is called *absolutely irreducible* if it remains irreducible in an algebraically closed field extending the coefficient field. I shall show below that, in contrast with irreducibility relative to a prescribed field, absolute irreducibility can be formulated algebraically. More precisely, the following theorem holds.

Theorem. To every polynomial in $n > 2$ variables with indeterminate coefficients one can attach a whole rational function of those coefficients and of further indeterminates — the *reducibility form* — in such a way that for every specialized polynomial of the same degree the necessary and sufficient condition for absolute irreducibility is the nonvanishing of the reducibility form. In other words: for every specialized system of coefficients for which the reducibility form is not identically zero in the auxiliary indeterminates, the polynomial is absolutely irreducible. Equivalently, the polynomial remains irreducible modulo every prime ideal of every finite algebraic field containing the coefficient field, with at most finitely many exceptional prime ideals. In particular, the containing field may be the rational numbers.

Further applications to the theory of relatively integral functions will be given elsewhere.

§1. The universal product relation

We first show that every polynomial in more than two variables with indeterminate coefficients is absolutely irreducible. Write

$$F(x) = \sum A_{i_1 \dots i_n} x_1^{i_1} \cdots x_n^{i_n}, \quad i_1 + \cdots + i_n = l, \quad (0.1)$$

$$G(x) = \sum B_{j_1 \dots j_n} x_1^{j_1} \cdots x_n^{j_n}, \quad j_1 + \cdots + j_n = m, \quad (0.2)$$

$$H(x) = F(x)G(x) = \sum C_{k_1 \dots k_n}(A, B) x_1^{k_1} \cdots x_n^{k_n}, \quad k_1 + \cdots + k_n \leq l + m. \quad (0.3)$$

Here the A and B are indeterminates, and the coefficients $C(A, B)$ are integral bilinear expressions in the A 's and B 's.

Consequently the quotients

$$D = \frac{C(A, B)}{C_{0 \dots 0}(A, B)}$$

are rational functions of the quotients $A/A_{0 \dots 0}$ and $B/B_{0 \dots 0}$. But the number of such functions is larger than the number of their arguments: namely, the three respective counts are

$$\binom{l+n}{n} - 1, \quad \binom{m+n}{n} - 1, \quad \binom{l+m+n}{n} - 1,$$

and one has the inequality

$$\binom{l+n}{n} + \binom{m+n}{n} \geq \frac{(l+m+n)!}{n!(l+m)!} = \binom{l+m+n}{n}.$$

Therefore there is at least one algebraic relation among the $D(A, B)$, and hence also among the coefficients $C(A, B)$:

$$\Phi(Z) \not\equiv 0 \quad (\text{identically in } Z), \quad \Phi(C(A, B)) = 0 \quad (\text{identically in } A, B). \quad (0.4)$$

Here Φ denotes an integral polynomial.

Hence a polynomial with indeterminate coefficients,

$$E(x) = \sum Z_{k_1 \dots k_n} x_1^{k_1} \cdots x_n^{k_n}, \quad k_1 + \cdots + k_n \leq l + m, \quad (0.5)$$

is necessarily absolutely irreducible for $n > 2$.

§2. The prime ideal of reducibility relations

The totality of all such polynomials $\Phi(Z)$ that vanish identically after the substitution $Z = C(A, B)$ forms an ideal in the polynomial ring; more precisely it forms a *prime ideal* $\mathfrak{B}$. Indeed, if a product $\Phi_1\Phi_2$ vanishes identically under the substitution, then at least one factor vanishes identically.

Since relation (0.4) holds identically in A, B , it remains valid for every specialized choice of A, B , where A, B , and therefore also $\Gamma = C(A, B)$, may be elements of an arbitrary field. Thus the coefficients Γ of every polynomial that becomes reducible in an extension field satisfy the relations

$$\Phi(\Gamma) = 0;$$

in other words, they are zeros of the prime ideal $\mathfrak{B}$, equivalently of finitely many basis polynomials for $\mathfrak{B}$. What remains is to prove the converse.

For this it is important to note that all relations among A, B, C remain valid if in the polynomials (0.1)–(0.3) we homogenize by adding a new variable y and pass to homogeneous forms $F(x, y)$, $G(x, y)$, $H(x, y)$ of dimensions $l, m, l + m$. Thus coefficients Γ may occur among the zeros of $\mathfrak{B}$ for which, say, $F(x, y)$ degenerates to a power of y ; in passing back to inhomogeneous polynomials this corresponds to a lowering of degree for $H(x)$ rather than to an actual factorization. It will be shown that, apart from this degree-lowering phenomenon, the converse always holds. In particular, for homogeneous forms the converse holds without exception, provided not all coefficients of F vanish; that is, every zero of $\mathfrak{B}$ is produced by a parameter representation

$$\Gamma = C(A, B)$$

with coefficients A, B in a suitable extension field.

Noether proves this by constructing finitely many special functions $\Phi(Z)$ directly. She associates to the polynomials (0.1)–(0.3) corresponding one-variable polynomials by means of Kronecker substitution, shows that gaps occur among the resulting exponents, and from the norms of the corresponding coefficients obtains the desired functions $\Phi(Z)$ and therefore the criterion.

§3. Kronecker substitution

Set

$$d = l + m + 1. \tag{0.6}$$

Then the Kronecker substitution associates to the polynomials F, G, H the one-variable polynomials

$$f(\xi) = \sum A_{i_1 \dots i_n} \xi^{i_1 + i_2 d + i_3 d^2 + \dots + i_n d^{n-1}}, \tag{0.7}$$

$$g(\xi) = \sum B_{j_1 \dots j_n} \xi^{j_1 + j_2 d + j_3 d^2 + \dots + j_n d^{n-1}}, \tag{0.8}$$

$$h(\xi) = \sum C_{k_1 \dots k_n}(A, B) \xi^{k_1 + k_2 d + \dots + k_n d^{n-1}}, \quad k_1 + \dots + k_n \leq l + m. \tag{0.9}$$

This correspondence is one-to-one, because from an induced exponent one can recover the exponents in the original monomial: the base- d expansion is unique, and since $d = l + m + 1$ no carrying occurs among the exponents that arise from multiplication.

Thus one has simply

$$h(\xi) = f(\xi)g(\xi), \tag{0.10}$$

and conversely, whenever a relation of the form (0.10) holds in such a way that no exponents other than the induced ones occur in f and g , one may recover the original multivariable factorization

$$H(x) = F(x)G(x). \tag{0.11}$$

Moreover, genuine gaps do appear among the induced exponents in $f(\xi)$ and $g(\xi)$. The highest exponent in $f(\xi)$ is ld^{n-1} , while the second highest is $(l-1)d^{n-1} + d^{n-2}$. Their difference is

$$d^{n-2}(d-1) > 1,$$

because $d = l + m + 1 > 3$ and $n > 2$. The same holds for $g(\xi)$.

The relations (0.10) and (0.11), first established for the indeterminates A, B , remain valid after every specialization of A, B . To preserve the degree one homogenizes (0.10) and (0.11) by means of a new variable η (and likewise y in the multivariable side). Thus a homogeneous form $H(x, y)$ factors in an algebraic extension field if and only if the corresponding binary form $h(\xi, \eta)$ splits into two factors in such a way that only the induced exponents appear in the factors.

This is the point at which Noether passes from the mere existence of algebraic relations to the construction of the reducibility form via norms. The later sections produce the explicit criterion from these one-variable relations.

1 An algebraic criterion for absolute irreducibility: completion

4. Norms of the auxiliary linear form

Let

$$p = ld^{n-1}, \quad q = md^{n-1},$$

so that these are the degrees of the one-variable polynomials $f(\xi)$ and $g(\xi)$ obtained above by Kronecker substitution. Write

$$\prod(\xi - a_i\eta) = \xi^p + r_1\xi^{p-1}\eta + \cdots + r_p\eta^p,$$

$$\prod(\xi - b_j\eta) = \xi^q + s_1\xi^{q-1}\eta + \cdots + s_q\eta^q,$$

and

$$\prod(\xi - c_k\eta) = \xi^{p+q} + t_1\xi^{p+q-1}\eta + \cdots + t_{p+q}\eta^{p+q}.$$

Thus r, s, t denote the homogenized elementary symmetric functions in the rows a and b ; equivalently, they are the symmetric functions, linear in each separate row, from which all integral symmetric functions that are homogeneous and of equal degree in each row are composed uniquely and integrally.

Introduce further indeterminates $u_{\mu\nu}$ and form the linear expression

$$\ell(u, a, b) = \sum u_{\mu\nu} a^\mu b^\nu, \quad (9)$$

where the summation is extended over prescribed index pairs (μ, ν) . Multiply $\ell(u, a, b)$ by all its conjugates obtained from permutations of the rows a, b , omitting repetitions. This product is the norm $N(\ell(u))$ of $\ell(u)$ with respect to the field of the $t(a, b)$, when $\ell(u)$ is regarded as a function in the field of the $r(a, b)$ and $s(a, b)$. It is an integral symmetric function of all rows a, b , homogeneous of the same degree in each row. Hence, by the preceding discussion, it is a uniquely determined integral polynomial in the $t(a, b)$ and the $u_{\mu\nu}$:

$$N(\ell(u, a, b)) = T(t(a, b), u), \quad (10)$$

identically in a, b, u .

The same argument applied to $z - \ell(u)$ gives

$$N(z - \ell(u)) = z^\delta + T_{\delta-1}(t, u)z^{\delta-1} + \cdots + T_0(t, u),$$

where all T_i are integral polynomials in t and u . Both sides vanish for $z = \ell(u)$. Because the elementary symmetric functions $r(a, b)$ and $s(a, b)$ are algebraically independent, the identity remains an identity in r, s, u once $t(a, b)$ is replaced by the corresponding integral bilinear expressions $w(r, s)$ and $\ell(u)$ is interpreted as a function of r and s . Thus one obtains

$$\ell(u)^\delta + T_{\delta-1}(w(r, s), u)\ell(u)^{\delta-1} + \cdots + T_0(w(r, s), u) = 0, \quad (11)$$

identically in r, s, u .

The identities (10) and (11) persist under every specialization of the systems a, b , and hence of r, s . Suppose in particular that the specialized systems are chosen so that all products $a^\mu b^\nu$ that occur in (9) vanish. Then $\ell(u)$ vanishes under the substitution. If $w(\rho, \sigma) = \tau$, equation (11) gives

$$T_0(\tau, u) = 0 \quad (12)$$

identically in u .

Conversely, let $\tau_1, \dots, \tau_{p+q}$ be a system, not all zero, for which (12) holds. In an algebraic extension field there is a factorization

$$\xi^{p+q} + \tau_1\xi^{p+q-1}\eta + \cdots + \tau_{p+q}\eta^{p+q} = \prod_i(\xi - \alpha_i\eta) \prod_j(\xi - \beta_j\eta),$$

where in no factor do both α and β vanish simultaneously, although some individual α or β may vanish. Substitution in (10), using (12), shows that $\ell(u, \alpha, \beta)$ or one of its conjugate factors vanishes identically in u . After a suitable numbering of the α 's and β 's, putting $r(\alpha, \beta) = \rho$ and $s(\alpha, \beta) = \sigma$ means exactly that

$$\tau(\xi, \eta) = \rho(\xi, \eta)\sigma(\xi, \eta), \quad (13)$$

with all products that occur in $\ell(u)$ vanishing, but not all coefficients of ρ or of σ vanishing.

5. Construction of the reducibility equations

Now take p and q to be the degrees of $f(\xi)$ and $g(\xi)$ in the Kronecker substitution. Split the indices μ, ν into two classes. In the first class, μ runs through the exponents missing from $f(\xi)$ and ν through the exponents occurring in $g(\xi, \eta)$; in the second class, ν runs through the exponents missing from $g(\xi)$ and μ through the exponents occurring in $f(\xi, \eta)$.

Let

$$e(\xi) = \sum Z_k \xi^k$$

be the one-variable polynomial corresponding to the original polynomial $E(x)$ under Kronecker substitution. Define $S(Z, u)$ to be the integral polynomial in Z and u obtained from the right hand side $T(t, u)$ of (10), with summation restricted to the index classes just described, by replacing the t by the coefficients Z of $e(\xi)$ and by zeros in the positions of exponents absent from e .

If now one substitutes the coefficients A, B of F, G and the corresponding zeros of f and g for r, s , then the linear form $\ell(u)$ vanishes by the chosen summation. Moreover $t = w(r, s)$ becomes the coefficient system $C(A, B)$, together with the corresponding zeros of $h(\xi)$. Hence $T(t, u)$ becomes $S(C(A, B), u)$. From (11) one obtains

$$S(C(A, B), u) = 0 \quad (14)$$

identically in A, B, u .

Since (14) remains true under specialization, the coefficient systems

$$\Gamma = C(A, B)$$

of all specialized forms $H(x)$, and more generally $H(x, y)$, that split into two factors in an extension field, satisfy

$$S(\Gamma, u) = 0 \quad (15)$$

identically in u .

Conversely, suppose that the coefficients Γ , not all zero, of a polynomial $H(x)$ or a form $H(x, y)$ satisfy (15). This is equivalent, by the preceding construction, to $T(\Gamma, u) = 0$ for the coefficients of the one-variable form $h(\xi, \eta)$. By (13) there is, in an algebraic extension field of the coefficient field, a factorization

$$h(\xi, \eta) = f(\xi, \eta)g(\xi, \eta),$$

where, by the restricted choice of indices, no exponents occur in f and g except those induced by Kronecker substitution. Therefore this one-variable factorization comes from a factorization of the original homogeneous form,

$$H(x, y) = F(x, y)G(x, y).$$

If no factor becomes of degree zero when $y = 1$, in particular if the coefficients Γ do not lower the degree of $H(x)$, one obtains also

$$H(x) = F(x)G(x).$$

Thus (15) is necessary and sufficient for $H(x, y)$, and, after exclusion of degree lowering, also $H(x)$, to split in an extension field.

It follows further that $S(Z, u)$ cannot vanish identically in Z and u . Section 1 showed that for $n > 2$ the polynomial with indeterminate coefficients, and therefore the corresponding homogeneous form in one further variable, is absolutely irreducible. Consequently

$$S(Z, u) = \sum_i \Phi_i(Z) U_i,$$

where the U_i are monomials in the u 's and, by (14), each $\Phi_i(Z)$ belongs to the prime ideal $\mathfrak{B}$ of all reducibility relations. We have therefore shown that $\mathfrak{B}$ has no zeros other than those represented by the parameterization $\Gamma = C(A, B)$ with A, B in an algebraic extension field. The converse asserted at the end of section 2 is proved.

6. The reducibility form

The condition for absolute irreducibility can now be formulated directly. Decompose the degree of $E(x)$ in every possible way as a sum of two positive integers $l + m$. For each such decomposition form the corresponding $S(Z, u)$. The product over all decompositions,

$$R(Z, u) = \prod S(Z, u), \quad (16)$$

is the reducibility form of $E(x)$. Indeed, by the preceding result every splitting of $E(x)$, respectively of $E(x, y)$, makes one factor of (16) vanish, and conversely. This proves the main theorem stated in the introduction, and at the same time shows how the reducibility form may actually be constructed in finitely many steps.

7. Ostrowski's theorem

The existence of the reducibility form (16) immediately gives the theorem of Ostrowski cited in the introduction. Let

$$E(x) = \sum \Gamma_i x^i$$

be an absolutely irreducible polynomial whose coefficients Γ_i are algebraic integers. Let K be a finite algebraic number field containing all the Γ_i , and let $\mathfrak{p}$ be a prime ideal of K .

For the exact degree of $E(x)$ the reducibility form $R(Z, u)$ does not vanish after the substitution $Z = \Gamma$; say

$$R(\Gamma, u) = \sum_j P_j U_j \neq 0$$

identically in u . The ideal $(P_1, P_2, \dots)$ is divisible only by finitely many prime ideals of K . Hence, except for these finitely many $\mathfrak{p}$, the reduction of $R(\Gamma, u)$ modulo $\mathfrak{p}$ is not identically zero. Since the residue classes modulo $\mathfrak{p}$ again form a field, it follows that $E(x)$ modulo $\mathfrak{p}$ is irreducible, indeed absolutely irreducible. The same conclusion holds for the homogenized form $E(x, y)$, so that no lowering of degree occurs modulo $\mathfrak{p}$.

If the coefficients of $E(x)$ are algebraic fractions rather than algebraic integers, one must also exclude the finitely many prime ideals that divide a common denominator. For every remaining prime ideal one may replace $E(x)$ by a scalar multiple with algebraic integral coefficients, and the preceding argument applies. This proves the theorem.

2 Formal calculus of variations and differential invariants

The generation of all differential invariants and the derivation of the principal theorems are most uniformly carried out by the formal methods of the calculus of variations. This principle

goes back to Riemann and received its principal development, in form-theoretic terms, through Lipschitz.

Begin with the variational problem belonging to a homogeneous form $f(dx)$,

$$\delta \int f(dx) = 0. \quad (140)$$

By partial integration one is led to the invariant system of equations

$$\psi_i = 0,$$

where the Lagrange expressions ψ_i are contragredient to the differentials dx_i . Formally this is seen most simply by defining the ψ_i through the integration-free identity corresponding to partial integration, the Lagrange central equation,

$$df - dh = - \sum_i D\psi_i dx_i.$$

Here both f and df are invariants, and consequently the right hand side is invariant as well.

For a quadratic form one obtains in particular

$$\sum g_{ij} dx_i dx_j - 2 \sum g_{ij} dx_i D dx_j = -2 \sum_i \psi_i(d, d) D x_i.$$

If d is replaced by the cogredient expression $d + \lambda \delta$ and powers of λ are compared, one obtains the corresponding system of identities

$$\sum Dg_{ij} dx_i dx_j - 2 \sum g_{ij} dx_i D dx_j = -2 \sum_i \psi_i(d, d) D x_i, \quad (141a)$$

$$\sum Dg_{ij} dx_i \delta x_j - \sum g_{ij} \delta x_i D dx_j - \sum g_{ij} dx_i D \delta x_j = -2 \sum_i \psi_i(d, \delta) D x_i, \quad (141b)$$

$$\sum Dg_{ij} \delta x_i \delta x_j - 2 \sum g_{ij} \delta x_i D \delta x_j = -2 \sum_i \psi_i(\delta, \delta) D x_i. \quad (141c)$$

It follows that $\psi_i(d, \delta)$, and in particular $\psi_i(d, d)$ and $\psi_i(\delta, \delta)$, are contragredient vectors. From (141) one computes

$$\psi_i(d, \delta) = \sum_j g_{ij} D \delta x_j + \sum_{j,k} \Gamma_{ijk} dx_j \delta x_k,$$

and from this the cogredient vector

$$p_i(d, \delta) = \sum_j g^{ij} \psi_j(d, \delta). \quad (142)$$

Setting $p(d, d) = 0$ gives, as is evident, the equations of the geodesics.

Knowledge of the cogredient vector $p(d, \delta)$ leads directly to covariant differentiation. If, for example, $h(d, \delta)$ is a form in the two rows dx and δx , then

$$\nabla h(dx, \delta x) = dh - \sum_i \frac{\partial h}{\partial dx_i} p_i(d, \delta) - \sum_i \frac{\partial h}{\partial \delta x_i} p_i(\delta, \delta) \quad (143)$$

(as written in Noether's notation) is a difference of invariants and hence itself an invariant; it is formed in such a way that second differentials cancel. Thus $\nabla h(dx, dx)$ is a form involving only first differentials: the covariant derivative of h . The same applies when h contains more rows of differentials.

The construction of the curvature form in Riemann and Lipschitz rests on the same principle. One passes from $\delta^2 f$ to the "normal form of the second variation"

$$Q = \sum D^2 g_{ij} dx_i dx_j - 2 \sum d\delta g_{ij} dx_i \delta x_j + \sum \delta^2 g_{ij} \delta x_i \delta x_j, \quad (144)$$

which, being an aggregate of invariants, is itself an invariant, and in which the third differentials have now been eliminated, generalizing the Lagrange central equation. The second differentials are eliminated in analogy with covariant differentiation by forming

$$K = Q - 2 \sum_i \{p_i(d, \delta)\psi_i(d, \delta) - p_i(\delta, d)\psi_i(d, d)\}. \quad (145)$$

This may also be written in the form

$$K = dD\psi(d, d) - 2D\psi(d, \delta)dx - 2\{p(d, \delta)\psi(d, \delta) - p(d, d)\psi(d, \delta)\}. \quad (146)$$

The law of construction of K may be phrased as follows: in $\delta^2 f$ the second differentials are eliminated by setting $p(d, d)$ and $p(d, \delta)$, equivalently the right hand sides of (141), equal to zero. This is Riemann's definition of the curvature form. It must be explicitly noted, however – and (146) makes this especially clear – that this zero substitution may be made only after the differentiation has been carried out. Lipschitz's formal construction avoids this difficulty. Correspondingly, the covariant derivative (143) may be defined by eliminating the second differentials in dh through setting the expressions $p(d, \delta)$ and the analogous ones equal to zero.

From the curvature form, by successive formation of covariant derivatives, one obtains a series of forms, Christoffel's series $[\Phi_0, \Phi_1, \dots]$. After adjoining f , their projective invariants exhaust, according to Christoffel and Ricci, the totality of the differential invariants of the quadratic form; and their equivalence with respect to linear transformations, without further integrability conditions, expresses the equivalence of two quadratic differential forms. The inner reason for this reduction theorem lies in the existence of the Riemann normal coordinates arising from the variational problem (140): they transform the geodesics through a point into straight lines and therefore transform linearly under an arbitrary transformation of the variables x . The formation of all invariants and the equivalence question are thereby reduced to the corresponding question for the homogeneous components in the expansion of f in normal coordinates; one shows that these components are linear combinations of $f, \Phi_0, \Phi_1, \dots$. For quadratic forms this is carried out in detail by Vermeil, following a note of Noether. According to that note, the theorems and methods remain valid when an arbitrary differential expression is taken as the starting point. Thus the reach of the methods of the formal calculus of variations, as compared with those of pure elimination theory, becomes apparent.

Levi-Civita gave a geometric interpretation of setting $p(d, \delta) = 0$ as the parallel displacement of a vector. This concept, later axiomatized by Weyl, is essentially restricted to quadratic forms, but it allows a generalization of another kind. It is preserved under enlargement of the group, namely multiplication of the g_{ij} by an arbitrary function, provided that, besides the quadratic form, a linear form is also taken as given. Then $p(d, \delta)$ is uniquely determined, constructions of invariants analogous to the preceding ones can be carried out, and geodesic coordinates can be defined. The equation $p(d, d) = 0$, however, no longer arises from a variational problem. Questions of reduction theorems and equivalence have not yet been attacked in full generality in this setting; only for invariants of lowest order has Weitzenböck given a reduction theorem.

Finally, one should mention the general "invariant variational problems", that is, those in which the simple or multiple integral J is invariant under a group in Lie's sense. Special cases of physical interest were studied by Hamel, Herglotz, Lorentz, Fokker, Weyl, and Klein. The principled formulation by Noether shows that invariance of J under a finite group G_ρ with ρ essential parameters corresponds to ρ linearly independent divergences; invariance under an infinite group containing ρ arbitrary functions and their derivatives up to order σ corresponds to ρ identities among the Lagrange expressions and their derivatives up to order σ . In both cases the converse holds. Since the Lagrange expressions become relative invariants of the group, this gives at the same time a process for generating invariants.

3 K. Hentzelt, revised by E. Noether: On polynomial ideals and resultants

The following concerns the general problem of elimination theory: to determine the common zeros of all polynomials in an ideal of a polynomial ring, or equivalently the zeros of any finite number of polynomials forming a basis of the ideal. At the same time the multiplicities that occur will receive a conceptual interpretation. The coefficients of the polynomials may belong to any field; the zeros then belong to the algebraically closed field derived from it. Moreover the ideal may, without loss of generality, be assumed to be transformed, in the sense introduced below. The essential result is then the following.

To every polynomial ideal $\mathfrak{m}$ there may be attached a resultant form

$$R_{\mathfrak{m}} = R^{(1)}(x_1, \dots, x_n) \cdots R^{(i)}(x_i, \dots, x_n) \cdots R^{(n)}(x_n) \equiv 0(\mathfrak{m}), \quad (\text{A})$$

so that $R_{\mathfrak{m}}$ vanishes for all zeros of $\mathfrak{m}$ and only for them. If $\mathfrak{n}$ is a divisor of $\mathfrak{m}$ and the resultant forms $R_{\mathfrak{n}}$ and $R_{\mathfrak{m}}$ agree, then the ideals $\mathfrak{n}$ and $\mathfrak{m}$ themselves agree.

The second part of the theorem says that the resultant form gives the zeros with their characteristic multiplicity. It is analogous to the theorem that two ideals in an algebraic number field, one of which divides the other, are equal if and only if their norms agree. The inner reason for both theorems is the same.

Indeed $\mathfrak{m}$ may be considered as a module M_{n-1} of linear forms: each polynomial is regarded as a linear form in the monomials in $x_1, \dots, x_{n-1}$, and x_n is adjoined to the coefficient domain. The resultant factor $R^{(n)}$ then becomes the norm of the corresponding fundamental module G_{n-1} after M_{n-1} . This at once gives an interpretation of the multiplicity – the product of degree and exponent of a factor of $R^{(n)}$ – as the number of linearly independent residue classes. This fact was previously known only in the special case in which the resultant can be defined for indeterminate coefficients.

The derivation proceeds as follows. Sections 1 and 2 give theorems on modules of linear forms whose coefficients are either rational integers or polynomials in one indeterminate. Section 3 introduces ideals of polynomials; the isomorphism theorem of section 4 connects these ideals with the preceding module theory. Section 5 proves the second part of the main theorem. Section 7 gives the theorem about zeros, after section 6 has developed a general elimination theory. In section 6, by adjoining new indeterminates, the decomposition of the resolvent into linear forms in these indeterminates is also proved. This supplies, for the elimination method used here, an unambiguous proof of a claim made by Kronecker.

3.1 1. Modules of linear forms

In the first two sections the phrase “integral quantities” means rational integers or polynomials in one indeterminate with coefficients in an arbitrary abstract field P . By a linear form $a(\xi)$ one understands a linear expression in finitely many indeterminates $\xi_1, \dots, \xi_k$ with integral quantities as coefficients.

A module M of linear forms is defined by the following closure conditions: together with $a(\xi)$ and $b(\xi)$ it contains $a(\xi) - b(\xi)$; together with $a(\xi)$ it contains $c a(\xi)$, where c is any integral quantity.

The rank of M is, as usual, the maximum number of linearly independent linear forms in M . The h -th determinant divisor d_h of M is the greatest common divisor of all determinants of order h formed from coefficient matrices of h forms of M . If ρ is the rank, then the non-zero determinant divisors are $d_1, \dots, d_{\rho}$, and the elementary divisors are

$$e_1 = d_1, \quad e_2 = d_2/d_1, \dots, \quad e_{\rho} = d_{\rho}/d_{\rho-1}.$$

The module is finite: it has a module basis consisting of exactly ρ linear forms $a_1(\xi), \dots, a_\rho(\xi)$ from which every form of M is represented linearly with integral coefficients. After an unimodular substitution $\xi = \Theta(\eta)$ the elementary divisor theorem gives the module equality

$$M = (e_1\eta_1, e_2\eta_2, \dots, e_\rho\eta_\rho). \quad (1)$$

Definition I. A module G is called a *fundamental module* if it has no proper divisor of the same rank.

Equivalently, G is a fundamental module precisely when from

$$cg(\xi) \equiv 0(G), \quad c \neq 0, \quad (2)$$

there follows $g(\xi) \equiv 0(G)$.

The relation between an arbitrary module and a fundamental module is given by the following theorem.

Theorem I. Every module M is divisible by one and only one fundamental module G of the same rank. This G is the smallest divisor of M of the same rank and, in suitable coordinates, is represented by

$$G = (\eta_1, \eta_2, \dots, \eta_\rho). \quad (3)$$

It will be called the fundamental module of M .

The assertion that G is the smallest divisor of the same rank means that G consists exactly of those linear forms $g(\xi)$ for which there is a relation

$$bg(\xi) \equiv 0(M), \quad b \neq 0. \quad (4)$$

From this description G is plainly a module. It is fundamental: if $cg(\xi) \equiv 0(G)$ with $c \neq 0$, then multiplying (4) by c gives $bcbg(\xi) \equiv 0(M)$, whence again $g(\xi) \equiv 0(G)$. No second fundamental divisor of the same rank can exist, for it would have to divide G and, being fundamental, could not be a proper divisor. The representation (3) is fundamental, since any proper divisor of it would have to contain an additional indeterminate and would therefore have higher rank.

A second concept needed later is Dedekind's quotient of two modules. If M and B are modules of linear forms, define the quotient M/B to be the set of integral quantities c satisfying

$$cB \equiv 0(M).$$

This is an ideal of integral quantities, hence a principal ideal in the present coefficient domains. Similarly, if $\mathfrak{b}$ is an ideal of integral quantities, define $M/\mathfrak{b}$ to be the set of linear forms $c(\xi)$ satisfying

$$c(\xi)\mathfrak{b} \equiv 0(M).$$

This is again a module of linear forms, and, provided $\mathfrak{b} \neq 0$, it has the same rank as M .

The quotient concept gives the reciprocal relation between a module's fundamental module and its highest elementary divisor.

Theorem II. If e_ρ is the highest elementary divisor of M and G its fundamental module, then

$$(e_\rho) = M/G, \quad G = M/(e_\rho). \quad (5)$$

Indeed, since every elementary divisor divides e_ρ , (1) and (3) give

$$e_\rho G \equiv 0(M), \quad (e_\rho)G \equiv 0(M). \quad (6)$$

Thus (e_ρ) is divisible by the quotient M/G . Conversely, from $bG \equiv 0(M)$ one obtains $b\eta_i \equiv 0(M)$ for each i , hence b is divisible by e_ρ .

For later use it is essential that the latter quotient representation of G can also be proved without recourse to the elementary divisors. This gives it greater generality.

Theorem III. If the integral quantities are polynomials in several indeterminates, then still

$$G = M/(d), \quad (7)$$

where d is any non-identically vanishing determinant of order ρ from M .

The notions of rank, fundamental module, and module quotient remain valid in this enlarged setting, although a basis representation need no longer be available. Let $a_1(\xi), \dots, a_\rho(\xi)$ be linearly independent forms of M and choose the indeterminates so that $d = |a_{ij}| \neq 0$. Then M contains special forms of the shape

$$d\xi_i + \sum_{j>\rho} b_{ij}\xi_j, \quad i = 1, \dots, \rho. \quad (8)$$

It cannot contain a non-zero form depending only on $\xi_1, \dots, \xi_\rho$, for then a determinant of order $\rho + 1$ would be non-zero. Since $M/(d)$ is a divisor of the same rank as M , it is divisible by G . Conversely, if $g \in G$, then $cg \in M$ for some $c \neq 0$; multiplying by d and using (8) shows that $dg \in M$, whence $g \in M/(d)$. This proves (7).

3.2 2. Decomposition theorems for norms and modules

We return to modules of linear forms over rational integers or over polynomials in one indeterminate over P .

Definition III. Group all linear forms congruent modulo M into residue classes. The symbol G/M denotes the system of residue classes modulo M that are represented by elements of G . The norm of G after M , written $N(G/M)$, is the determinant of the transition substitution from G to M , that is, the determinant of a substitution expressing any linearly independent module basis of M through one of the fundamental module G of M .

From (1) and (3), or from Theorem II, one obtains

$$N(G/M) = e_1 e_2 \cdots e_\rho, \quad G = M/(N(G/M)). \quad (9)$$

If the coefficients are rational integers, $N(G/M)$ is the number of residue classes of G modulo M . If the coefficients are polynomials in one indeterminate, the number of residue classes is generally infinite; the degree of $N(G/M)$ is then the number of residue classes linearly independent over P .

If B is a divisor of M of the same rank, then B contains either all representatives of a residue class or none. It is obtained by adjoining finitely many residue classes and their linear combinations to M . Consequently:

Theorem IV. If B is a divisor of M of the same rank, so that B and M have the same fundamental module G , and if

$$N(G/B) = N(G/M),$$

then $B = M$.

Theorem V. Suppose

$$N(G/M) = rs, \quad (r, s) = 1. \quad (10)$$

Then there is one and only one module R , and one and only one module S , both divisors of M of the same rank, such that

$$r = N(G/R), \quad s = N(G/S).$$

They are

$$R = M/(s), \quad S = M/(r), \quad (11)$$

and M is the least common multiple of R and S in the module sense. Moreover, if $r_i = (r, e_i)$ and $s_i = (s, e_i)$, then the elementary divisors of R and S are respectively the r_i and s_i .

The proof is the usual elementary-divisor argument. In coordinates where $M = (e_1\eta_1, \dots, e_\rho\eta_\rho)$ and $G = (\eta_1, \dots, \eta_\rho)$, the coprimeness of r and s separates the elementary divisors into relatively prime parts. Conversely, any modules satisfying the norm conditions have the same elementary divisors and the same fundamental module as the constructed modules; Theorem IV then forces equality.

3.3 3. Ideals of polynomials

Let $\mathfrak{m}$ be an ideal of polynomials in n indeterminates $x_1, \dots, x_n$ with coefficients in the field P . Such an ideal is, in the usual sense, a set closed under subtraction and under multiplication by arbitrary polynomials.

An ideal $\mathfrak{m}$ is called *transformed* when it remains unchanged, in the relevant sense, under a sufficiently general triangular linear transformation of the variables. By adjoining new indeterminates to the field P one may always pass to a transformed ideal. The purpose of doing so is to avoid exceptional positions in elimination: the successive resultants then have the expected degrees and the associated factors separate correctly.

The basic constructions are these. Regard the polynomials of $\mathfrak{m}$ as linear forms in the monomials in $x_1, \dots, x_i$ with coefficients in the polynomial ring $P[x_{i+1}, \dots, x_n]$. This gives a module M_i . Its fundamental module is denoted G_i . The highest elementary divisor of M_i is denoted $E^{(i+1)}$; in the elimination setting these elementary divisors are the individual resultant factors. Their product over the successive steps gives the resultant form $R_{\mathfrak{m}}$.

The highest elementary divisor is also characterized as a greatest common divisor of the maximal determinants of the coefficient matrix of the module. Thus the construction is independent of the chosen basis of the ideal. This independence is crucial: the resultant form belongs to the ideal and not to an arbitrarily selected system of generators.

For a transformed ideal, the fundamental ideals $\mathfrak{g}_i$ of the successive stages are again transformed. This is a special case of the following assertion: if $\mathfrak{m}$ is transformed, then the fundamental ideal obtained from the associated module is transformed. The proof is a direct application of the determinant criterion and of the invariance of the determinant divisors under the admissible transformations.

3.4 4. The isomorphism theorem

The relation between polynomial ideals and modules of linear forms is governed by an isomorphism theorem. Let M_i be the module obtained from $\mathfrak{m}$ by treating the polynomials as linear forms in the monomials in $x_1, \dots, x_i$, with coefficients in the remaining variables. Then the residue class system of the module agrees with the corresponding residue class system of the ideal after adjoining the remaining variables. In symbols, the construction gives an isomorphism of the form

$$G_i/M_i \simeq \mathfrak{g}_i/\mathfrak{m}$$

within the appropriate coefficient domain.

This theorem is proved by induction on i . For $i = 1$ the assertion is immediate, because one is dealing with ordinary linear forms. In the induction step one first writes the module at stage i in terms of the previous stage, then separates coefficients with respect to the new variable. The determinant divisors and the fundamental module behave functorially under this passage. Hence the quotient system of the polynomial ideal is faithfully reflected by the quotient system of the associated module.

The practical consequence is that statements about norms, elementary divisors, and residue classes of modules can be transported to polynomial ideals. Thus, when one says that a resultant factor is the norm of a fundamental module, this is simultaneously a statement about the residue classes of the corresponding polynomial ideal.

3.5 5. Norms, elementary factors, and equality of ideals

We now prove the second part of the main theorem. Let $\mathfrak{n}$ be a divisor of $\mathfrak{m}$ in the ideal sense, and suppose that the resultant forms of $\mathfrak{n}$ and $\mathfrak{m}$ agree. Passing successively to the associated modules, the equality of the resultant factors says precisely that the corresponding norms agree at each stage. By Theorem IV the modules at each stage are equal. By the isomorphism theorem the associated fundamental ideals and residue class systems are therefore equal. Running the construction back through the variables gives

$$\mathfrak{n} = \mathfrak{m}.$$

Equivalently, the resultant form does not merely locate the zero set; it records the characteristic multiplicity with which the zero set occurs. If a factor occurs with exponent a and degree d , then the product ad is the number of linearly independent residue classes represented by the corresponding component. This is the precise sense in which the resultant form supplies the multiplicities.

3.6 6. Successive elimination

The elimination theory is now developed by eliminating the variables one after another. Start from a transformed ideal $\mathfrak{a}$ and form, at each stage, the greatest common divisor of those polynomials that remain after viewing the preceding variables as coefficients. The successive greatest common divisors are denoted, in Noether's notation, by expressions such as $D^{(i)}$ and $H^{(i)}$; they are primitive in the new indeterminates and regular in the variable to be eliminated.

If a polynomial of the ideal has a zero $\alpha = (\alpha_1, \dots, \alpha_n)$, then the first elimination polynomial vanishes at the corresponding value of the eliminated variable. Conversely, because the ideal is transformed and because the determinant divisors are chosen in the general position secured by the transformation, every linear factor of the elimination polynomial extends in one and only one way to a complete zero system of the ideal after adjoining the remaining parameters.

In concrete terms, after introducing new indeterminates $u_1, \dots, u_n$ and setting

$$z = u_1x_1 + \dots + u_nx_n,$$

one obtains a resolvent polynomial whose factors have the form

$$z - (u_1\alpha_1 + \dots + u_n\alpha_n).$$

Noether's argument shows that this factorization exhausts the resolvent: no residual factor remains which is irreducible in z and the u 's. If such a factor existed, it would acquire a linear factor in an algebraic extension; the corresponding zero would have to coincide with one already obtained from the ideal, contradicting the assumption that a further factor remained.

Thus the explicit decomposition of the resolvent is

$$H(z, u) = \prod (z - (u_1\alpha_1 + \dots + u_n\alpha_n))^{e_\alpha}, \quad (36)$$

where the exponents are determined by the corresponding residue-class multiplicities. The construction differs from Kronecker's method in that it depends only on the given ideal and not on a choice of basis of that ideal.

3.7 7. Resultant form and elimination theory

To connect the resultant form with elimination theory, apply the successive elimination just described to the quotient of the ideal by the fundamental ideal of stage $n - 1$:

$$\mathfrak{a} = \mathfrak{m}/\mathfrak{g}_{n-1}.$$

First one must verify that this quotient is again transformed. Since $\mathfrak{m}$ is transformed, and since the fundamental ideals of transformed ideals are transformed, the quotient inherits the transformation property.

The determinant construction shows that the highest elementary divisor $E^{(n)}(x)$ divides the corresponding elimination polynomial $D^{(n)}(x)$. Conversely, the divisibility properties of the quotient and the regularity conditions show that $D^{(n)}(x)$ divides the resultant factor, up to the multiplicities already accounted for by the elementary divisors. The same reasoning remains valid after the auxiliary variables u are introduced, so that the factorizations of the resolvent and of the resultant agree term by term.

The conclusion is:

Theorem XIII. If the resultant form $R^{(n)}(x)$ is decomposed in an algebraic extension field of $P(u, x_{i+1}, \dots, x_n)$ into linear factors

$$x_i - \alpha_i,$$

then each α_i can be completed in one and only one way to a complete zero system

$$\alpha_1, \dots, \alpha_n$$

of $\mathfrak{m}/\mathfrak{g}_{n-1}$, and hence of $\mathfrak{m}$. The finitely many zeros of $\mathfrak{m}$ arising in this way are summarized by the explicit decomposition

$$R(z, u) = \prod (z - (u_1\alpha_1 + \dots + u_n\alpha_n))^{e_\alpha}. \quad (37)$$

This decomposition remains valid, because of the regularity of the resultant in z , after the adjoined parameters are specialized to arbitrary values in $P(u)$ or in its algebraic closure.

This also separates the zeros of $\mathfrak{m}$ according to the individual factors of the resultant form. The number of indeterminates that must be adjoined to $P(u)$ is called the dimension of the corresponding algebraic set.

Finally group, in the decomposition of $R(z, u)$, those linear factors that are conjugate over $P(u, x_{i+1}, \dots, x_n)$. One obtains a decomposition of the resultant into powers of irreducible polynomials

$$R^{(n)}(x) = S_1(x)^{\lambda_1} \dots S_r(x)^{\lambda_r}.$$

To this decomposition corresponds a representation of the relevant fundamental ideal as an intersection, or least common multiple in the ideal sense, of components. Each S_j is the norm of the corresponding component, and the exponent λ_j has the residue-class meaning already described: if γ_j is the degree of S_j , then the multiplicity $\lambda_j\gamma_j$ is the number of residue classes linearly independent over the appropriate coefficient field.

When the coefficient field P has characteristic zero, the auxiliary indeterminates u_i may be specialized to elements of P so that the required regularity of the determinants is preserved. One may even choose the specialization so that the resultant remains the greatest common divisor of finitely many maximal determinants. Then the whole elimination theory is retained after specialization; the factorization is obtained simply by substituting the chosen numerical values for the u_i in the general formula.

4 Algebraic and differential invariants

If one is to report on the development of algebraic and differential invariants, it is natural in both areas to restrict attention to the critical period which, according to Hilbert's remark, follows the naive and formal period. This critical period is characterized, for algebraic invariants, by Hilbert himself; for differential invariants, by Riemann. In substance, it is characterized in the first case by the arithmetic methods of algebra, which developed essentially in connection with these questions and here acquired their sharpest form; in the second, by the methods of the formal calculus of variations.

I shall therefore discuss the arithmetic methods and their significance beyond the special subject, while in the second part I shall restrict myself to the subordination of differential invariants to algebraic invariants, precisely as achieved in connection with the methods of the calculus of variations.

4.1 1. The algebraic invariant problem

The algebraic invariants are based, as is well known, on the general linear group in $x_1, \dots, x_n$:

$$x_i = \sum_j s_{ij} x'_j, \quad \text{or shortly } x = S(x'), \quad (1)$$

where the s_{ij} are indeterminates and $\Delta = |s_{ij}|$ is non-zero. If $f(a, x)$, $g(b, x), \dots$ are forms in x of dimensions $\alpha, \beta, \dots$ with indeterminate coefficient rows $a, b, \dots$, then (1) induces transformations of $a, b, \dots$ by substituting into the forms and comparing coefficients. The transformed coefficients $a', b', \dots$ are linear homogeneous functions of $a, b, \dots$, their coefficients depending on the s_{ij} .

An invariant is a whole rational function of the indeterminates $a, b, \dots, x, y, \dots$ which, under the transformations of the variables and of the coefficient rows, reproduces itself up to a factor, namely a power of the substitution determinant:

$$J(a', b', \dots, x', y', \dots) = \Delta^\omega J(a, b, \dots, x, y, \dots). \quad (3)$$

The coefficients of J may be assumed to be rational numbers, or even integers, because every invariant with coefficients in an arbitrary number field P can be expressed linearly, with coefficients in P , through finitely many such special invariants. Likewise it is no restriction to assume J separately homogeneous in each row; any invariant is a finite sum of such components.

Conversely, the requirement (3) determines the transformations (1) and the induced transformations of the coefficients. Ostrowski and Schur have recently shown that the induced transformations can also be characterized as the totality of separate linear transformations in each row which, with certain explicit exceptions, leave a given invariant not involving the variables $x, y, \dots$ invariant.

The product of two invariants is again an invariant. A sum is an invariant if and only if the weights, the exponents ω in (3), agree. If one restricts to substitutions of determinant one, however, arbitrary sums are again invariants, and these exhaust the invariants of the restricted group. Thus one obtains an integral domain, indeed a homogeneous integral domain: if a polynomial belongs to it, then its homogeneous components belong to it individually.

Here arises the fundamental problem at which invariant theory and arithmetic algebra as a whole developed: is this integral domain finite? That is, does it possess a finite integral basis $I_1, \dots, I_k$ such that every invariant can be represented rationally and integrally by them,

$$I = G(I_1, \dots, I_k)?$$

Hilbert's solution of the problem rests on reducing the question of an integral basis to that of an ideal basis. Gordan's proof in the binary case did not extend because of the intractable symbolic calculations, and the Mertens-Hilbert proof relied on decomposing a binary form into linear factors. I sketch Hilbert's train of thought, emphasizing the arithmetic ideas, and then mention three related questions: actual construction of the basis in finitely many steps, finiteness questions for subgroups, and finiteness questions with integrality of coefficients taken into account.

Recall the ideal definition in a finite algebraic number field. A system $\mathfrak{a}$ of numbers of the field is called an ideal if it belongs to the domain $\mathfrak{o}$ of all algebraic integers of the field; if with α and β it contains $\alpha - \beta$; and if with α it contains $A\alpha$ for every $A \in \mathfrak{o}$. Every ideal has an ideal basis,

$$\mathfrak{a} = (\alpha_1, \dots, \alpha_r),$$

meaning that every element of $\mathfrak{a}$ is of the form $A_1\alpha_1 + \cdots + A_r\alpha_r$.

This definition uses only the fact that $\mathfrak{o}$ is an integral domain. It therefore carries over word for word when $\mathfrak{o}$ is replaced by any integral domain, as do the notions of ideal and ideal basis.

Hilbert's finiteness proof decomposes into three statements.

1. In the ring of all polynomials in finitely many indeterminates over a rationality domain, every ideal has a finite ideal basis.
2. A homogeneous integral domain of polynomials is finite if and only if the ideal-basis theorem holds in it.
3. For the invariant domain the ideal-basis theorem holds in the sharpened form that every ideal basis formed relative to the full polynomial ring is at the same time an ideal basis relative to the invariant domain.

In Hilbert's proof the last assertion follows from a known theorem concerning the Ω -process. E. Fischer later showed that the assertion follows already from the characteristic property of the general linear group that together with each transformation it contains the contragredient, respectively the conjugate-complex one.

4.2 2. Function fields and actual construction

The actual construction of the basis depends on a further arithmetic transformation of the finiteness concept, using function-field theory.

An integral domain $\mathcal{J}$ of polynomials is finite if and only if it contains a finite subdomain $\mathcal{H}$ such that $\mathcal{J}$ depends algebraically and integrally on $\mathcal{H}$. Any integral basis of $\mathcal{H}$ is then supplemented by a fundamental system of the algebraic integral elements to give a basis of $\mathcal{J}$.

In the case of homogeneous domains $\mathcal{J}$, such as invariant domains, for which the ideal-basis theorem holds in the sharpened form above, such a finite subdomain $\mathcal{H}$ can be obtained from finitely many polynomials $J_1, \dots, J_s$ with the property that the vanishing of these polynomials forces the vanishing of all polynomials of $\mathcal{J}$. This is again based on a general theorem of ideal theory: to every ideal of polynomials one can attach a finite set of functions that controls its zeros and hence the integral dependence of other functions over the generated subdomain.

From this upper bound K. Hentzelt obtained an upper bound for the degrees of the invariants needed for a basis. This makes the construction finite in principle, although not necessarily practical in the older symbolic sense.

4.3 3. Subgroups and integrality

The second adjoining question concerns invariants of subgroups. If one replaces the full linear group by a subgroup, then the invariant domain need not automatically satisfy Hilbert's original argument. Nevertheless, for many groups the arithmetic method persists. In particular, if the invariant domain is integral over a finite polynomial subdomain, then the same finiteness conclusion follows. This applies to finite groups and to many of the classical groups appearing in invariant theory.

The third question concerns integrality. One asks not only for rational representation of invariants by a finite basis, but for integral rational representation, with integer coefficients when the original invariant has integer coefficients. This requires replacing mere field arguments by ideal-theoretic divisibility and congruence arguments. Noether's own work on integral rational representation shows that, after suitable choice of the basic invariants and of an appropriate finite subring, integral dependence and ideal bases give the required representation. Thus the arithmetic method not only proves finiteness over fields; it also explains the integrality phenomena in invariant theory.

4.4 4. Differential invariants

I turn now to differential invariants and to the question raised at the beginning: their reduction to algebraic invariant theory.

The underlying group is, as usual, given by transformations

$$x_i = x_i(y_1, \dots, y_n). \quad (4)$$

The differentials transform by

$$dx_i = \sum_j \frac{\partial x_i}{\partial y_j} dy_j,$$

and the higher differentials transform by formulas obtained by differentiating repeatedly. If one starts with a differential expression

$$f(x, dx, d^2x, \dots, d^r x),$$

then a differential invariant is an expression that reproduces itself, possibly up to a relative factor, under the induced transformations of all the differential quantities.

The key observation is that these transformation formulas are algebraic in the derivatives of the coordinate transformation. If the derivatives of x with respect to y are treated as independent algebraic variables subject only to the non-vanishing of the Jacobian, then the problem of differential invariants becomes an algebraic invariant problem for a suitable algebraic group acting on the finite jet variables up to a prescribed order.

For each order r one obtains a finite algebraic problem. The higher differentials that occur in differentiating the invariant are eliminated by the formal variational methods described earlier. Thus the theory of differential invariants is subordinated to algebraic invariant theory: the differential invariants up to a fixed order are algebraic invariants of a finite system of forms or tensors built from the jets.

This reduction theorem shows its full force for a homogeneous differential expression $f(x, dx)$. The sequence of forms obtained from the curvature form and its covariant derivatives supplies the algebraic data whose projective invariants exhaust the differential invariants. Conversely, equivalence of the corresponding algebraic systems expresses the equivalence of the differential forms, subject to the integrability conditions already built into their construction.

Therefore the critical development of differential invariant theory parallels that of algebraic invariant theory. The old formal computations are replaced by structural, arithmetic-algebraic methods: finite generation, integral dependence, ideals, and elimination. In this way the theory of differential invariants enters the same conceptual circle as the theory of algebraic invariants.

5 Elimination theory and general ideal theory

Introduction

In what follows the task is to place elimination theory, in the arithmetical form that I gave to Hentzelt's exposition and that may be called an ideal theory in a polynomial domain, inside general ideal theory. At the same time the parts of the theory concerning zeros will be founded anew. The basic notions of the two theories, together with those of field theory, are collected first, so that the later argument can refer back to them.

The ordering and the new proof are grouped around four questions. First, there is an arithmetical formulation of the notion of dimension. Secondly, there is the theory of zeros. Thirdly, there is the connection between the decomposition theorems for norms and elementary divisors and the decomposition theorems of general ideal theory. Fourthly, there is the characterization of prime ideals and primary ideals by the norm and by the elementary-divisor form.

The arithmetical form of the dimension concept is given by chains of prime ideals. It is the arithmetical equivalent of the geometric fact that surfaces contain curves and curves contain points. This arithmetical formulation is shown to be identical with the parameter definition used in elimination theory. With a slight modification it can be transferred to arbitrary rings; there, together with the corresponding transfer of parts of the other questions, it gives a precise view of the structure of ideals.

For zeros Hentzelt and Noether attack the given ideal directly, as is usual, by going back to successive elimination. This does not completely avoid the character of a verification. The method followed here is the one always used for polynomials in one indeterminate. One writes the given polynomial as a product of powers of prime functions, or primary functions, and for each single prime function constructs the field of zeros as a field isomorphic to the residue-class field. The degree of the prime function gives the degree of this field; the degree of the primary function, whose zeros are the same as those of the prime function, gives the degree of the residue-class ring modulo that primary function.

Passing for each prime function to the Galois field gives the decomposition into linear factors. Thus the problem of zeros, linear factorization, and multiplicity is settled for the original polynomial. Quite analogously, for a prime ideal the system of residue classes is enlarged by forming quotients to the residue-class field, and a zero-field is constructed that is isomorphic to it. This zero-field contains a subfield generated by adjoining certain indeterminates. The degree of the norm gives the degree relative to this subfield; and by passing to the Galois field one obtains the decomposition of the elementary-divisor form, and therefore of the norm, into linear factors. For the corresponding primary ideal the zeros are the same; the decomposition is already supplied, since the norm becomes a power of the elementary-divisor form of the prime ideal. The degree of the norm gives the degree of the residue-class ring of the primary ideal relative to the subfield derived from the indeterminates.

The question for an arbitrary ideal is thereby reduced to the connection between the decomposition theorems for norms and elementary divisors and those of general ideal theory. The fact that the zeros of an ideal are assembled from the zeros of its associated prime ideals corresponds to the decomposition of the norm into greatest primary factors. These greatest primary factors become powers of the elementary-divisor forms of the individual associated prime ideals. This gives, in general, the factorization of the norm into linear factors. The multiplicity receives its interpretation, in Hentzelt–Noether, by means of fundamental ideals and their components. The degree of a greatest primary factor of the norm is equal to the number of linearly independent residue classes, after the indicated adjunction of variables, of the fundamental ideal of level $i - 1$ modulo a component of the fundamental ideal of level i .

The fundamental ideal of level $i - 1$ will be identified with the isolated component of the decomposition that is defined by all prime ideals of dimension higher than $n - i$. A component of the fundamental ideal of level i will be identified with the isolated component defined by the prime ideal corresponding to the factor of the norm and by all prime ideals of higher dimension. The degree of the corresponding factor of the norm is then the degree of that subring of residue classes of this component which consists only of classes of the fundamental ideal of level $i - 1$.

The characterization of prime ideals and proper primary ideals is obtained as a direct consequence of the consideration of residue-class fields. If the original coefficient domain is a perfect field, then to prime ideals there correspond, as norm and elementary-divisor form, prime functions; and to proper primary ideals there correspond proper primary functions, and conversely. If the coefficient field is imperfect, one can state only necessary conditions or sufficient conditions. Examples show that the norm of a prime ideal may become properly primary, and that the elementary-divisor form of a proper primary ideal may be a prime function. More precisely, in characteristic p the norm of a prime ideal becomes the p^f -th power, $f \geq 0$, of its elementary-divisor form. For proper primary ideals whose elementary-divisor form is a prime function, the norm may become an arbitrary power, as examples again show.

Finally absolute prime ideals are considered, that is, ideals which remain prime ideals after extension of the coefficient domain to an algebraically closed field. For absolute prime ideals with coefficients in a finite algebraic number field, the characterization gives a transfer of a theorem first formulated by A. Ostrowski for absolute prime functions: the property of being a prime ideal is preserved modulo every prime ideal of the number field, with at most finitely many exceptions.

The last question is analogous to ideal theory in algebraic number fields. As the elementary-divisor form of an ideal of such a field one may take the least rational integer divisible by the ideal. For a prime ideal this is always a prime number; conversely an ideal is prime as soon as its norm is prime. The proofs run in parallel. The examples show that over an imperfect field there occur analogues of prime ideals of higher degree and of ramification ideals, whereas over a perfect field there are only prime ideals of first degree and no ramification ideals.

5.1 1. Basic notions from elimination theory, general ideal theory, and field theory

Let $\mathfrak{S}$ denote the integral domain of all polynomials in variables $y_1, \dots, y_n$ with coefficients in an abstractly given field P . The variables y are subjected to a linear transformation with indeterminate coefficients u_{rs} ,

$$y_r = \sum_s u_{rs} x_s,$$

with inverse transformation

$$x_s = \sum_r t_{sr} y_r.$$

The u_{rs} , or equivalently the t_{sr} , are adjoined to P , giving the coefficient field $P(u) = P(t)$. Let $\mathfrak{S}'$ denote the polynomial domain in $x_1, \dots, x_n$ with coefficients in this enlarged coefficient field. The domains $\mathfrak{S}$ and $\mathfrak{S}'$ are the domains underlying elimination theory. Ideals in these domains are denoted by $\mathfrak{a}, \mathfrak{b}, \dots$ and $\mathfrak{a}', \mathfrak{b}', \dots$. In any ring an ideal is defined in the usual way: together with α and β it contains $\alpha - \beta$, and together with α it contains $\lambda\alpha$ for arbitrary ring element λ . The unit ideal is denoted by $\mathfrak{o}$.

An ideal $\mathfrak{m}'$ in $\mathfrak{S}'$ is called transformed if it arises from an ideal $\mathfrak{m}$ in $\mathfrak{S}$ by the transformation and by adjoining the u 's, or equivalently the t 's. Thus $\mathfrak{m}'$ consists of the linear combinations, with coefficients in the enlarged polynomial domain, of the transformed polynomials $f(x) = f(y)$ belonging to $\mathfrak{m}$. If $f(y)$ runs through all polynomials of $\mathfrak{m}$, then $\mathfrak{m}'$ contains all corresponding $f(x)$, and because the coefficients are symmetric under the relevant changes of rows and columns, it also contains the forms obtained from them by the corresponding permutations of the variables. Unless the contrary is expressly stated, every ideal occurring below is to be assumed transformed.

If an ideal is not transformed, one may make it transformed by composing the preceding transformation with an additional linear substitution with new indeterminate coefficients. For an already transformed ideal this amounts only to replacing the old transformation coefficients by bilinear combinations of the old and new coefficients.

Throughout, expressions such as $f^{(i)}, g^{(i)}, a^{(i)}, b^{(i)}$ denote polynomials in the transformed domain that are free of $x_1, \dots, x_{i-1}$.

To each ideal $\mathfrak{m}$ there are associated n fundamental ideals

$$\mathfrak{g}_0, \mathfrak{g}_1, \dots, \mathfrak{g}_{n-1}.$$

The fundamental ideal of level $i - 1$, $\mathfrak{g}_{i-1}$, consists precisely of those polynomials $G(x)$ for which there is a non-zero polynomial $B^{(i)}$ such that

$$B^{(i)} G(x) \equiv 0 \pmod{\mathfrak{m}}.$$

The polynomial $B^{(i)}$ may depend on G , but the existence of a finite ideal basis shows that one may choose a single such $B^{(i)}$ that works for all $G(x)$ under consideration. The fundamental ideals of

transformed ideals are again transformed ideals. Equivalently, $\mathfrak{g}_i$ is the ideal obtained from $\mathfrak{m}$ by adjoining $x_{i+1}, \dots, x_n$ to the coefficient field and then restricting attention to polynomials integral in $x_1, \dots, x_i$. In particular the last fundamental ideal $\mathfrak{g}_n$ is always the unit ideal.

Every polynomial $f(x)$ can be regarded as a linear form in the monomials of $x_1, \dots, x_{i-1}$, with coefficients that are polynomials in $x_i, \dots, x_n$. In this way the ideal $\mathfrak{m}$ becomes a module M_{i-1} of linear forms over the coefficient domain of these coefficients: with two linear forms it contains their difference, and with one linear form it contains every multiple by a coefficient. The fundamental module G_{i-1} consists of the linear forms $g(\xi)$ for which some non-zero coefficient $B^{(i)}$ satisfies $B^{(i)}g(\xi) \equiv 0 \pmod{M_{i-1}}$.

The elementary divisors of these modules are understood in the usual determinantal sense. If M has rank ρ , the elementary divisors $E_1, \dots, E_\rho$ are obtained from the determinant divisors; all but finitely many of them are equal to one in the situations at hand. The highest elementary divisor $E^{(i)}$ is the greatest common divisor, in the polynomial sense, of the appropriate highest minors. It may be chosen integral and primitive in the remaining variables, and regular in the leading variable.

The product $R^{(i)}$ of the non-unit elementary divisors occurring in this module representation is called the norm of G_{i-1} modulo M_{i-1} , or the individual norm of the ideal at level i . Symbolically,

$$R^{(i)} = N(G_{i-1} \mid M_{i-1}) = N(\mathfrak{g}_{i-1} \mid \mathfrak{g}_i).$$

After adjoining the remaining variables, $R^{(i)}$ is the determinant of the transition substitution from G_{i-1} to M_{i-1} . Its degree gives the number of residue classes of $\mathfrak{g}_{i-1}$ modulo $\mathfrak{g}_i$ that are linearly independent over the relevant coefficient field.

The product of the highest elementary divisors

$$E_{\mathfrak{m}} = E^{(1)}E^{(2)} \dots E^{(n)}$$

is called the elementary-divisor form of $\mathfrak{m}$. The product

$$R_{\mathfrak{m}} = R^{(1)}R^{(2)} \dots R^{(n)}$$

is called the norm, or resultant form, of $\mathfrak{m}$. Both the elementary-divisor form and the norm are divisible by the ideal. The norm is divisible by the elementary-divisor form, and a power of the elementary-divisor form is divisible by the norm. More generally, successive products of the individual factors give congruences modulo the successive fundamental ideals, and hence finally modulo $\mathfrak{m}$.

If $\mathfrak{n}$ divides $\mathfrak{m}$ and their norms $R_{\mathfrak{n}}$ and $R_{\mathfrak{m}}$ agree, then $\mathfrak{n} = \mathfrak{m}$. Indeed, for a divisor of $\mathfrak{m}$, equality of the successive individual norms implies equality of the corresponding fundamental ideals; and since the last fundamental ideal is the unit ideal, the induction proceeds backwards. Equivalently, if $\mathfrak{n}$ is a proper divisor of $\mathfrak{m}$, then the first individual norm of $\mathfrak{n}$ that differs from the corresponding one of $\mathfrak{m}$ is a proper divisor.

A prime function means a polynomial irreducible over $P(u)$; a primary function is a power of a prime function; a proper primary function is a power higher than the first. A decomposition of a polynomial into greatest primary factors is a decomposition in which every factor is primary and the product of any two factors is no longer primary. To the decomposition of an individual norm into greatest primary factors corresponds a representation of a fundamental ideal as a least common multiple of ideals

$$\mathfrak{g}_i = [\mathfrak{t}_1, \dots, \mathfrak{t}_s].$$

The factors of the individual norm determine the components $\mathfrak{t}_j$ of the fundamental ideals. The highest elementary divisor of such a component is the greatest common divisor of the corresponding factor and the corresponding elementary-divisor factor; it is therefore a greatest primary factor of the elementary-divisor form. These components are uniquely determined

by their behavior with respect to the fundamental ideals and by the requirement that their individual norm, or their highest elementary divisor, be the prescribed greatest primary factor.

The highest dimension of an ideal is defined as follows. If $R^{(i)}$ is the first individual norm different from one, then the ideal is said to have highest dimension $n - i$. The unit ideal is assigned dimension -1 . If only one individual norm different from one occurs, the highest dimension is simply called the dimension of the ideal. If $\mathfrak{m}$ contains a non-zero polynomial free of $x_1, \dots, x_{i-1}$, then the ideal has highest dimension at most $n - i$. If $\mathfrak{m}$ has highest dimension $n - i$ and $\mathfrak{n}$ divides $\mathfrak{m}$, then $\mathfrak{n}$ has highest dimension at most $n - i$.

The zeros of an ideal are all systems of values in a suitable algebraic extension field of $P(u; x_{i+1}, \dots, x_n)$ for which every polynomial of the ideal vanishes. To each factor $R^{(i)}$ of the norm there correspond, after the indicated adjunction, as many zeros of $\mathfrak{m}$ as the degree of $R^{(i)}$ gives. Writing the factor in bilinear combinations of the transformation coefficients and further indeterminates gives an explicit decomposition, whose factors correspond to associated systems of zeros. Thus among the zeros of an ideal of highest dimension $n - i$ there are zeros depending on $n - i$ parameters, but no zeros depending on more parameters. This proves the agreement between the arithmetical dimension just defined and the usual parameter definition of dimension in elimination theory.

The ring used in general ideal theory is a commutative ring in which the finite chain condition holds: every chain of ideals

$$\mathfrak{a}_1 \supset \mathfrak{a}_2 \supset \dots$$

in which each is a proper divisor of the preceding one stops after finitely many steps. This condition is equivalent to the condition that every ideal possess a finite basis; in particular it holds in the polynomial domains just considered.

In such a ring an ideal $\mathfrak{p}$ is prime if divisibility of a product by $\mathfrak{p}$ implies divisibility of at least one factor by $\mathfrak{p}$. An ideal $\mathfrak{q}$ is primary if divisibility of a product by $\mathfrak{q}$ implies divisibility of one factor by $\mathfrak{q}$ or divisibility of a power of the other factor by $\mathfrak{q}$:

$$ab \equiv 0 \pmod{\mathfrak{p}} \Rightarrow a \equiv 0 \pmod{\mathfrak{p}} \text{ or } b \equiv 0 \pmod{\mathfrak{p}},$$

$$ab \equiv 0 \pmod{\mathfrak{q}} \Rightarrow a \equiv 0 \pmod{\mathfrak{q}} \text{ or } b^r \equiv 0 \pmod{\mathfrak{q}} \text{ for some } r.$$

To every primary ideal $\mathfrak{q}$ there is a unique associated prime ideal $\mathfrak{p}$: it divides $\mathfrak{q}$, and some power of it is divisible by $\mathfrak{q}$; moreover $\mathfrak{p}$ consists of all elements of the ring some power of which is divisible by $\mathfrak{q}$. Every prime ideal is primary; proper primary ideals are those not themselves prime.

The decomposition theorem says that every ideal admits a representation

$$\mathfrak{a} = [\mathfrak{q}_1, \dots, \mathfrak{q}_s]$$

as a least common multiple of primary ideals. It may be chosen shortest, meaning that no $\mathfrak{q}_i$ occurs in the least common multiple of the others. Then the primary components $\mathfrak{q}_i$ need not be uniquely determined, but their associated prime ideals are determined by $\mathfrak{a}$. Components belonging to isolated prime ideals, that is, to associated prime ideals which contain no other associated prime ideal, are themselves uniquely determined.

Finally, from Steinitz's field theory, recall that a field is perfect if every prime function with coefficients in it splits in a suitable extension field into distinct linear factors. Otherwise it is imperfect. Every field of characteristic zero is perfect; a field of characteristic p is perfect exactly when together with every element it contains its p -th root. Over an imperfect field a prime function can be a function of x^{p^f} and split into p^f -th powers of distinct linear factors. The integer f is the exponent. A prime function is of the first kind if the exponent is zero, i.e. if it splits into distinct linear factors in a suitable extension. An extension is of the first kind if every element is of the first kind. In a finite extension of an imperfect field there is a subfield of first kind; its degree is the reduced degree, and the exponent measures the inseparable p -power part.

5.2 2. Elementary-divisor form and norm of prime and primary ideals

We now work throughout with the polynomial domain defined above. The first lemmas state that passage to the transformed domain does not change primeness or primaryness.

Lemma I. The transform of a prime ideal is prime; the transform of a primary ideal is primary. In other words, the property of being prime or primary is preserved when the indeterminates u_{rs} are adjoined to the coefficient field.

For prime ideals the proof is based on the fact that the residue classes modulo a prime ideal form a domain without zero divisors, a property preserved by adjoining indeterminates. If $a(x)b(x)$ is divisible by the transformed ideal and if neither factor is so divisible, then, after writing the factors in terms of the inverse transformation, the highest non-vanishing coefficients with respect to a lexicographic order give a product not divisible by the original prime ideal, a contradiction.

For primary ideals the argument is parallel, but uses the Dedekind–Mertens theorem. If a is not divisible by $\mathfrak{q}$ and no power of b is divisible by $\mathfrak{q}$, one chooses transformed representatives $a(x), b(x)$ with the same property. The ideals generated by their coefficients then satisfy the Dedekind–Mertens relation for a sufficiently high exponent; this prevents the coefficient ideal of the product from being divisible by $\mathfrak{q}$, and therefore the product itself is not divisible by the transformed ideal.

Lemma II. If

$$\mathfrak{m} = [\mathfrak{c}_1, \dots, \mathfrak{c}_s]$$

is a representation in the original polynomial domain, then the transformed ideal satisfies

$$\mathfrak{m}' = [\mathfrak{c}'_1, \dots, \mathfrak{c}'_s].$$

Consequently, in a shortest representation by greatest primary components, all components may be assumed transformed.

The proof is direct. Every transformed polynomial of $\mathfrak{m}$ is, after multiplication by a suitable denominator from the transformation coefficients, a linear combination of transformed polynomials from the original ideal. Conversely, a polynomial divisible by every transformed component is divisible by the transformed least common multiple. Applying Lemma I to a primary decomposition in the original domain gives a primary decomposition in the transformed domain.

The first characterization of prime and primary ideals by elementary-divisor form and norm is then the following.

Theorem I. The elementary-divisor form of a prime ideal is a prime function; its norm is a power of this prime function. For a primary ideal, elementary-divisor form and norm are primary functions, and indeed powers of the same prime function, namely the elementary-divisor form of the associated prime ideal. For prime and primary ideals the highest dimension agrees with the dimension in the ordinary sense, and this dimension is the same for a primary ideal and its associated prime ideal.

For the unit ideal the assertion is clear. Suppose $\mathfrak{p}$ is a prime ideal of highest dimension $n - i$. The congruences defining the fundamental ideals show that $\mathfrak{p}$ coincides with the appropriate fundamental ideal. All individual norms before level i are therefore one, and the elementary-divisor form is the single factor $E^{(i)}$. If $E^{(i)}$ split into a product of two non-units, the defining greatest-common-divisor property would imply that one factor already gives divisibility by $\mathfrak{p}$; hence $E^{(i)}$ is a prime function. The norm is a power of it because a power of the elementary-divisor form is divisible by the norm and the norm is divisible by the elementary-divisor form.

For a primary ideal $\mathfrak{q}$ the same argument gives a single elementary-divisor form at the relevant level. If $\mathfrak{p}$ is the associated prime ideal, then some power of $\mathfrak{p}$ is divisible by $\mathfrak{q}$, and $\mathfrak{q}$ divides $\mathfrak{p}$. The comparison of dimensions follows from these mutual divisibilities; the elementary-divisor form of $\mathfrak{q}$ and its norm are powers of the elementary-divisor form of $\mathfrak{p}$.

The dimension criterion for prime ideals is stated next.

Theorem II. A prime ideal has no proper divisor of the same highest dimension; conversely, every ideal with this property is prime.

The proof uses the following elementary fact.

Lemma III. If a prime ideal in a polynomial domain over a field has only finitely many residue classes linearly independent over that field, then it has no proper divisor different from the unit ideal. Equivalently, its residue-class system is already a field.

The hypothesis gives a finite number of residue classes such that every residue class is linearly dependent on them. Multiplication by a non-zero residue class is then an injective linear transformation of a finite-dimensional vector space over the ground field, hence also surjective. Thus every non-zero class has an inverse; the residue classes form a field.

To prove Theorem II, localize the situation at the stage determined by the highest dimension. A proper divisor of a prime ideal of the same highest dimension would, after adjoining the remaining parameters, give a proper divisor of a zero-dimensional prime ideal, contradicting Lemma III. Conversely, if an ideal $\mathfrak{q}$ has no proper divisor of the same highest dimension and nevertheless is not prime, choose a, b with $ab \equiv 0 \pmod{\mathfrak{q}}$ but neither factor divisible by $\mathfrak{q}$. The ideal $[\mathfrak{q}, a]$ is then a proper divisor; the dimension comparison, together with the finite-chain condition, contradicts the defining property. Hence $\mathfrak{q}$ is prime.

5.3 3. Residue-class fields and fields of zeros

Let $\mathfrak{p}$ be a prime ideal. The residue classes modulo $\mathfrak{p}$ form a domain. By forming quotients one obtains the residue-class field $(\mathfrak{R})$ of $\mathfrak{p}$. We shall construct a field of zeros isomorphic to $(\mathfrak{R})$ and use this to connect the norm with the usual splitting into linear factors.

Let $\mathfrak{p}$ be of dimension $n - i$. The residue classes of the last $n - i$ variables may be taken as parameters. After adjoining them, the remaining residue classes are algebraic over the field $P(u, x_{i+1}, \dots, x_n)$. The elementary-divisor form $P^{(i)}$ is the prime function of a suitable class, say the class of a fundamental linear form. The residue class of this form is algebraic over the parameter field and generates, together with the parameters, a field isomorphic to the quotient field of the residue-class domain.

Theorem III. To the residue-class field of a prime ideal $\mathfrak{p}$ there corresponds an isomorphic field of zeros. If $\mathfrak{p}$ has dimension $n - i$, this zero-field is a finite algebraic extension of the parameter field $P(u, x_{i+1}, \dots, x_n)$. Conversely every zero-field of $\mathfrak{p}$ of this transcendence degree is isomorphic to the residue-class field.

A zero $\alpha = (\alpha_1, \dots, \alpha_n)$ is obtained by sending each residue class (x_j) to the corresponding element α_j in an extension field. Since exactly the congruences of $\mathfrak{p}$ become zero, this gives an isomorphism of the residue-class field with the field $P(\alpha_1, \dots, \alpha_n)$. Conversely any zero-field gives such a residue-class homomorphism and hence the same abstract field.

Passing from the zero-field to the corresponding Galois field over the parameter field gives the factorization of the elementary-divisor form. If α_1 is a chosen primitive element, then the associated prime function in α_1 splits in the Galois field into linear factors. Returning from the transformed variables x to the original variables y , the element α_1 becomes the fundamental form

$$u_1\beta_1 + \dots + u_n\beta_n,$$

where $\beta = (\beta_1, \dots, \beta_n)$ runs through systems of zeros in the original variables depending algebraically on the parameters. Thus the linear factors of the norm are precisely the factors corresponding to the conjugate systems of zeros.

In modern terms, the residue-class field is the function field of the irreducible component determined by $\mathfrak{p}$, and the zero-field is a concrete algebraic realization of that function field in which the coordinates of the generic point are displayed. Noether's proof keeps the older elimination terminology, but the construction is this one.

Lemma IV. The elementary-divisor form of a prime ideal is the prime function attached to a primitive element of its residue-class field; its norm is the product of the elementary divisors obtained in the module representation, and hence a power of that prime function. If the coefficient field is perfect, the power is the first power.

In an imperfect field the prime function may be inseparable. Then the norm records the full degree, while the elementary-divisor form records the reduced degree. This is the source of the possible power in Theorem I.

Theorem IV. In the Galois field determined by a prime ideal $\mathfrak{p}$, the norm $N(\mathfrak{p})$ decomposes into linear factors corresponding to the conjugate zeros of $\mathfrak{p}$. The degree of the norm equals the number of residue classes linearly independent over the parameter field, counted with the inseparable exponent when the coefficient field is imperfect.

For a primary ideal $\mathfrak{q}$ associated with $\mathfrak{p}$ the zero-field is the same as that of $\mathfrak{p}$, because $\mathfrak{p}$ and $\mathfrak{q}$ have the same zeros. The norm of $\mathfrak{q}$ is a power of the elementary-divisor form of $\mathfrak{p}$; the exponent measures the length of the residue-class ring of $\mathfrak{q}$ over the residue-class field of $\mathfrak{p}$ at the corresponding level.

5.4 4. Dimension

We now compare the arithmetical dimension defined by the first non-unit norm with the usual parameter dimension.

Theorem V. The dimension of a prime ideal is given by the transcendence degree of its residue-class field over $P(u)$. If $\mathfrak{p}$ is of dimension $n - i$, then its residue-class field contains $n - i$ algebraically independent parameters and is algebraic over the field generated by these parameters.

The proof follows from the construction of the zero-field. The last $n - i$ variables can be chosen so that their residue classes are algebraically independent, while every further residue class is algebraic over them. No set of $n - i + 1$ residue classes can be algebraically independent, because the relevant elementary-divisor form gives an algebraic equation. Conversely the selected parameter classes are independent because otherwise a non-zero polynomial in them would belong to $\mathfrak{p}$, forcing a non-unit individual norm at an earlier level.

This proves the equivalence of the dimension in elimination theory with the arithmetical chain definition. Noether's formulation is deliberately independent of geometric language: an ideal has dimension $n - i$ when the first non-trivial elimination norm occurs at level i ; it has the same dimension when its residue-class field has transcendence degree $n - i$; and the same number is characterized by the maximum length of chains of prime ideals lying above it.

Theorem VI. Let

$$\mathfrak{p}_0 \supset \mathfrak{p}_1 \supset \cdots \supset \mathfrak{p}_r$$

be a chain of prime ideals. Then the dimensions strictly decrease along a proper chain. Conversely, a prime ideal of dimension $n - i$ can be placed in a chain of prime ideals of length $n - i$ ending at a maximal ideal. Thus the dimension equals the maximum length of a chain from the given prime ideal down to a maximal ideal, counted in the appropriate direction.

The geometric translation is that passing from a variety to a proper subvariety lowers the number of independent parameters. In the polynomial domain, the arithmetical proof uses divisibility of norms: a proper divisor of a prime ideal cannot have the same highest dimension by Theorem II; hence the dimension must decrease. Conversely, the existence of parameters and of algebraic equations in the remaining variables allows one successively to specialize a parameter and obtain a proper prime divisor at each step until a zero-dimensional prime ideal is reached.

Theorem VII. A prime ideal $\mathfrak{p}$ is of dimension $n - i$ if and only if it contains a prime chain of exactly i algebraic elimination steps, or equivalently if the first non-unit individual norm attached to $\mathfrak{p}$ occurs at level i .

This theorem supplies the promised arithmetical formulation of the dimension concept. It is invariant under the transformations used in elimination theory and does not depend on a distinguished basis of the ideal.

5.5 5. Connection with the decomposition theorems of general ideal theory

The components introduced in Hentzelt–Noether through fundamental ideals are now identified with the isolated components of the primary decomposition of general ideal theory. This is the main comparison theorem.

Let the associated prime ideals of $\mathfrak{m}$ be arranged by dimension:

$$\mathfrak{p}_{0,1}, \dots; \quad \mathfrak{p}_{1,1}, \dots; \quad \mathfrak{p}_{2,1}, \dots; \quad \dots,$$

where $\mathfrak{p}_{\lambda,\mu}$ denotes a prime ideal of dimension $n - \lambda$. Let $\mathfrak{q}_{\lambda,\mu}$ be the corresponding primary ideals in a fixed primary decomposition of $\mathfrak{m}$.

For a given level i , the fundamental ideal of level i is, by definition, the unit ideal for those primary components whose associated prime ideals have dimension less than the corresponding threshold; for components of higher dimension it remains the given primary component. Thus the fundamental ideals are least common multiples of precisely those primary components whose associated primes are sufficiently large in dimension.

Lemma V. The fundamental ideal of level i is the isolated component of the primary decomposition defined by all associated prime ideals of dimension greater than $n - i$. More precisely, it is the least common multiple of those primary components whose associated prime ideals have dimension greater than $n - i$.

This follows from the defining congruence for the fundamental ideals. Multiplication by a suitable polynomial in the eliminated variables removes the components in smaller dimensions, while components of higher dimension remain. Since isolated components are uniquely determined in general ideal theory, the fundamental ideals are independent of the chosen primary decomposition.

Theorem VIII. If all associated prime ideals of $\mathfrak{m}$ have the same dimension $n - i$, then the norm of $\mathfrak{m}$ decomposes into greatest primary factors corresponding one-to-one to the associated prime ideals. Each factor is a power of the elementary-divisor form of the corresponding prime ideal.

In this unmixed case the relevant fundamental ideals reduce directly to the primary components. The individual norm at level i is the only non-unit norm. Its factorization into greatest primary factors therefore mirrors the primary decomposition of $\mathfrak{m}$.

Theorem IX. In the general case, where associated prime ideals of different dimensions occur, the decomposition of the norm of $\mathfrak{m}$ is obtained by grouping the contributions according to dimension. The greatest primary factors of the norm at the first non-trivial level correspond to the isolated components of greatest dimension; after their removal the next individual norm gives the components of the next dimension, and so on.

This is precisely the arithmetical form of the statement that the zeros of an ideal are the union of the zeros of its associated prime ideals. The norm does not merely vanish at those zeros; its factorization displays the associated components in the order of their dimension.

Theorem X. The degree of a greatest primary factor of the norm corresponding to an associated prime ideal $\mathfrak{p}$ is equal to the number of residue classes, linearly independent over the parameter field, of the component of the fundamental ideal at level $i - 1$ modulo the component of the fundamental ideal at level i determined by $\mathfrak{p}$.

This theorem gives the multiplicity. Instead of assigning multiplicity by a geometric convention at the zeros, Noether assigns it by a length, or number of independent residue classes, in a quotient of isolated components. For polynomials in one indeterminate this is exactly the ordinary multiplicity of a root: the primary component is the power of the corresponding prime

ideal, and the number of residue classes is the exponent. In higher-dimensional elimination theory, the same algebraic length replaces the root multiplicity.

The theorem also explains the role of the “basic ideals” in Hentzelt–Noether. They are not additional arbitrary constructions; they are the isolated components selected by dimension in the primary decomposition. Their quotient lengths are the multiplicities recorded by the norm.

Theorem XI. The decomposition of an ideal into primary components and the decomposition of its norm into greatest primary factors determine each other at the level of associated prime ideals and multiplicities. In particular, two ideals one of which divides the other and which have the same norm must be equal.

This recovers, in the polynomial domain, the arithmetical theorem analogous to the theorem on norms of ideals in algebraic number fields.

5.6 6. Characterization of prime ideals and proper primary ideals by elementary-divisor form and norm

The preceding theorems characterized primary ideals in a way that did not yet separate prime ideals from proper primary ideals. We now obtain the separation.

Theorem XII. A necessary condition for an ideal to be prime is that its elementary-divisor form be a prime function; a sufficient condition is that its norm be a prime function. Equivalently, the elementary-divisor form of a prime ideal is always a prime function, and the norm of a proper primary ideal is always a proper primary function.

The first assertion has already been proved in Theorem I. For the sufficient condition, suppose the norm of an ideal is a prime function. If the ideal had a proper divisor of the same highest dimension, then the first individual norm in which the two ideals differ would be a proper divisor of the given prime function, impossible unless it is a unit. Thus every proper divisor has lower highest dimension, and Theorem II shows that the ideal is prime.

A second proof uses the following lemma.

Lemma VII. The residue-class system represented by polynomials modulo the elementary-divisor form $E^{(i)}$ of an ideal whose associated prime ideals all have dimension $n - i$ is isomorphic to a subsystem of the residue-class system modulo the ideal. The degree of the elementary-divisor form is the degree of this subsystem over the quotient field $P(u, x_{i+1}, \dots, x_n)$.

Indeed, classes represented by the same polynomial correspond to each other. The correspondence is injective because exactly those polynomials divisible by the elementary-divisor form represent the zero class in the subsystem. If the norm is a prime function and hence agrees in degree with the elementary-divisor form, this subsystem exhausts all residue classes. Since a residue-class system modulo a prime function has no zero divisors, the ideal is prime.

In general Theorem XII cannot be sharpened to necessary and sufficient conditions. Over imperfect fields examples show two failures: the norm of a prime ideal can be a proper power, and the elementary-divisor form of a proper primary ideal can be a prime function. However the perfect-field case is clean.

Theorem XIII. If the coefficient field P is perfect, then the norm and the elementary-divisor form of a prime ideal are prime functions and are identical. The norm and the elementary-divisor form of a proper primary ideal are proper primary functions. Thus, over a perfect field, the condition that the elementary-divisor form be a prime function is necessary and sufficient for primeness; the same condition may be stated using the norm.

It remains only to show that over a perfect field the ideal is prime whenever the elementary-divisor form is a prime function, and that the norm then equals this prime function. Since P is perfect, a prime elementary-divisor form is of first kind. The residue-class field obtained by adjoining a root is therefore separable; by the usual Lagrange formula all the necessary residue classes lie in the subsystem described in Lemma VII. That subsystem exhausts the residue classes of the ideal. It has no zero divisors, so the ideal is prime. The equality of degrees gives equality of norm and elementary-divisor form.

For imperfect coefficient fields the following sharper statements hold.

Theorem XIV. For a prime ideal over a field of characteristic p , the norm is a p^f -th power of its elementary-divisor form, where f is the exponent of the finite extension of residue-class fields. If the elementary-divisor form of a proper primary ideal is a prime function, then its norm is a power of that prime function; the exponent need not be a power of p in general.

The proof follows the field-theoretic decomposition into reduced degree and inseparable exponent. The elementary-divisor form measures the reduced degree; the norm measures the full degree. Their quotient is therefore the inseparable p^f part for prime ideals. For proper primary ideals the nilpotent, or primary, multiplicity contributes an additional exponent.

Noether gives explicit examples in characteristic 2 and 3. If P is obtained by adjoining independent indeterminates to a finite field, then an ideal such as

$$(y_1^2 + \lambda, y_2^2 + \mu)$$

can be prime, because its residue-class field is purely inseparable of degree four; its elementary-divisor form has reduced degree while its norm is the square or fourth power accordingly. Specializing parameters can produce a proper primary ideal whose elementary-divisor form is still prime. In characteristic 3, a similar construction shows that the norm exponent for a proper primary ideal need not equal the characteristic exponent predicted for prime ideals.

These examples demonstrate exactly where imperfectness enters. In a perfect coefficient field, no inseparable power is hidden in the field extension, and the elementary-divisor form and norm detect primeness without ambiguity.

5.7 7. Absolute prime ideals

A prime ideal with coefficients in a field P is called an absolute prime ideal if it remains prime after extending the coefficient field to the algebraically closed field A belonging to P . Correspondingly, a prime function is absolute if it remains irreducible, or prime, over A . The characterization is the following.

Theorem XV. A necessary and sufficient condition for a prime ideal to be absolute is that its elementary-divisor form, chosen integral and primitive in the transformation coefficients u , be an absolute prime function as a polynomial in the x and u . The elementary-divisor form then coincides with the norm, so the condition may also be stated for the norm.

The proof starts from the observation that norm and elementary-divisor form are preserved under algebraic extension of the coefficient field. Each individual factor is defined as a greatest common divisor in a polynomial ring, and this property is unchanged by algebraic extension. Passing from P to the algebraically closed field A passes in particular to a perfect field. By Theorem XIII, over A primeness is equivalent to the elementary-divisor form being a prime function. Hence the original prime ideal is absolute exactly when that form is absolutely prime. Since over A the norm equals the elementary-divisor form, and equality descends back to P by the preceding invariance, the assertion follows.

For absolute prime functions with coefficients in a finite algebraic number field, Ostrowski proved that the function remains absolute prime modulo every prime ideal of the number field with only finitely many exceptions. The transfer to ideals uses the following finite-denominator statement.

Theorem XVI. Let the coefficient domain be the ring $\mathfrak{o}^*$ of algebraic integers in a finite algebraic number field, rather than a field. Then the norm and elementary-divisor form of a polynomial ideal remain the norm and elementary-divisor form after reduction modulo every prime ideal of $\mathfrak{o}^*$, with at most finitely many exceptions.

The proof modifies the polynomial domain. Its coefficients now lie in $\mathfrak{o}^*$, while the transformation coefficients u are adjoined polynomially. In forming the module representation for the individual norms, one reduces powers of the eliminated variable modulo a polynomial regular in

that variable. The possible denominators that occur are products of finitely many polynomials in the u with coefficients in $\mathfrak{o}^*$. Their coefficients generate an ideal of $\mathfrak{o}^*$ divisible by only finitely many prime ideals. Avoiding these prime ideals preserves the module representation after reduction.

The same is done for the greatest-common-divisor property defining the norm and elementary-divisor form. One chooses the norm and elementary-divisor form so that they are divisible by the ideal over $\mathfrak{o}^*[u]$. For all but finitely many prime ideals of $\mathfrak{o}^*$ the relevant ranks of the modules, the relevant determinants, and the greatest-common-divisor relations survive reduction. Thus, modulo such a prime ideal, the reduced forms are still exactly the norm and elementary-divisor form of the reduced ideal.

Combining Theorems XV and XVI with Ostrowski's theorem gives the final result.

Theorem XVII. If $\mathfrak{p}$ is an absolute prime ideal with coefficients that are algebraic integers in a finite algebraic number field, then $\mathfrak{p}$ remains a prime ideal, more precisely an absolute prime ideal, modulo every prime ideal of the number field, with at most finitely many exceptions.

Choose the norm $R_{\mathfrak{p}}$ as in Theorem XVI so that it is divisible by $\mathfrak{p}$ over $\mathfrak{o}^*[u]$. Then

$$R_{\mathfrak{p}} = T(u)P(x),$$

where $P(x)$ may be chosen integral and primitive in the u . By Theorem XV, regarded as a polynomial in u and x with coefficients in the number field, P is an absolute prime function. Ostrowski's theorem says that P remains absolute prime modulo all but finitely many prime ideals. Avoiding also the finite exceptional set in Theorem XVI, this reduced P is the norm of the reduced ideal. Theorem XV then implies that the reduced ideal is again an absolute prime ideal.

This completes the transfer of the arithmetic theorem on absolute prime functions to absolute prime ideals.

6 Elimination theory and ideal theory

I want to show how elimination theory can be built up in complete parallel with the theory of zeros for polynomials in one indeterminate. This construction rests on a connection between the elimination theory given by Hentzelt and the methods of field theory as developed by Steinitz, and of ideal theory as I have developed it.

First consider the case of polynomials in one indeterminate. The coefficient domain is a fixed field P , and notions such as prime function are to be understood relative to P . The field may be abstractly defined in Steinitz's sense; in special cases it may be the field of rational numbers or the field of all complex numbers, as was formerly usual in elimination theory. For polynomials in one indeterminate the theory of zeros falls into four steps.

First comes the decomposition theorem, which reduces the question to prime functions, that is, to polynomials irreducible over P . From a factorization

$$a(x) = q_1(x)q_2(x) \cdots q_s(x),$$

where the underlying $p_i(x)$ are distinct prime functions and the $q_i(x)$ are their powers, hence greatest primary functions, it follows that $a(x)$ vanishes exactly at the zeros of the prime functions $p_i(x)$.

Second, one constructs the zero-field for a prime function $p(x)$. Such a field

$$K = P(\alpha), \quad p(\alpha) = 0,$$

can always be constructed, by Steinitz, as an extension field of P isomorphic to the residue-class field of $p(x)$. Conversely, every zero-field lying in an extension field of P is isomorphic to that residue-class field. Thus zero-field and residue-class field are abstractly identical.

Third, one decomposes $p(x)$ into linear factors in the Galois field determined by $p(x)$. Repetition of the construction just described leads to an essentially uniquely determined Galois field $P(\alpha_1, \dots, \alpha_r)$ large enough for $p(x)$ to split into linear factors.

Fourth, one interprets multiplicity for the original polynomial $a(x)$. By the preceding steps the zeros and the corresponding linear factorization are known for an arbitrary polynomial. The multiplicity may be taken to be the degree of the corresponding primary function $q(x)$, equivalently the number of residue classes modulo $q(x)$ that are linearly independent over P . If P is algebraically closed and the prime functions are linear, this number is exactly the usual multiplicity of the zero.

Now pass to general elimination theory. Let $\mathfrak{S}$ be the domain of all polynomials in $x_1, \dots, x_n$ with coefficients in P . Elimination theory may then be defined as the theory of the zeros of an ideal in this polynomial domain. An ideal $\mathfrak{a}$ is a system of polynomials such that with f and g it contains $f - g$, and with f it contains Af for every polynomial A in $\mathfrak{S}$. A zero of $\mathfrak{a}$ is a system of values $\alpha_1, \dots, \alpha_n$ in an extension field of P for which every polynomial f in $\mathfrak{a}$ vanishes.

If $f_1, \dots, f_s$ is an ideal basis, so that every $f \in \mathfrak{a}$ has the form

$$f = A_1 f_1 + \dots + A_s f_s,$$

then the zeros of $\mathfrak{a}$ are exactly the common zeros of $f_1, \dots, f_s$. Conversely, any finite system of polynomials may be regarded as a basis of the ideal it generates. Thus the ideal-theoretic definition of zeros is the usual one, but avoids the arbitrariness of singling out a particular basis. In one variable this difficulty did not appear, because only principal ideals occur and the basis polynomial is determined up to a unit. Even there, however, the ideal viewpoint gives a clearer interpretation.

The theory again falls into four steps.

I. The decomposition theorem represents an ideal as a least common multiple of greatest primary components and thereby reduces the question to prime ideals. An ideal $\mathfrak{p}$ is prime if divisibility of a product by $\mathfrak{p}$ implies divisibility of one factor; an ideal $\mathfrak{q}$ is primary if divisibility of a product by $\mathfrak{q}$ implies divisibility of one factor or of some power of the other factor. To every primary ideal $\mathfrak{q}$ there belongs a unique associated prime ideal $\mathfrak{p}$ which divides $\mathfrak{q}$ and some power of which is divisible by $\mathfrak{q}$. One has a representation

$$\mathfrak{a} = [\mathfrak{q}_1, \dots, \mathfrak{q}_s],$$

where the $\mathfrak{q}_i$ are greatest primary components. Although the primary components themselves need not be uniquely determined, their associated prime ideals are uniquely determined by $\mathfrak{a}$. Since every associated prime ideal divides $\mathfrak{a}$, and conversely a product of powers of the associated primes is divisible by $\mathfrak{a}$, the zeros of the ideal agree with the zeros of its associated prime ideals.

II. The residue classes modulo a prime ideal form a ring without zero divisors. If the prime ideal is not the unit ideal, this ring contains a non-zero element, and by adjoining quotients it becomes the residue-class field of the prime ideal. There is always an extension field of P isomorphic to this residue-class field; it becomes the zero-field of the prime ideal. Thus

$$K = P(\alpha_1, \dots, \alpha_n),$$

where $(\alpha_1, \dots, \alpha_n)$ is a zero of $\mathfrak{p}$. The field has transcendence degree k , $0 \leq k < n$, over P : it contains k algebraically independent elements over P , while every $k + 1$ elements are algebraically dependent. Conversely, every zero-field of transcendence degree k is isomorphic to the residue-class field. Therefore zero-field and residue-class field are abstractly the same object.

III. The norm of $\mathfrak{p}$ is decomposed into linear factors in the Galois field determined by $\mathfrak{p}$. If the variables $x_1, \dots, x_n$ are obtained from original variables $y_1, \dots, y_n$ by a linear transformation with indeterminate coefficients, then the zero-field of transcendence degree k can be viewed as a finite algebraic extension of $P(x_{k+1}, \dots, x_n)$. Passing to its Galois field, the prime function

for a primitive element splits into linear factors. This prime function plays, after returning to the y variables, the role of the fundamental equation: its root is the fundamental linear form in a zero system β . The factorization therefore gives the product over all zeros. The prime function, or a power of it, is also the norm $N(\mathfrak{p})$ of the prime ideal, obtained by regarding $\mathfrak{p}$ as a module of linear forms in the monomials in the eliminated variables and defining the norm as the product of the elementary divisors. Over a perfect coefficient field this norm is exactly the prime function; over an imperfect field it may be a power.

IV. For the original ideal $\mathfrak{a}$, one regards it successively as a module of linear forms in the monomials of $x_1, \dots, x_{i-1}$ for $i = 1, \dots, n$. At each step, with coefficient field $P(x_i, \dots, x_n)$, only finitely many elementary divisors differ from one. Their product gives the individual norm, and the norm $N(\mathfrak{a})$ is the product of all individual norms. The greatest primary factors of this norm correspond one-to-one to the associated prime ideals $\mathfrak{p}$ of $\mathfrak{a}$. Each such factor is a power of the norm of $\mathfrak{p}$ or, in the inseparable case, of its highest elementary divisor. Thus the norm of $\mathfrak{a}$ splits according to the associated primes, and the preceding steps give the linear factorization for all zeros of $\mathfrak{a}$.

The multiplicity attached to the zeros corresponding to an associated prime ideal $\mathfrak{p}$ is the degree of the corresponding primary factor of the norm. This degree is the number of linearly independent residue classes, over the appropriate parameter field, of the isolated component determined by $\mathfrak{p}$ and by the prime ideals of higher dimension. In the one-variable case this is exactly the degree of the primary factor and therefore the usual multiplicity.

The case of one indeterminate is thus fully included by passing from a polynomial to its principal ideal. Its factorization may be read either as a least common multiple of primary components or as a decomposition of the norm into norms of prime ideals. The zero-field belongs to the prime ideal defined by $p(x)$, while the norm gives the linear factorization. The multiplicity is the number of residue classes in the relevant quotient, and the lowest fundamental ideal is the unit ideal whereas the first fundamental ideal is already the ideal itself.

In this form the usual elimination theory is incorporated in ideal theory: the norm of the ideal has the meaning of a resultant. The old elimination problem becomes the part of ideal theory in the polynomial domain that treats, alongside ideal decompositions, the decomposition of norms and the relation between the two decompositions.

7 Abstract construction of ideal theory in algebraic number fields

The following abstract properties were indicated as completely equivalent to the ideal theory in an algebraic number field, and hence as characterizing all rings that possess such an ideal theory. They are domains with identity element and without zero divisors in which the double chain condition holds: every divisor chain of ideals stops after finitely many steps, and so does every multiple chain modulo every ideal different from the zero ideal. In addition the ring is integrally closed in its quotient field. If the last assumption is omitted, one obtains theorems on ideal theory in a finite order, some of which are already found without proof in Dedekind's third edition. A detailed exposition will appear in the *Mathematische Annalen*.

8 Hilbert numbers in ideal theory

By Hilbert numbers one generally understands numbers that concern the enumeration of residue classes of ideals. Hilbert himself gave, in his "characteristic function", a function which for ideals in a polynomial domain gives, from a certain bound on, the exact number of linearly independent residue classes. For lower degrees Macaulay supplemented this function by a generating function, which Ostrowski could evaluate explicitly.

Macaulay also introduced numbers of another kind. These have proved most essential for general ideal theory because they admit an abstract formulation. In accordance with the general decomposition theorems it suffices to define them for primary ideals $\mathfrak{q}$. They count the number of linearly independent residue classes with respect to the residue-class field of the associated prime ideal $\mathfrak{p}$.

These numbers decompose into subsystems given by the numbers of irreducible components of

$$\mathfrak{q}, \quad \mathfrak{q}/\mathfrak{p}, \quad \mathfrak{q}/\mathfrak{p}^2, \quad \dots$$

The number of the i -th subsystem is also characterized as the number of terms in a composition series running from $\mathfrak{q}/\mathfrak{p}^i$ to $\mathfrak{q}/\mathfrak{p}^{i-1}$. The total composition series and its Hilbert numbers form the substitute for the representation of primary ideals as powers of prime ideals, a representation which is no longer valid in general, for example in finite orders of an algebraic number field.

9 Group characters and ideal theory

Let G be a finite group and let K be a splitting field for G . The group algebra $K[G]$ decomposes into simple components corresponding to the irreducible characters. The arithmetical questions concern what happens when one replaces K by a subring, or by the ring of algebraic integers in a number field, and considers ideals or modules stable under the group operation.

The character equations give idempotent decompositions after extension to the splitting field. Ideal theory supplies the descent from these decompositions to integral modules. In this way the elementary divisors attached to a group representation can be read in parallel with the elementary divisors of ideals. The irreducible characters describe the simple constituents over the field; the corresponding integral ideal theory describes how these constituents combine when denominators and divisibility are taken into account.

This viewpoint places the arithmetic of group characters beside the arithmetic of polynomial ideals. In both cases a decomposition over a field gives a formal splitting, while ideal theory records the integral and modular data lost in the field decomposition alone. The connection is especially useful when studying representations modulo prime ideals of the coefficient ring: the character-theoretic decomposition may remain valid, refine, or coalesce according to the behavior of the associated ideals after reduction.

10 Abstract construction of ideal theory in algebraic number and function fields: first installment

In the following an abstract characterization is given of all rings whose ideal theory agrees with the ideal theory of all algebraic integers in an algebraic number field, that is, rings in which every ideal is uniquely expressible as a product of powers of prime ideals. Among the best known examples are also the ring of all integral elements of an algebraic function field in one indeterminate, and more generally, for several indeterminates, the rings obtained by restricting oneself, in Kronecker's sense, to ideals of highest dimension and passing to the corresponding quotient ring, the functional domain.

The axioms completely equivalent to this ideal theory, for the commutative ring under consideration, are the following.

I. Divisor chain condition. Every chain of ideals in which each ideal is a proper divisor of the preceding one terminates after finitely many steps; equivalently, in every divisor chain of ideals there is an index from which on all ideals are equal.

II. Multiple chain condition modulo each non-zero ideal. Every chain of ideals, all of which divide a fixed ideal different from the zero ideal, and in which each ideal is a proper multiple of the preceding one, terminates after finitely many steps.

III. There is an identity element for multiplication.

IV. The ring has no zero divisors.

V. Integral closedness in the quotient field. Every element of the quotient field integral over the ring belongs to the ring.

From these axioms Noether develops step by step an increasingly restricted ideal theory, ending with the desired theory, and conversely shows that all five axioms are actually satisfied in the classical examples.

From the divisor chain condition I follows the representation of an ideal as a least common multiple of finitely many primary ideals belonging to distinct prime ideals. Noether repeats the proof because it is simplified by using the ideal quotient and because a gap in the earlier proof is to be corrected: the old proof used, at one point, the existence of a multiplicative identity, which had not been assumed.

The multiple chain condition II implies that in the residue-class ring modulo every ideal different from the zero ideal, a prime ideal can have no proper divisor other than the unit ideal containing all elements. Hence the primary components of such ideals are uniquely determined except for components belonging to the unit ideal. In a somewhat less sharp form this result already occurs in work of Masazo Sono. Sono's hypothesis of the existence of a composition series in the residue-class ring is equivalent to the double chain condition in that ring. By the double chain condition Noether means the simultaneous validity of the divisor chain condition and the multiple chain condition, without restricting to non-zero ideals.

If axiom III, the existence of the unit element, is added, then the unit ideal does not occur as an associated prime ideal. The primary components are therefore uniquely determined; they are pairwise relatively prime, and their least common multiple is their product. Axioms I, II, and III hold for all finite orders of an algebraic number field. Dedekind had already stated, without a complete proof, the unique representation of an ideal as a product of pairwise relatively prime one-type ideals, i.e. primary components.

Axiom IV, the absence of zero divisors, implies that the different powers of an ideal different from both the zero ideal and the unit ideal are all distinct, and that the quotient field exists. If finally axiom V is added, integral closedness in the quotient field, then the primary ideals, which by IV are uniquely determined, become powers of prime ideals. This gives the usual decomposition theorem. The proof uses a consequence of integral closedness already drawn by Dedekind in the number-field case: a genuinely fractional element can be written as a quotient of integral elements in such a way that even the square of the numerator is not divisible by the denominator.

In later treatments this consequence is often replaced by the less transparent generalized Gauss theorem, or by corresponding module theorems that give somewhat more. Those results require basis elements in their application; the proof here works with ideals themselves.

In the converse direction the axioms other than V follow almost directly from unique factorization of ideals. Axiom V follows from the proof that every genuinely fractional element can be represented so that no power of the numerator is divisible by the denominator. That is even stronger than Dedekind's original consequence of integral closedness. The proof is obtained only after drawing the customary conclusions from the ideal factorization: principal-ideal representation modulo a fixed ideal, and the theory of fractional ideals, i.e. ideal quotients in the field. Dedekind had already known that integral closedness follows from the representation by prime-ideal powers.

The axioms I–V imply that four decompositions which are generally distinct coincide: decomposition into least pairwise relatively prime ideals, into mutually prime ideals, into greatest primary components, and into irreducible ideals. Conversely, this coincidence together with axioms I–IV implies integral closedness. If zero divisors are allowed in the ring, then the remaining axioms, with V interpreted as integral closedness in the total quotient ring, do not necessarily imply this coincidence; the residue-class ring modulo certain ideals of a finite order gives examples.

Section 1 sketches the theory of elements integral over a ring, through Dedekind's consequences of integral closedness. Section 2 proves the transfer of chain conditions from a ring to modules of linear forms over it. Section 3 applies this to passage to finite extension fields of first kind: axioms I–IV are preserved for every order, and axiom V holds for the order consisting of all integral elements. This places both number fields and function fields under the same scheme. In particular, once it is shown directly that the functional domain derived from integral polynomials in several indeterminates satisfies the axioms, Kronecker's ideal theory is fully incorporated. The later ideal theory does not depend on these classificatory sections; sections 4 and 5 give the foundations independent of axioms I–V and show more clearly which parts of the theory do not require finiteness hypotheses.

10.1 1. Theory of integral elements

Let T be a commutative ring without zero divisors and with multiplicative identity e , and let R be a fixed subring of T containing e . The words module, order, and integral are throughout to be understood relative to this fixed ring R .

A system M of elements of T is called an R -module, or simply a module, if with two elements α and β it contains $\alpha + \beta$, and with α it contains $r\alpha$ for arbitrary $r \in R$. If every element of M also belongs to N , then M is said to be divisible by N , written $M \equiv 0 \pmod{N}$; equivalently, M is a multiple and N a divisor. R -modules all of whose elements lie in R are ideals of R .

The intersection, or least common multiple, of any family of modules is again a module, and so is T . Thus the module generated by any system of elements is defined uniquely as the intersection of all modules containing that system. The sum, or greatest common divisor, of a family of modules is the module generated by their union. The product MN of two modules is the module generated by all products $\mu\nu$ with $\mu \in M$ and $\nu \in N$. This product is associative and commutative because the ring multiplication is; distributivity follows from distributivity in T .

Since R itself is a module, saying that R is the multiplier domain is expressed by $RM \equiv 0 \pmod{M}$. Since R contains the identity element, also $M \equiv 0 \pmod{RM}$, hence $M = RM$.

A module is finite if it is generated by finitely many elements. If

$$M = (\mu_1, \dots, \mu_s),$$

then the μ_i form a module basis. Sums and products of finitely many finite modules are again finite; for products this follows from writing every element of M as an R -linear combination of the basis elements and using commutativity of multiplication in T .

An extension ring of R lying in T , that is, a subring of T containing all elements of R , is called an R -order, or simply an order. The intersection of arbitrary orders is again an order, and T is an order; hence the order generated by any set of elements is uniquely defined. Every order is also an R -module. An order is finite if it is finite as a module. Sums and products of finite orders are finite orders. There is also a transitivity law: if A is a finite R -order and B a finite A -order, then B is a finite R -order, since products of an R -basis of A with an A -basis of B give an R -basis of B .

An element $\alpha \in T$ is called integral over R if the order $R[\alpha]$ generated by it is finite.

Equivalently, the chain of modules

$$R_1 = (1), \quad R_2 = (1, \alpha), \quad R_3 = (1, \alpha, \alpha^2), \dots$$

terminates. Equivalently again, there exists a non-zero principal module $C = (\gamma)$ such that $CR[\alpha]$ is finite; that is, the chain CR_i terminates. Finally this is identical with the usual equation formulation: α is integral if it satisfies an equation

$$\alpha^m + r_1\alpha^{m-1} + \dots + r_m = 0, \quad r_i \in R.$$

These formulations are equivalent. If $R[\alpha]$ is finite, a basis can be chosen among the powers of α , and termination of the chain gives the monic equation. Conversely such an equation expresses every high power of α in terms of lower powers. The principal-module formulation is equivalent because multiplying by a non-zero principal module in a domain does not destroy the eventual equality of the chains.

The ring property of the integral elements and the transitivity of integral dependence rest on the following lemma.

Lemma. If $\mathcal{O}$ is a finite order in T and $\alpha \in \mathcal{O}$, then the order $R[\alpha]$ generated by α is finite. In other words, all elements of a finite order are integral.

Indeed, let $\omega_1, \dots, \omega_s$ be a module basis of $\mathcal{O}$. Since $\alpha\omega_i$ again lies in $\mathcal{O}$, there are relations

$$\alpha\omega_i = \sum_j r_{ij}\omega_j.$$

The determinant of the corresponding system gives a monic equation for α . The use of the determinant is legitimate because T , and hence $\mathcal{O}$, has no zero divisors.

The system $\mathcal{G}$ of all elements of T integral over R is an order. It contains R , because the elements of R are integral with basis e . It is closed under addition and multiplication: if α and β are integral, the order generated by them is the product of two finite orders and is finite, so $\alpha + \beta$ and $\alpha\beta$ are elements of a finite order and hence integral by the lemma.

The transitivity law for integral dependence is this: if $\mathcal{O}$ is an order consisting of elements integral over R , then every element of T integral over $\mathcal{O}$ is already integral over R . For such an element α satisfies a monic equation with coefficients in $\mathcal{O}$; the order generated over R by those finitely many coefficients is finite, and α is integral over that finite order. By the transitivity of finite orders, $R[\alpha]$ is finite.

The ring R is called integrally closed in T if every element of T integral over R already belongs to R . By the second formulation of integrality, an element $\alpha \in T$ belongs to R exactly when there is a non-zero principal module C such that $CR[\alpha]$ is finite.

Dedekind drew two important consequences from integral closedness.

Dedekind consequence I. Suppose R is integrally closed in T , and suppose further that the divisor chain condition for ideals of R holds. If β is not integral over R and $c \neq 0$ is an element of R , then there is an exponent $\rho > 0$ such that in the sequence

$$c, \quad c\beta, \quad c\beta^2, \quad \dots, \quad c\beta^3, \dots$$

the terms $c, \dots, c\beta^{\rho-1}$ are integral and all subsequent terms are not integral.

The point is that if all $c\beta^r$ were integral, then by integral closedness they would all belong to R . The ideals generated by the initial segments would form a divisor chain and would therefore stop. That would imply finiteness of $R[\beta]$ after multiplication by c , hence integrality of β , contrary to assumption. Once a non-integral term occurs, every later term remains non-integral; otherwise a power relation would make the earlier term integral. Thus there is a first non-integral term.

Specializing T to the quotient field of R gives the second consequence.

Dedekind consequence II. Suppose R is integrally closed in its quotient field and satisfies the divisor chain condition for ideals. If β is a non-integral element of the quotient field, then it has a representation

$$\beta = m/n, \quad m, n \in R,$$

such that m^2/n is also not integral.

Indeed, write $\beta = b/c$ with $c \neq 0$. In the sequence $c, c\beta = b, c\beta^2, \dots$ the first two terms are integral, so the exponent ρ of the preceding consequence is greater than one. Put

$$m = c\beta^\rho, \quad n = c\beta^{\rho-1}.$$

Then $m/n = \beta$, while $m^2/n = c\beta^{\rho+1}$ is not integral.

Finally, integral closedness is itself transitive: if $\mathcal{G}$ is the system of elements of T integral over R , then $\mathcal{G}$ is also the system of elements integral over $\mathcal{G}$. Any element integral over $\mathcal{G}$ is integral over R by transitivity, hence belongs to $\mathcal{G}$.

10.2 2. Chain conditions in finite module domains

The module theorem used in the application says that chain conditions pass from the coefficient ring to finite modules. A module domain M over a ring R is an abelian group with scalar multiplication by elements of R satisfying the associative and distributive laws. All module definitions from the preceding subsection that do not use multiplication of module elements remain valid.

A module domain is finite if finitely many elements $\varepsilon_1, \dots, \varepsilon_k$ generate it; if R has a unit, its elements are all linear forms

$$r_1\varepsilon_1 + \dots + r_k\varepsilon_k.$$

Module theorem. Let M be a finite module domain over a commutative ring R with identity. If the divisor chain condition, respectively the multiple chain condition, holds for ideals of R , then the corresponding divisor, respectively multiple, chain condition holds for submodules of M .

The proof assigns to each submodule U of M a finite system of ideals $a_1, \dots, a_k$ of R . An element of M is said to have length i if it can be expressed using $\varepsilon_1, \dots, \varepsilon_i$ but not using fewer generators. Let U_i be the elements of U of length at most i , and let a_i be the set of coefficients of ε_i occurring in elements of U_i . If V is a proper divisor or proper multiple of U , then at least one of the ideals assigned to V is a proper divisor or proper multiple of the corresponding a_i . Thus an infinite chain of submodules would produce an infinite chain among finitely many ideal chains in R , impossible under the corresponding chain condition.

A useful consequence is the following. If in R the multiple chain condition holds modulo every non-zero ideal, then in a finite module domain M the multiple chain condition holds modulo every submodule whose associated ideals $c_1, \dots, c_k$ are all non-zero. The proof is the same, applied modulo those non-zero ideals.

10.3 3. Passage to finite extension fields

The preceding module theorem is used to bring algebraic number fields and function fields under the axioms.

Assume that in R all axioms I–V except possibly the multiple chain condition II hold. Let K be the quotient field of R , and let L be a finite extension field of first kind over K . Let $\mathcal{G}$ be the system of elements of L integral over R . Then $\mathcal{G}$ satisfies the same axioms, and every order contained in $\mathcal{G}$ satisfies all of them except possibly integral closedness.

The axioms III and IV hold because the integral elements form a ring without zero divisors and with the same identity. Integral closedness of $\mathcal{G}$ in L follows from the transitivity of integral dependence. It remains to see the divisor chain condition. Since the extension is of first kind,

the field L is generated by a single element α integral over R . Its conjugates in the Galois field are integral, and the square D of the product of their differences is a non-zero element of K , hence, by integral closedness of R , an element of R . The usual discriminant argument shows that the integral closure $\mathcal{G}$ is contained in a finite R -module generated by elements of the form α^j/D . Therefore the module theorem gives the divisor chain condition for ideals of $\mathcal{G}$ and for ideals of every order in $\mathcal{G}$.

If all axioms I–V hold in R , then all hold in $\mathcal{G}$, and every order in $\mathcal{G}$ satisfies all but integral closedness. The only remaining point is axiom II. By the consequence of the module theorem it suffices to show that a non-zero ideal of $\mathcal{G}$, regarded as an R -module inside the finite module containing $\mathcal{G}$, has all associated coefficient ideals non-zero. This follows because multiplication by a suitable non-zero element of R carries each basis element of the finite module into the given ideal. The same argument applies to orders of full rank; for lower-rank orders one passes to their quotient fields, intermediate between K and L and again of first kind.

It remains only to verify the axioms for the basic domains: the rational integers, polynomials in one indeterminate, and the functional domain of polynomials in several indeterminates. For the rational integers and for polynomials in one indeterminate over a field, the axioms follow from unique factorization into irreducible elements and from the fact that every ideal is principal. Integral closedness follows because a non-integral quotient can be written with coprime numerator and denominator, so that no power of the numerator is divisible by the denominator.

For the ring of polynomials in several indeterminates, axioms III, IV, and V again follow from unique factorization of elements. The divisor chain condition is Hilbert's basis theorem. The multiple chain condition need not hold. Passing to the functional domain restores it. One adjoins a new indeterminate u and considers rational functions whose denominators are primitive polynomials in u . The elements with primitive numerator and denominator become units, and every element of the functional domain has a unique expression as an element of the original polynomial ring multiplied by a unit. The factorization theory of the original ring therefore transfers to the functional domain. In that domain every ideal is principal: starting with an element of an ideal and another not divisible by it, the greatest common divisor also lies in the ideal and is a proper divisor; after finitely many repetitions one obtains a basis element. Hence unique factorization of ideals and the axioms I and II follow.

10.4 4. Isomorphism theorems and direct sums

In the following only the ring or module structure is assumed; no chain axiom is required.

Let M and $\bar{M}$ be module domains over R . We say that M is homomorphic to $\bar{M}$ if to every element of M there corresponds one and only one element of $\bar{M}$, so that all of $\bar{M}$ is reached, and such that difference and multiplication by the same element of R correspond. Thus from

$$\beta \sim \bar{\beta}, \quad \gamma \sim \bar{\gamma}$$

one obtains

$$\beta - \gamma \sim \bar{\beta} - \bar{\gamma}, \quad r\beta \sim r\bar{\beta}.$$

Every R -module B in M then corresponds to an R -module $\bar{B}$ in $\bar{M}$. Conversely, if $\bar{C}$ is a module in $\bar{M}$, the totality C^* of elements of M to which elements of $\bar{C}$ correspond is an R -module in M uniquely determined by $\bar{C}$; it is the module associated with $\bar{C}$. If A is the module corresponding to the zero element of $\bar{M}$, then C^* is a divisor of M and decomposes into congruence classes modulo A , these classes corresponding one-to-one to the elements of $\bar{C}$. Starting from a module B in M , passing to $\bar{B}$ in $\bar{M}$, and coming back gives

$$B^* = (B, A),$$

the greatest common divisor of B and A . If B divides M , then $B^* = B$.

If the correspondence is reversible, the module domains are isomorphic. If A is any submodule of M , the residue-class module M/A is obtained by taking congruence modulo A as the equality relation. Equivalently, one replaces equal elements modulo A by one residue class. Every homomorphism is obtained in this way: if M maps homomorphically onto $\bar{M}$ and A is the kernel, then $\bar{M}$ is isomorphic to M/A .

The two isomorphism theorems follow. First, if $\bar{M} = M/A$ and C divides A , then

$$\bar{M}/C \simeq M/C.$$

Indeed, equality modulo A is already equality modulo C after the indicated passage to residue classes. Secondly, for two submodules B and A of M one has

$$(B, A)/A \simeq B/[B, A],$$

where (B, A) denotes the greatest common divisor and $[B, A]$ the least common multiple. This is the module form of the familiar second isomorphism theorem.

The same considerations apply to commutative rings when modules are replaced by ideals. If R maps homomorphically onto $\bar{R}$, difference and product correspond. If $\mathfrak{a}$ is an ideal, the residue-class ring $R/\mathfrak{a}$ is formed by using congruence modulo $\mathfrak{a}$ as equality; products of residue classes are then well-defined. Every ring homomorphism is obtained in this way, and the isomorphism theorems become

$$(R/\mathfrak{a})/(\mathfrak{c}/\mathfrak{a}) \simeq R/\mathfrak{c} \quad (\mathfrak{c} \mid \mathfrak{a}),$$

and

$$(\mathfrak{b}, \mathfrak{a})/\mathfrak{a} \simeq \mathfrak{b}/[\mathfrak{b}, \mathfrak{a}].$$

Assume now that the commutative ring R has a multiplicative identity. Two ideals $\mathfrak{a}, \mathfrak{b}$ are relatively prime if their greatest common divisor is the unit ideal,

$$(\mathfrak{a}, \mathfrak{b}) = \mathfrak{o}.$$

The basic rules are most simply stated without writing identity elements explicitly. If $(\mathfrak{a}, \mathfrak{b}) = \mathfrak{o}$, then

$$(\mathfrak{a}, \mathfrak{b}\mathfrak{c}) = (\mathfrak{a}, \mathfrak{c}).$$

Therefore from $\mathfrak{c}\mathfrak{b} \equiv 0 \pmod{\mathfrak{a}}$ and $(\mathfrak{a}, \mathfrak{b}) = \mathfrak{o}$ it follows that $\mathfrak{c} \equiv 0 \pmod{\mathfrak{a}}$. If every $\mathfrak{a}_i$ is relatively prime to every $\mathfrak{b}_j$, then the product of the $\mathfrak{a}_i$ is relatively prime to the product of the $\mathfrak{b}_j$. In particular, if $\mathfrak{b}_1, \dots, \mathfrak{b}_s$ are pairwise relatively prime, their least common multiple is their product.

The Chinese-remainder decomposition is then immediate. If $\mathfrak{b}_1, \dots, \mathfrak{b}_s$ are pairwise relatively prime and

$$\mathfrak{v} = [\mathfrak{b}_1, \dots, \mathfrak{b}_s] = \mathfrak{b}_1 \cdots \mathfrak{b}_s,$$

then the residue-class ring modulo $\mathfrak{v}$ is the direct sum of the residue-class rings modulo the $\mathfrak{b}_i$:

$$R/\mathfrak{v} \simeq R/\mathfrak{b}_1 \oplus \cdots \oplus R/\mathfrak{b}_s.$$

The idempotents giving the direct summands are obtained by choosing, for each i , an element congruent to 1 modulo $\mathfrak{b}_i$ and congruent to 0 modulo the other $\mathfrak{b}_j$.

Conversely, a decomposition of a ring or module into a direct sum is expressed ideal-theoretically by such idempotent decompositions and by pairwise relatively prime annihilator ideals. This is the form in which Noether uses direct sums later: it is not an additional structure, but the ideal-theoretic expression of pairwise relatively prime components.

10.5 5. Primary ideals without chain assumptions

Noether distinguishes weak and strong primary ideals in a general ring before imposing chain conditions. An ideal $\mathfrak{q}$ is weakly primary if, whenever $ab \equiv 0 \pmod{\mathfrak{q}}$ and a is not divisible by $\mathfrak{q}$, every sufficiently high power of b is divisible by $\mathfrak{q}$. Equivalently, the zero divisors of the residue-class ring $R/\mathfrak{q}$ are precisely those residue classes all of whose sufficiently high powers vanish. The associated prime ideal $\mathfrak{p}$ consists of all elements whose powers are divisible by $\mathfrak{q}$.

The strong version is obtained by replacing elements by ideals. Thus $\mathfrak{q}$ is strongly primary if from

$$\mathfrak{a}\mathfrak{b} \equiv 0 \pmod{\mathfrak{q}}$$

and from the non-divisibility of $\mathfrak{a}$ by $\mathfrak{q}$, it follows that some power of $\mathfrak{b}$ is divisible by $\mathfrak{q}$. In rings with the divisor chain condition the distinction disappears; without that hypothesis it is real. Noether records an example, due to R. Hölzer, showing that a weak primary ideal need not be strong.

The least common multiple of finitely many weak, respectively strong, primary ideals with the same associated prime ideal is again weak, respectively strong, primary with that same associated prime ideal. But a shortest representation by finitely many primary ideals belonging to different prime ideals is not primary. Indeed, if

$$\mathfrak{m} = [\mathfrak{q}_1, \dots, \mathfrak{q}_r]$$

is a shortest representation with distinct associated primes $\mathfrak{p}_i$, one can choose elements, or ideals in the strong case, which vanish modulo all but one component and not modulo the remaining prime. Multiplying them gives a product divisible by $\mathfrak{m}$ while no suitable power condition required for primaryness can hold. This proves non-primaryness of such a mixed component.

The quotient of modules is now introduced. Let T be an extension ring of R , and let $M, B, C, \dots$ be R -modules in T . The quotient

$$C = M : B$$

is the smallest module satisfying

$$CB \equiv 0 \pmod{M},$$

in the sense that if $DB \equiv 0 \pmod{M}$, then $D \equiv 0 \pmod{C}$. Equivalently, C is the greatest common divisor of all modules D with $DB \equiv 0 \pmod{M}$. The zero module always gives existence. From the definition one obtains the formal rules: if M is a multiple of N , then $M : B$ is a multiple of $N : B$; if B is a multiple of C , then $M : B$ is a divisor of $M : C$; and

$$M : BC = (M : B) : C = (M : C) : B.$$

If the extension ring is the original ring, the module quotient becomes the ideal quotient $\mathfrak{a} : \mathfrak{b}$, with the same rules. Since $\mathfrak{a}\mathfrak{b} \equiv 0 \pmod{\mathfrak{a}}$, one also has

$$\mathfrak{a} \equiv 0 \pmod{\mathfrak{a} : \mathfrak{b}}.$$

An ideal $\mathfrak{b}$ is called prime to $\mathfrak{a}$ if

$$\mathfrak{a} : \mathfrak{b} = \mathfrak{a}.$$

This says exactly that $d\mathfrak{b} \equiv 0 \pmod{\mathfrak{a}}$ always implies $d \equiv 0 \pmod{\mathfrak{a}}$. From the quotient rules it follows that if $\mathfrak{b}$ and $\mathfrak{c}$ are prime to $\mathfrak{a}$, then both their product and their least common multiple are prime to $\mathfrak{a}$. If $\mathfrak{b}$ is prime to $\mathfrak{a}$ and $\mathfrak{a}$ is prime to $\mathfrak{b}$, the ideals are mutually prime. Relatively prime ideals in the sense of greatest common divisor one are mutually prime, but the quotient formulation is more flexible and is the one needed for the proofs under chain conditions.

10.6 6. Ideal theory under the divisor chain condition

Assume now that the commutative ring R satisfies axiom I, the divisor chain condition. Fix once and for all a well-ordering of all elements of R . This also gives a well-ordering of the ideals. Indeed, the divisor chain condition implies that every ideal has a finite basis; ordering finite subsets quasi-lexicographically and assigning to each ideal its first possible basis well-orders the set of ideals.

Theorem I. Under the divisor chain condition every ideal of R can be represented as a least common multiple of finitely many irreducible ideals, i.e. ideals not representable as a least common multiple of two proper divisors.

If the theorem failed for an ideal $\mathfrak{m}$, then $\mathfrak{m}$ would be reducible; writing $\mathfrak{m} = [\mathfrak{a}, \mathfrak{b}]$, at least one of $\mathfrak{a}$ or $\mathfrak{b}$ would again fail the theorem. Choose the first such proper divisor in the well-ordering. Repeating the construction gives a definite infinite chain of proper divisors, contradicting the divisor chain condition. This is the only part of the argument where the well-ordering is used; the statement is set-theoretic in nature and applies equally to module domains with the corresponding chain condition.

With the divisor chain condition, primary ideals in the weak and strong sense coincide, and one may simply speak of primary ideals. The relation between primary and irreducible is the key.

Theorem II. Under the divisor chain condition every irreducible ideal is primary. Equivalently, every non-primary ideal is reducible.

If $\mathfrak{m}$ is not primary, then there are elements, or equivalently ideals through the quotient formalism, giving a product divisible by $\mathfrak{m}$ while neither factor satisfies the required power-divisibility condition. The ideal quotient then furnishes two proper divisors of $\mathfrak{m}$ whose least common multiple is $\mathfrak{m}$ itself. Thus $\mathfrak{m}$ is reducible. Combining this with Theorem I gives the representation of every ideal as a least common multiple of finitely many primary ideals.

Continuing the same proof, once every irreducible ideal is known to be primary, the representation theorem takes the following explicit form. Every ideal $\mathfrak{a}$ admits a shortest representation

$$\mathfrak{a} = [\mathfrak{q}_1, \dots, \mathfrak{q}_s]$$

by primary ideals. The associated prime ideals are obtained by taking, for each $\mathfrak{q}_i$, the totality of elements whose powers are divisible by $\mathfrak{q}_i$. The primary components may still fail to be unique at this stage. What is fixed by the divisor chain condition alone is the existence of a finite primary representation and the possibility of choosing it shortest.

Noether's reason for introducing the ideal quotient now becomes visible. If $\mathfrak{b}$ is prime to $\mathfrak{a}$, the equality $\mathfrak{a} : \mathfrak{b} = \mathfrak{a}$ is an exact algebraic substitute for saying that $\mathfrak{b}$ does not meet the component defined by $\mathfrak{a}$. Multiplying by ideals prime to a component cannot introduce new divisibility modulo that component. Thus the quotient calculus supplies the separation of components without choosing elements or bases.

The next step, which belongs to the multiple chain condition, is the uniqueness of isolated components. Passing to the residue-class ring modulo a non-zero ideal, the multiple chain condition prevents a prime ideal from having proper divisors other than the unit ideal. Consequently the primary component corresponding to an isolated associated prime is forced: it is recovered as the quotient of the given ideal by all ideals prime to that associated prime. This is the abstract version of separating one primary component by multiplying away all the others.

When the ring has an identity, the unit ideal no longer appears as an associated prime for non-zero ideals. Hence the ambiguity just described disappears for non-zero ideals once both chain conditions hold. The primary components become unique; they are pairwise relatively prime, and their least common multiple is their product. Thus the theory has reached the form familiar from orders in number fields: an ideal is represented as a product of one-type components, and after integral closedness is added these one-type components become powers of prime ideals.

This order of proof is important. The primary decomposition, its uniqueness, and the passage from primary ideals to prime powers are not bundled together. They are separated according to the axioms that force them: divisor chains give existence, multiple chains give uniqueness of the components, the identity and absence of zero divisors remove exceptional unit phenomena and create a quotient field, and integral closedness turns primary components into prime powers. The classical Dedekind ideal theory is therefore not a single indivisible phenomenon but the conjunction of these four structural facts.

For the continuation of the paper this separation of the axioms is the controlling device. In the next sections Noether proves the converse direction: from the prime-power factorization of ideals one recovers the chain conditions and integral closedness. The proof is again conducted with ideal quotients. Fractional ideals are introduced as quotients in the quotient field, their classes form a group precisely when every non-zero ideal factors into prime powers, and Dedekind's criterion for integral closedness is recovered from the impossibility of writing a genuinely fractional element with all powers of the numerator divisible by the denominator. Thus the abstract axioms and the classical factorization theorem determine one another.

This is the point at which the paper passes from mere primary decomposition to Dedekind ideal theory. The previous sections account for the decomposition of ideals into components; the subsequent ones show why, in integrally closed domains satisfying the double chain condition, those components have the special and much stronger form of powers of prime ideals. The proof is deliberately intrinsic: it does not rely on the existence of bases for ideals in a number field, but only on divisibility, quotients, chain conditions, and integrality.

The distinction is also why the paper can include both algebraic number fields and algebraic function fields without changing the formal proof. In the number-field case the integral closure of the rational integers in a finite extension field is the standard order of all algebraic integers. In the function-field case the functional domain plays the same formal role after the passage that removes the lower-dimensional ideals. Once the axioms are verified for the base domain and shown to survive finite extensions of first kind, the subsequent ideal theory proceeds without referring to the concrete origin of the ring.

The practical mathematical point is that Noether is not merely restating Dedekind's theory abstractly. She is isolating exactly which parts of the classical proof require which finiteness or integrality assumptions. This separation lets later commutative algebra treat primary decomposition, isolated components, ideal quotients, integral closure, and unique factorization of ideals as separate structural layers rather than as one monolithic theorem inherited from algebraic number theory.

11 Abstract Construction of Ideal Theory in Algebraic Number and Function Fields

11.1 Continuation of §6. Ideal theory under the divisor-chain condition

To obtain a representation by greatest primary components it is only necessary to replace, wherever possible, the representation by irreducible ideals whose existence is supplied by Theorem I - and which by Theorem II are primary - by a shortest such representation. If the primary ideals belonging to the same prime ideal are then collected together, the desired representation follows from §5, 3. The primary components may be called greatest components, since the least common multiple of any several of them is no longer primary.

11.2 §7. Ideal theory under the double chain condition

The simplifications, compared with the theory developed so far, rest on the two auxiliary results stated below.

First, if the multiple-chain condition holds in a commutative ring without zero divisors, then the ring is already a field. It is enough to show that, for $a \neq 0$, the equation

$$ax = b$$

always has a solution in the ring. It cannot have more than one solution, because the ring has no zero divisors. Let $\mathfrak{a}$ be the principal ideal generated by a . By hypothesis the sequence of powers of $\mathfrak{a}$ eventually stops; say $\mathfrak{a}^n$ is equal to all later powers. If $\mathfrak{b}$ denotes the principal ideal generated by b , then

$$\mathfrak{a}^n \mathfrak{b} = \mathfrak{a}^{n+1} \mathfrak{b}.$$

In particular, the element $a^n b$ has a representation of the form

$$a^n b = ka^{n+1}b + na^{n+1}b = a^{n+1}c,$$

where k, n are ordinary integers and c is again an element of the ring. Since zero divisors do not occur, this gives $b = ac$. Thus the field property is proved.

The preceding argument immediately gives the following lemma. If the multiple-chain condition holds in a commutative ring, then a prime ideal has no proper divisor different from the zero ideal. Likewise, if only axiom II is assumed - the multiple-chain condition modulo every ideal different from the zero ideal - then every prime ideal different from the zero ideal has no proper divisor different from 0. Indeed, the residue class ring modulo a prime ideal is, by definition, a ring without zero divisors; by homomorphy it too satisfies the multiple-chain condition, and hence is a field. By the one-to-one correspondence between divisors of the prime ideal and ideals of the residue class ring, the prime ideal has no proper divisor different from 0.

Theorem IV. If the double chain condition holds in a commutative ring, then in every

shortest representation of an ideal the primary components which do not belong to the unit ideal are uniquely determined.

Supplement. Under axioms I and II the same assertion holds for every ideal different from the zero ideal.

Let

$$\mathfrak{m} = [\mathfrak{q}, \mathfrak{q}_1, \dots, \mathfrak{q}_r] = [\bar{\mathfrak{q}}, \bar{\mathfrak{q}}_1, \dots, \bar{\mathfrak{q}}_s]$$

be two shortest representations of $\mathfrak{m}$ by greatest primary components, and let

$$0, \mathfrak{p}_1, \dots, \mathfrak{p}_r \quad \text{and} \quad 0, \bar{\mathfrak{p}}_1, \dots, \bar{\mathfrak{p}}_s$$

be the corresponding prime ideals. The component $\mathfrak{q}$, respectively $\bar{\mathfrak{q}}$, is omitted if no primary ideal belonging to 0 occurs. Since

$$0 \cdot \mathfrak{p}_1 \cdots \mathfrak{p}_r \equiv 0 \pmod{\mathfrak{m}},$$

each $\bar{\mathfrak{p}}_j$ must be contained in one of the $\mathfrak{p}_i$, and hence, by the lemma, must be equal to it. Conversely each $\mathfrak{p}_i$ must agree with one of the $\bar{\mathfrak{p}}_j$. Thus the prime ideals different from 0 coincide.

Next, for instance,

$$\mathfrak{q}\mathfrak{q}_2 \cdots \mathfrak{q}_r \equiv 0 \pmod{\mathfrak{q}_1},$$

and therefore $\bar{\mathfrak{q}}_1 \equiv 0 \pmod{\mathfrak{q}_1}$, since the other components are not divisible by $\mathfrak{p}_1$ and are therefore prime to $\mathfrak{q}_1$. The reverse divisibility is obtained in the same way; hence the primary component belonging to each non-zero prime ideal is unique. If a component belonging to 0 occurs, then, because the representation is shortest, the corresponding component occurs in the other representation as well; this gives the remaining uniqueness. The same proof also shows that every representation containing no primary component belonging to 0 is itself shortest.

If, in addition to the double chain condition, the existence of a unit element is assumed, Theorem IV sharpens as follows.

Theorem V. In a commutative ring with unit element satisfying the double chain condition, every ideal can be represented uniquely as a product of finitely many pairwise relatively prime primary ideals.

Supplement. Under axioms I, II, III, Theorem V holds for every ideal different from the zero ideal.

Because the unit element exists, $0^0 = \mathfrak{o}$; hence every primary ideal belonging to 0 is equal to 0. Such a component cannot occur in a shortest representation of a non-zero ideal. The primary components are therefore uniquely determined by Theorem IV. At the same time every representation becomes shortest after omitting the zero component. Distinct prime ideals are, by the lemma, relatively prime; hence by the calculation rules of §4, 4, the same is true of the corresponding primary components. The least common multiple is therefore equal to their product.

The same hypotheses also give the following lemma. In a commutative ring with unit element and double chain condition, the prime ideals which are prime to an ideal $\mathfrak{m}$ are precisely all prime ideals, except the unit ideal, which are not contained in $\mathfrak{m}$. The notions “prime to” and “relatively prime” coincide.

Under axioms I, II, III the ideal $\mathfrak{m}$ is to be assumed different from the zero ideal.

If $\mathfrak{p}$ is not contained in $\mathfrak{m}$, then it is different from all prime ideals belonging to $\mathfrak{m}$. Hence, by the lemma above, it is divisible by none of these prime ideals, and so is prime to each primary component and therefore to $\mathfrak{m}$. It is also relatively prime to each component and hence relatively prime to $\mathfrak{m}$.

If $\mathfrak{p} \neq 0$ is contained in $\mathfrak{m}$, then it must coincide with one of the prime ideals belonging to $\mathfrak{m}$. Suppose it belongs to the component $\mathfrak{q} = \mathfrak{q}_1$. Then $\mathfrak{q} : \mathfrak{p}$ is a proper divisor of $\mathfrak{q}$, since $\mathfrak{p}^{e-1} \equiv 0 \pmod{\mathfrak{q} : \mathfrak{p}}$ but not modulo $\mathfrak{q}$. At the same time $\mathfrak{q} : \mathfrak{p}$, as a divisor of $\mathfrak{p}^{e-1}$, is divisible only by one non-zero prime ideal and is therefore primary. By unique factorization the product

$$(\mathfrak{q} : \mathfrak{p})\mathfrak{q}_2 \cdots \mathfrak{q}_r,$$

which is divisible by $\mathfrak{m} : \mathfrak{p}$, is a proper divisor of $\mathfrak{m}$. Thus $\mathfrak{p}$ is not prime to $\mathfrak{m}$.

Consequently an ideal $\mathfrak{t} \neq \mathfrak{p}$ is prime to $\mathfrak{m}$ precisely when none of the prime ideals belonging to $\mathfrak{t}$ is contained in $\mathfrak{m}$; in that case $\mathfrak{t}$ is also relatively prime to $\mathfrak{m}$. Since, conversely, two relatively prime ideals are always mutually prime (§5, 5), the two notions coincide under the hypotheses above.

11.3 §8. Ideal theory under integral closedness in the quotient field

The ideal theory developed up to this point is now to be sharpened to the usual one by adjoining the last two axioms.

First lemma. Let R be a commutative ring without zero divisors and with unit element, and assume the divisor-chain condition in R (axioms I, III, IV). Then from

$$\mathfrak{a} = \mathfrak{a}\mathfrak{b}, \quad \mathfrak{a} \neq (0),$$

it always follows that $\mathfrak{b} = \mathfrak{o}$. Thus

$$\mathfrak{c} \neq \mathfrak{c}\mathfrak{t}$$

for every ideal different from the zero and unit ideals.

The proof is the same as in the lemma of §1, 3, since $\mathfrak{a}\mathfrak{b}$ may be regarded as a finite module with respect to $\mathfrak{b}$. If $\alpha_1, \dots, \alpha_n$ is a basis of the ideal $\mathfrak{a}$, which exists by axiom I, then from $\mathfrak{a} = \mathfrak{a}\mathfrak{b}$ one obtains a system of equations

$$\alpha_i = b_{i1}\alpha_1 + \cdots + b_{in}\alpha_n, \quad b_{ij} \in \mathfrak{b}.$$

Since R has no zero divisors, the determinant $|b_{ij} - e_{ij}|$ is zero, where $e_{ij} = 0$ for $i \neq j$ and $e_{ii} = 1$. The equation obtained from this determinant has the form

$$1 + b_1 + \cdots + b_n = 0, \quad b_i \in \mathfrak{b},$$

and hence $1 \in \mathfrak{b}$, so $\mathfrak{b} = \mathfrak{o}$.

It remains to show that, once integral closedness in the quotient field is added to axioms I-IV (axiom V), every primary ideal different from zero is a power of its associated prime ideal. The uniqueness of the resulting representation

$$\mathfrak{m} = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r}$$

for all ideals different from zero has already been proved by Theorem V and the preceding lemma; at the same time those prime ideals have no proper divisors except the unit ideal. The proof will first be given for exponent two, using Dedekind's second consequence from §1, 4, and then in general by complete induction.

Lemma. Under axioms I-V there are no primary ideals of exponent two except $\mathfrak{q} = \mathfrak{p}^2$.

It must be shown that from

$$\mathfrak{p}^2 \equiv 0 \pmod{\mathfrak{q}}, \quad \mathfrak{q} \equiv 0 \pmod{\mathfrak{p}}, \quad \mathfrak{q} \not\equiv 0 \pmod{\mathfrak{p}^2}$$

one necessarily gets $\mathfrak{q} = \mathfrak{p}^2$. Choose

$$\mathfrak{c} \equiv 0 \pmod{\mathfrak{q}}, \quad \mathfrak{c} \equiv 0 \pmod{\mathfrak{p}}, \quad \mathfrak{c} \not\equiv 0 \pmod{\mathfrak{p}^2}.$$

Then by the lemma of §7, 3, the ideal $\mathfrak{c} : \mathfrak{p}$ is a proper divisor of $\mathfrak{c}$. Thus there is an element b such that

$$b\mathfrak{p} \equiv 0 \pmod{\mathfrak{c}} \equiv 0 \pmod{\mathfrak{q}}, \quad b\mathfrak{o} \not\equiv 0 \pmod{\mathfrak{c}}.$$

Consequently $\gamma = b/c$ is not integral: it belongs to the quotient field but not to R .

We shall prove $b \equiv 0 \pmod{\mathfrak{p}}$; then, since $\mathfrak{q}$ is primary and belongs to $\mathfrak{p}$, it follows that $\mathfrak{p} \equiv 0 \pmod{\mathfrak{q}}$ and hence $\mathfrak{q} = \mathfrak{p}^2$. By Dedekind's second consequence there exist elements m, n of R with

$$\gamma = b/c = m/n$$

such that m^2/n is still not integral. Equivalently,

$$bn = mc, \quad m \not\equiv 0 \pmod{nR}, \quad m^2 \not\equiv 0 \pmod{nR}.$$

Multiplying $b\mathfrak{p} \equiv 0 \pmod{\mathfrak{c}}$ by n gives

$$mb\mathfrak{p} \equiv 0 \pmod{n\mathfrak{c}},$$

and hence $m\mathfrak{p} \equiv 0 \pmod{nR}$, because the ring has no zero divisors. From $m^2 \equiv 0 \pmod{\mathfrak{p}nR}$ and $n \not\equiv 0 \pmod{\mathfrak{p}}$ it follows that $m \equiv 0 \pmod{\mathfrak{p}}$; here one again uses that $nR : \mathfrak{p}$ is a proper divisor of nR . The four relations

$$m \equiv 0 \pmod{\mathfrak{p}}, \quad n \not\equiv 0 \pmod{\mathfrak{p}}, \quad c \equiv 0 \pmod{\mathfrak{p}}, \quad cot \equiv 0 \pmod{\mathfrak{p}^2}, \quad bn = mc$$

then force $b \equiv 0 \pmod{\mathfrak{p}}$. Indeed, since $\mathfrak{p}^2$ is primary, first $mc \equiv 0 \pmod{\mathfrak{p}^2}$, and from $bn = mc$ one obtains $b \equiv 0 \pmod{\mathfrak{p}}$. This proves the lemma.

A second lemma removes the restriction on the exponent.

Lemma. Under axioms I-V there are no primary ideals of exponent ρ except $\mathfrak{q} = \mathfrak{p}^\rho$.

This follows from the preceding lemma without further use of the axioms. First one proves:

$$\text{if } c \equiv 0 \pmod{\mathfrak{p}}, \quad cot \equiv 0 \pmod{\mathfrak{p}^2}, \quad \text{then } \mathfrak{p}^\rho = [c^\rho, \mathfrak{p}^{\rho+1}]$$

for every ρ . For $\rho = 1$ this is just the exponent-two lemma. If

$$\mathfrak{p}^{\rho-1} = [c^{\rho-1}, \mathfrak{p}^\rho],$$

then multiplication by $\mathfrak{p}$ and the distributive law give

$$\mathfrak{p}^\rho = [c^\rho, c^{\rho-1}\mathfrak{p}^{\rho+1}, \mathfrak{p}^{\rho+1}] = [c^\rho, \mathfrak{p}^{\rho+1}].$$

The assertion can be put in the following equivalent form. If

$$\mathfrak{q} \equiv 0 \pmod{\mathfrak{p}^\rho}, \quad qot \equiv 0 \pmod{\mathfrak{p}^{\rho+1}},$$

and if

$$\mathfrak{p}^{\rho+k} \equiv 0 \pmod{\mathfrak{q}}, \quad k \geq 1,$$

then already

$$\mathfrak{p}^{\rho+k-1} \equiv 0 \pmod{\mathfrak{q}}.$$

Indeed, for every primary ideal there is an exponent $\rho > 1$ with $\mathfrak{q} \equiv 0 \pmod{\mathfrak{p}^\rho}$ but not modulo $\mathfrak{p}^{\rho+1}$; and $qot \equiv 0 \pmod{\mathfrak{p}^{\rho+1}}$ follows from $\mathfrak{p}^\rho e \mathfrak{p}^{\rho+1}$ by the first lemma of this section. Repeated application of the second formulation gives $\mathfrak{p}^\rho \equiv 0 \pmod{\mathfrak{q}}$ and hence $\mathfrak{q} = \mathfrak{p}^\rho$.

From the hypotheses of the second formulation, and from the above representation of $\mathfrak{p}^\rho$, one obtains for every element g of $\mathfrak{q}$ a representation

$$g = bc^\rho \pmod{\mathfrak{p}^{\rho+1}}$$

with either $b \equiv 0 \pmod{\mathfrak{p}}$ or

$$bc^\rho \equiv 0 \pmod{[\mathfrak{q}, \mathfrak{p}^{\rho+k}]}.$$

Since $[\mathfrak{q}, \mathfrak{p}^{\rho+1}]$ is primary, it follows in the latter case that

$$c^\rho \equiv 0 \pmod{[\mathfrak{q}, \mathfrak{p}^{\rho+1}]},$$

or else

$$c^{\rho+k-1} \equiv 0 \pmod{[\mathfrak{q}, \mathfrak{p}^{\rho+k}]}. \quad \square$$

In either case one obtains

$$\mathfrak{p}^{\rho+k-1} \equiv 0 \pmod{\mathfrak{q}}.$$

This proves the general lemma.

Combining the results gives:

Theorem VI. If axioms I-V hold in a commutative ring, then every ideal different from the zero and unit ideals is uniquely representable as a product of finitely many prime ideals different from the zero and unit ideals. These prime ideals have no proper divisors except the unit ideal.

11.4 §9. The axioms as consequences of the assumed decomposition

We now prove conversely that the usual ideal factorization implies the axioms. It is necessary to assume the factorization in the following sharpened form.

Assumption. In the commutative ring R , every ideal not generated by the zero or unit element is uniquely representable as a product of powers of prime ideals. These prime ideals are simple ideals not generated by the unit element; that is, they have no proper divisor different from 0. Conversely, all simple ideals are prime ideals. The uniqueness is to be understood sharply: for every ideal not generated by the zero or unit element,

$$\mathfrak{a}^\rho \neq \mathfrak{a}^{\rho+1}.$$

The existence of a unit element is not presupposed. If no unit element exists, the condition “not generated by the unit element” imposes no restriction.

First we prove axiom III, the existence of a unit element. Suppose axiom III fails. We shall show that R^2 is then a simple ideal without being a prime ideal, contradicting the assumption. By the uniqueness assumption $R \neq R^2$; hence

$$R \equiv 0 \pmod{R^2}, \quad R^2 o t \equiv 0 \pmod{R^2},$$

and so R^2 is not a prime ideal. But R^2 is simple. Indeed, it can be divisible by no non-zero prime ideal, and therefore, by the assumed product representation, it has no divisors except 0 and R^2 . This proves the existence of a unit.

Next comes axiom IV, absence of zero divisors. If a is a zero divisor, so that

$$ab = 0, \quad a \neq 0, \quad b \neq 0,$$

then multiplication of the representations of the principal ideals generated by a and b gives

$$(0) = \mathfrak{p}_1^{\rho_1} \cdots \mathfrak{p}_r^{\rho_r},$$

where the $\mathfrak{p}_i$ are different from the zero and unit ideals. The representation may be assumed shortest in the sense that omission of any factor gives a proper divisor of zero; this need not be automatic, since no uniqueness assumption has been made for the zero ideal. If the representation contains only one prime ideal, then

$$(0) = \mathfrak{p}^\rho = \mathfrak{p}^{\rho+1},$$

contrary to the sharpened uniqueness condition. If it contains more than one prime ideal, then because the unit element exists $\mathfrak{p}_1$ is relatively prime to all the other prime ideals; hence again the sharpened uniqueness condition is contradicted. Thus zero divisors do not occur.

The chain conditions, axioms I and II, follow as well. The assumptions imply that for every non-zero ideal, divisibility is equivalent to the occurrence of a product representation. More precisely, from

$$\mathfrak{m} \equiv 0 \pmod{\mathfrak{a}}, \quad \mathfrak{a} \neq 0,$$

it follows that

$$\mathfrak{m} = \mathfrak{a}\mathfrak{b}, \quad \mathfrak{b} \equiv 0 \pmod{\mathfrak{m}}.$$

If

$$\mathfrak{m} = \mathfrak{p}_1^{\rho_1} \cdots \mathfrak{p}_r^{\rho_r}, \quad \mathfrak{a} = \mathfrak{q}_1^{\sigma_1} \cdots \mathfrak{q}_s^{\sigma_s},$$

then each $\mathfrak{q}_j$ must be identical with one of the $\mathfrak{p}_i$, and the corresponding exponent must not exceed ρ_i . Hence the divisors of $\mathfrak{m}$ are only the finitely many ideals corresponding to the combinations

$$0 \leq \sigma_i \leq \rho_i.$$

The residue class ring modulo $\mathfrak{m}$ therefore satisfies the double chain condition. This proves axioms I and II; the divisor-chain condition holds in R itself, since every proper divisor of the zero ideal is different from the zero ideal.

Before proving axiom V, integral closedness in the quotient field, we derive the usual consequences of ideal factorization.

First, the residue class ring modulo every non-zero ideal is a principal ideal ring. The ideal may be assumed different from the unit ideal, since for the residue ring consisting only of the zero element the assertion is clear. If the ideal is primary, say $\mathfrak{q} = \mathfrak{p}^\rho$, and if

$$\mathfrak{c} \equiv 0 \pmod{\mathfrak{p}}, \quad \mathfrak{c} \not\equiv 0 \pmod{\mathfrak{p}^2},$$

then the argument of §8 gives

$$\mathfrak{p}^\rho = [\mathfrak{c}^\rho, \mathfrak{p}^{\rho+1}]$$

for every ρ . Thus in the residue class ring modulo a primary ideal every ideal is principal. In general, let

$$\mathfrak{m} = \mathfrak{q}_1 \cdots \mathfrak{q}_r$$

be the decomposition into pairwise relatively prime primary ideals, and let

$$R/\mathfrak{m} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_r$$

be the corresponding direct-sum decomposition. Each $\mathfrak{a}_i$ is isomorphic to $R/\mathfrak{q}_i$, and every ideal in $\mathfrak{a}_i$ is principal. If $\mathfrak{c}$ is an arbitrary ideal of the residue class ring, then $\mathfrak{a}_i\mathfrak{c}$ is a principal ideal, say $\mathfrak{a}_i c_i$. Setting

$$c = c_1 + \cdots + c_r$$

one obtains

$$\mathfrak{c} = Rc.$$

The ideal is therefore principal.

The principal-ideal-ring result may also be stated in the following form. If $\mathfrak{c}$ is any ideal, then it can be transformed into a principal ideal by multiplication with an ideal relatively prime to a prescribed ideal $\mathfrak{b}$. Indeed, set $\mathfrak{m} = \mathfrak{b}\mathfrak{c}$. Then $\mathfrak{c}$ becomes principal modulo $\mathfrak{m}$, i.e.

$$\mathfrak{c} = (cR, \mathfrak{m}) = cR + c\mathfrak{b} = c(R, \mathfrak{b}),$$

and hence $\mathfrak{c}\mathfrak{a} = cR$ with $(\mathfrak{a}, \mathfrak{b}) = R$.

We now pass to fractional ideals. A fractional ideal is by definition a finite R -module in the quotient field K .

Theorem. The non-zero finite R -modules in K form an abelian group under multiplication.

The product of two finite R -modules is again finite; multiplication is associative and commutative; and, since $R = R \cdot R$, the unit ideal is the identity element. It remains only to show that the equation

$$\mathfrak{A}X = R$$

has a solution in the system of finite R -modules.

We use the following preliminary remark. If the equation

$$\mathfrak{A}T = \mathfrak{B}$$

has one and only one solution, then

$$T = \mathfrak{B} : \mathfrak{A}.$$

For $T \equiv 0 \pmod{\mathfrak{B} : \mathfrak{A}}$, and conversely

$$\mathfrak{A}(\mathfrak{B} : \mathfrak{A}) = \mathfrak{B}.$$

If first $\mathfrak{A}$ is a principal module, $\mathfrak{A} = (\alpha)$, then $X = (1/\alpha)$ solves $\mathfrak{A}X = R$; it is again a finite R -module. It follows that every finite R -module is the quotient of two ideals of R , which justifies the term fractional ideal. Indeed, if $\tau_1, \dots, \tau_n$ is a module basis of T , write $\tau_i = t_i/\alpha$ and let $\mathfrak{t}$ be the ideal generated by the t_i in R . Then

$$\alpha T = \mathfrak{t},$$

so by the preceding remark

$$T = \mathfrak{t} : (\alpha).$$

The inverse of a general finite module can now be written down. Suppose

$$\mathfrak{A} = \mathfrak{a} : \mathfrak{c},$$

and choose $\mathfrak{b}$ so that $\mathfrak{a}\mathfrak{b}$ is a principal ideal, say (α) . Then

$$\mathfrak{A}\mathfrak{c} = \mathfrak{a}, \quad \mathfrak{A}\mathfrak{b}\mathfrak{c} = (\alpha),$$

and therefore

$$\mathfrak{A}(\mathfrak{b}\mathfrak{c} : (\alpha)) = R.$$

The module $\mathfrak{b}\mathfrak{c} : (\alpha)$ is, as the quotient of an ideal by a principal ideal, again a finite R -module. The group property is proved. It also follows that the quotient of any two ideals $\mathfrak{a} : \mathfrak{b}$, as the solution of $\mathfrak{b}X = \mathfrak{a}$, is a finite R -module.

The group property has the following consequences.

First, cancellation is possible:

$$T = AM : BM \iff T = A : B.$$

Indeed, from $TBM = TAM$ follows $TB = TA$, and conversely.

Second, every fractional ideal admits a representation

$$\mathfrak{C} = \mathfrak{a} : \mathfrak{b}$$

where $\mathfrak{a}$ and $\mathfrak{b}$ are relatively prime; under this condition $\mathfrak{a}$ and $\mathfrak{b}$ are uniquely determined. Existence follows from the previous paragraph and cancellation. If

$$\mathfrak{C} = \mathfrak{a} : \mathfrak{b} = \bar{\mathfrak{a}} : \bar{\mathfrak{b}},$$

then $\mathfrak{a}\bar{\mathfrak{b}} = \mathfrak{b}\bar{\mathfrak{a}}$; uniqueness follows when both pairs are assumed relatively prime.

Third, every principal module generated by a non-integral element - an element belonging to the quotient field but not to R - has a representation as a quotient of principal ideals such that no power of the numerator is divisible by the denominator. Let

$$\eta R = \mathfrak{b} : \mathfrak{c}$$

be the reduced representation, so $(\mathfrak{b}, \mathfrak{c}) = R$. Choose, by the principal-ideal statement above, $\mathfrak{a}$ such that

$$\mathfrak{b}\mathfrak{a} = (b), \quad (\mathfrak{a}, \mathfrak{c}) = R.$$

Then

$$\eta R = \mathfrak{a}\mathfrak{b} : \mathfrak{a}\mathfrak{c} = (b) : (c).$$

No power of b is divisible by c , because $(\mathfrak{a}, \mathfrak{c}) = R$ and $(\mathfrak{b}, \mathfrak{c}) = R$.

Now axiom V follows. Every non-integral element η of the quotient field admits a quotient representation

$$\eta = a/c$$

such that no power of the numerator is divisible by the denominator. Indeed, from $\eta R = (b) : (c)$ it follows that η differs from b/c only by a unit of R . Such an element can satisfy no equation characterizing integrality over R ; for from

$$\eta^n + r_1\eta^{n-1} + \cdots + r_n = 0$$

one obtains

$$a^n \equiv 0 \pmod{cR},$$

contrary to the construction. Thus R is integrally closed in its quotient field.

A useful corollary is the following. If axioms I-IV hold in a commutative ring and every primary ideal is irreducible, then axiom V also holds. Indeed, by axioms I-III every power $\mathfrak{p}^\rho$ of a prime ideal is primary, in particular $\mathfrak{p}^2$ is primary. By assumption $\mathfrak{p}^2$ is irreducible. It remains to prove a representation

$$\mathfrak{p} = (cR, \mathfrak{p}^2),$$

for then §8, 3 shows that every primary ideal is a power of its prime ideal, and this, together with the remaining hypotheses, implies integral closedness by the preceding paragraph.

By the divisor-chain condition one can write

$$\mathfrak{p} = (c_1R, \dots, c_kR),$$

where the c_i are chosen linearly independent over the residue class field modulo $\mathfrak{p}$. If $k > 1$, this

independence gives a decomposition

$$\mathfrak{p}^2 = [\mathfrak{q}_1, \mathfrak{q}_2],$$

where

$$\mathfrak{q}_1 = (c_1 R, \mathfrak{p}^2), \quad \mathfrak{q}_2 = (c_2 R, \dots, c_k R, \mathfrak{p}^2).$$

Both $\mathfrak{q}_1$ and $\mathfrak{q}_2$ are proper divisors of $\mathfrak{p}^2$, contradicting irreducibility. Therefore $k = 1$ and the required representation follows.

Conversely, under axioms I-V every primary ideal is irreducible. A power $\mathfrak{p}^\rho$ of a prime ideal cannot be written as the least common multiple of proper divisors of the form $\mathfrak{p}^\alpha$ and $\mathfrak{p}^\beta$.

If zero divisors are allowed in the ring, axioms I-III together with integral closedness in the quotient ring do not imply that every primary ideal is irreducible. Let T be a ring in which all axioms I-IV except integral closedness hold; such rings exist, for example finite orders different from the full ring of integral elements in a finite extension of the quotient field of a ring satisfying I-V. By the preceding corollary there is in T a reducible primary ideal $\mathfrak{q}$. Let $\mathfrak{p}^\rho$ be a proper multiple of $\mathfrak{q}$, and let

$$R = T/\mathfrak{p}^\rho$$

be the residue class ring. Then the ideal corresponding to $\mathfrak{q}$ becomes in R a non-zero reducible primary ideal. The ring R satisfies axioms I-III and is nevertheless integrally closed in its quotient ring. For every element of T not divisible by $\mathfrak{p}$ becomes relatively prime to $\mathfrak{p}^\rho$, so in R every regular element is a unit; hence R coincides with its quotient ring.

11.5 §10. Double chain condition and composition series

We shall now show that, for arbitrary well-ordered module domains (§2), the validity of the double chain condition - every divisor chain and every multiple chain of modules terminates - is equivalent to the existence of a composition series. Specializing the module domain to a commutative ring then gives the corresponding facts for the system of all ideals of the ring. In that case “module isomorphism” below is to be replaced throughout by “ring isomorphism”.

A module domain G is called **simple** if there are in G no R -modules other than G itself and the zero module 0. A module A is called **simple in G** if G/A is simple.

A multiple chain

$$G, U_1, \dots, U_r, 0$$

is called a composition series of G of length r if all modules in the chain are distinct and each module is simple in the preceding one. Thus the residue modules, or quotient groups,

$$U_{i-1}/U_i$$

are simple module domains, and similarly G/U_1 and $U_r/0$. By forming the residue module G/A the case of a composition series from G down to a module $A \neq 0$ is reduced to the absolute case.

Assume first that the double chain condition holds in G . Then there exists a composition series in G , provided $G \neq 0$. From the assumed well-ordering of G and the divisor-chain condition, a well-ordering of the system of all modules is obtained by the quasi-lexicographic arrangement used in §6. By this well-ordering and the divisor-chain condition, at least one module simple in G exists. Let G_1 be the first proper divisor of G in the well-ordering, and, in general, let G_i be the first proper divisor of G_{i-1} . The well-defined chain

$$G, G_1, G_2, \dots$$

terminates by the divisor-chain condition and so necessarily reaches a module A simple in G . If $A \neq 0$, the same procedure gives a module simple in A , and so on. We obtain a well-defined multiple chain which, by the multiple-chain condition, must terminate. It is therefore a composition series of G .

Conversely, suppose that a composition series exists in G . Then the double chain condition holds in G . The proof proceeds by complete induction, and no well-ordering of G is needed for this direction. During the induction one also proves the Jordan-Hölder theorem under the sole assumption that a composition series exists.

If G is simple, so that the only composition series is $G, 0$, the following assertions are immediate:

- (a) Through every module simple in G one can draw a composition series.
- (b) Jordan-Hölder theorem: every composition series has the same length and the same system of quotient groups, up to order and isomorphism.
- (c) Through every module of G different from 0 one can draw a composition series.
- (d) The multiple-chain condition holds.
- (e) The divisor-chain condition holds.

Assume, then, that the assertions (a)-(e) have been proved for every module domain possessing a composition series of length $r < r+1$. We prove them for a module domain G with a composition series

$$G, U, U_1, \dots, U_r, 0$$

of length $r + 1$.

First, if there is no module simple in G other than U , there is nothing to prove. Let B be a module simple in G and different from U . Since U and B are both simple in G and are distinct, necessarily

$$(U, B) = G.$$

Put

$$M = [U, B].$$

The second isomorphism theorem (§4, 2) gives

$$G/B \simeq U/M, \quad G/U \simeq B/M.$$

Thus M is simple in both U and B . Since U has a composition series of length $r < r + 1$, the induction hypotheses (a) and (d) give a composition series in U passing through M , say

$$U, M, M_1, \dots, M_{r-1}, 0.$$

Then

$$G, B, M, M_1, \dots, M_{r-1}, 0$$

is a composition series of G passing through B .

To prove the Jordan-Hölder theorem, let

$$G, A, A_1, \dots, A_s, 0 \quad \text{and} \quad G, B, B_1, \dots, B_t, 0$$

be two composition series of G . If $A = B$, the result follows immediately from the induction hypothesis for length $r < r + 1$. If $A \neq B$, comparison with the series constructed above,

$$G, A, M, M_1, \dots, M_{r-1}, 0 \quad \text{and} \quad G, B, M, M_1, \dots, M_{r-1}, 0,$$

shows the following. By the induction hypothesis the systems of quotient groups of the first given series and of the first constructed series agree; likewise for the second given series and the second constructed series, since the latter passes through B and has length $r + 1$. Hence the lengths are equal. Moreover, the isomorphism noted above gives the required agreement between the quotients arising in the two constructed series. Therefore the two original composition series have the same quotient groups up to order and isomorphism. This proves the Jordan-Hölder theorem.

We next need a preliminary observation. Let A, B be modules in G and let B be simple in A . Then the modules (B, H) and (A, H) are either equal, or (B, H) is simple in (A, H) . Indeed, by the second isomorphism theorem,

$$(A, B, H)/(B, H) \simeq A/[A, (B, H)].$$

Since $B = 0(A)$, this may also be written with

$$C = [A, (B, H)].$$

Now C lies between A and B , and hence, by the simplicity assumption, is equal either to A or to B . The assertion follows.

Let

$$G, U, U_1, \dots, U_r, 0$$

again be a composition series of G , and let H be any module of G different from 0. Form the series

$$G, (U, H), (U_1, H), \dots, (U_r, H), H.$$

By the preliminary observation, the distinct modules among these form a composition series from G down to H . Therefore G/H has a composition series, and hence, by the induction hypothesis, one can draw a composition series through H in G/H . Adding H gives a composition series of G through H . Repetition shows moreover that such a series may be taken as an extension of the one just constructed from G down to H .

The multiple-chain condition is now immediate. Suppose again that G has a composition series of length $r + 1$, and let

$$G, B, B_1, \dots, B_n, \dots$$

be a multiple chain, each module being a proper multiple of the preceding one. There is no loss in beginning the chain with G , since G may always be adjoined as first term. By the preceding paragraph there is a composition series through B in G , and therefore B has a composition series of shorter length. By the induction hypothesis the chain

$$B, B_1, \dots, B_n, \dots$$

terminates; hence the chain with G adjoined terminates as well.

Finally, the divisor-chain condition is proved similarly. Let

$$G, G_1, G_2, \dots, G_n, \dots$$

be a divisor chain in which every module is a proper divisor of the preceding one. Again it is no restriction that the chain begins with G . By the preceding paragraph there is a composition series through G_1 in G ; hence G/G_1 has a composition series of shorter length. By the induction hypothesis the chain

$$G/G_1, G_2/G_1, \dots$$

terminates in G/G_1 . By the one-to-one correspondence supplied by the first isomorphism theorem (§4, 2), the original chain beginning with G_1 also terminates; therefore the chain with G adjoined terminates. The equivalence of the two assumptions - existence of a composition series and the double chain condition - is proved.

12 The Discriminant Theorem for the Orders of an Algebraic Number or Function Field

The Dedekind discriminant theorem says, as is well known, that a prime number divides the field discriminant if and only if it is divisible by at least the square of a prime ideal.

I show here that this theorem remains valid for all finite orders of a finite algebraic number field - that is, for all rings of algebraic integers which contain the ring of rational integers - in the following form: a prime number p divides the discriminant of an order if and only if, in the ideal decomposition valid in that order of the principal ideal (p) into greatest primary components, at least one proper primary ideal occurs, that is, a primary ideal which is not itself a prime ideal. Since in the principal order - the system of all algebraic integers of the field - there are no primary ideals other than powers of prime ideals, Dedekind's theorem follows as a special case.

The proof is obtained by passing from the order modulo (p) to the residue class ring. The assertion is thereby reduced to the decomposition theory of rings of finite rank over a field contained in the ring. Such a ring is a commutative system of hypercomplex quantities with principal unit and with coefficients from that field. In the present case the coefficient field is the residue class field of the rational integers modulo p , and the decomposition of (p) corresponds in the residue class ring to the decomposition of the zero ideal.

Call such a finite-rank ring **completely reducible** if its zero ideal can be represented as the least common multiple of finitely many prime ideals. We are then concerned with criteria for complete reducibility. First, if the coefficient field is perfect, the ring is completely reducible if and only if the extension ring obtained by passing to an algebraically closed coefficient field is completely reducible. For this extension ring the non-vanishing of the discriminant is an especially simple necessary and sufficient criterion of complete reducibility. Since the discriminant of the order modulo every prime number passes into the discriminant of the residue class ring, the discriminant theorem follows.

For function fields in place of number fields the order must contain the ground domain of all integral rational functions; for algebraic functions in several indeterminates or for integral algebraic functions it must contain the ground domain of all integral functionals with integral-rational coefficients. In this situation the imperfect coefficient-field case may occur in the residue class ring, for example when, in the integral functional domain case, one passes to the residue class ring modulo a prime number. With an imperfect coefficient field, prime ideals of the first and second kind must be distinguished according as the residue class field modulo the prime ideal is an extension field of the first or second kind. Correspondingly, complete reducibility of the first and second kind must be distinguished. The criteria just stated concern only complete reducibility of the first kind, which alone occurs in the perfect case.

In its most general form, then, the discriminant theorem says: a prime element of the ground domain divides the discriminant of an order if and only if in the ideal decomposition of (p) either at least one proper primary component or a prime ideal of the second kind occurs.

For the principal order this fact was first observed in examples by Kronecker, after the earlier Festschrift statement had still contained, mistakenly, the form valid for the principal order of a number field, though without a complete proof. For the principal order Ostrowski gave a proof of this discriminant theorem and further ramification theorems connected with it, along with a detailed account of the literature. The discriminant theorem for all orders of relative fields follows directly by passing to the quotient ring with respect to each prime ideal of the ground domain. The idea of this passage, known in special cases, has been emphasized in principle by W. Krull; a systematic theory of quotient rings in a more general context is given by H. Grell.

The uniform treatment here of all orders, and of both perfect and imperfect coefficient fields, rests not only on ideal theory but also on taking from the start the theory of rings of finite rank, and hence the general concept of finite extension, as fundamental, rather than the concept of an extension generated by a primitive element. The module basis of an order, together with the relations among its elements, takes the place of the powers of a primitive element and its defining equation, or the place of the powers of a fundamental form and its fundamental equation.

Such a module basis passes modulo every prime element of the ground domain into a basis of the residue class ring. This need not be true, even for the principal order, of a basis consisting of powers of a primitive element; nor of a basis consisting of powers of the fundamental form, once integral algebraic functions of several indeterminates are involved. Equivalently, the residue class ring of the polynomial ring in one indeterminate, with coefficients in the ground domain, modulo an arbitrary defining equation - even the fundamental equation - need not be isomorphic modulo p to the residue class ring of the principal order modulo (p) . The main difficulties in the usual representations lie precisely in this failure of isomorphism. Since the auxiliary polynomial ring does not occur here, these difficulties are automatically avoided. At the same time the additive decomposition theory of the residue class ring becomes the equivalent of the multiplicative decomposition of the equation; when the equation decomposes into relatively prime factors, a corresponding additive decomposition occurs in the polynomial residue class ring.

It should be added that the discriminant theorem can only be the first theorem of a general ramification theory for orders, just as for the principal order the discriminant theorem is accompanied by the finer theory of the different or ramification ideal, which there permits a more precise determination of exponents. This determination of exponents, resting on the power representation of the primary factors of equations, may be interpreted as determination of the length of a composition series, from p to $\mathfrak{q} = p^t$, and will presumably appear in this form in the general ramification theory.

12.1 §1. Extension rings of fields

We begin with a few remarks about arbitrary rings. If T is a ring and S is a system of elements of T , the “ring derived from S in T ” means the intersection of all rings in T which contain S . Correspondingly, one defines the “ideal derived from S in T ”, or the “ P -module derived from S in T ” when P is a subring of T .

If R is a subring of T and S is a system of elements of T , the ring derived from R and S in T is denoted by

$$R[S]$$

and is called the ring obtained by ring-adjointing S to R . It consists of all polynomials in the elements of S with coefficients in R , where equality and the operations are fixed by the equality and operation relations already present in T . Because these relations have already been fixed, in special cases it is enough to form only a subsystem of all polynomials - for example, all linear forms in S with coefficients in R . Such a case occurs, for instance, when R is a ring with unit element and S is a field with the same unit element. Then all linear combinations

$$\sum c_i y_i,$$

with $c_i \in S$ and $y_i \in R$, form the ring $R[S]$.

One can also construct the extension ring $R[S]$ when only the ring R is given and, in addition, a system S of elements not lying in R is given with equality relations and perhaps mutual operation relations. One shows that there is always a ring containing R and S , namely the ring formally formed from all polynomials in S with coefficients in R , subject to the usual operation rules. Apart from these operation rules, there remains complete freedom in defining equality, except that the equality definitions already valid in R and S must be included. Thus one is forced to identify only those polynomials which, by equality in R and S and by the stated operations, are built in the same way from equal elements of R and S . The ring T so constructed, containing R and S , is then identical with the ring derived in T from R and S .

From now on, rings R are assumed to have a unit element and to contain a field P as subring; they are thus over-rings of fields. The unit element e of R is also the unit element of P , and R is a P -module.

If $\bar{P}$ is an overfield of P , the extension ring

$$\bar{R} = R[\bar{P}]$$

obtained by ring-adjointing $\bar{P}$ to R is to be understood as follows. Assume that the set-theoretic intersection of R and $\bar{P}$ is exactly P , and that no operation relations exist between elements of R not in P and elements of $\bar{P}$ not in P . This assumption can always be achieved by replacing $\bar{P}$ by an equivalent extension field of P , or R by an equivalent extension ring.

The elements of $\bar{R}$ are all formally formed linear combinations

$$c_1 y_1 + \dots + c_n y_n,$$

where the c_i run through the elements of $\bar{P}$ and the y_i through the elements of R . The definition is to be unambiguous: if the y_i are replaced by equal elements of R , and the c_i by equal elements of $\bar{P}$, the element of $\bar{R}$ is to be the same.

The sum is defined in the evident way,

$$\sum c_i y_i + \sum d_i y_i = \sum (c_i + d_i) y_i,$$

where zeros may occur among the c_i, d_i so that formally the same y_i occur. The product is likewise defined by distributivity,

$$\left(\sum c_i y_i \right) \left(\sum d_j y_j \right) = \sum c_i d_j y_i y_j.$$

For equality one declares the following. If

$$y_{i_1}, \dots, y_{i_s}$$

are linearly independent over P , then two elements

$$\sum c_j y_{i_j}, \quad \sum d_j y_{i_j}$$

are equal if and only if $c_j = d_j$ for all j in $\bar{P}$. In other words, elements of R linearly independent over P are declared linearly independent over $\bar{P}$.

This gives equality for all elements of $\bar{R}$, because every element can be expressed by linearly independent elements of R . Indeed, from a relation

$$u = c_1 y_1 + \dots + c_n y_n$$

in $\bar{R}$, multiplication by $b = be$ gives

$$bu = bc_1 y_1 + \dots + bc_n y_n,$$

so that all elements are expressible in terms of linearly independent elements of R . Equality is therefore coefficient equality over $\bar{P}$. For elements lying simultaneously in R and $\bar{R}$, or in P and $\bar{P}$, this agrees with the old equality. For the remaining elements of $\bar{R}$, the equality in R and $\bar{P}$ says no more than the required unambiguity of the definition.

Nor does the requirement that sum and product be unambiguous introduce further relations: equal elements may be assumed coefficientwise equal, and hence they give formally equal sums and products. One checks at once that all ring axioms hold. Thus the extension ring $R[\bar{P}]$ generated by ring adjunction always exists.

Now let $\mathfrak{a}$ be an ideal of R . The module product

$$\bar{R}\mathfrak{a}$$

is an ideal $\bar{\mathfrak{a}}$ of $\bar{R}$, the extension ideal of $\mathfrak{a}$. It is also the ideal derived from $\mathfrak{a}$ in $\bar{R}$; it consists of

all finite linear combinations

$$\sum c_i a_i,$$

where $a_i \in \mathfrak{a}$ and $c_i \in \bar{P}$.

If $\bar{\mathfrak{b}}$ is an ideal of $\bar{R}$, then the intersection

$$[\bar{\mathfrak{b}}, R]$$

is an ideal $\mathfrak{b}$ of R , the contraction ideal of $\bar{\mathfrak{b}}$. Every ideal $\mathfrak{a}$ of R is a contraction ideal, namely the contraction of its own extension ideal. For an element a of $[R, \bar{\mathfrak{a}}]$ has the form $\sum c_i a_i$ with the a_i chosen linearly independent over P and hence over $\bar{P}$. It follows that the elements a, a_i are linearly dependent in $\bar{R}$, and therefore, by the equality definition, already in R ; hence a belongs to $\mathfrak{a}$. The converse is immediate.

If $\bar{\mathfrak{p}}$ is a prime ideal of $\bar{R}$, then

$$\mathfrak{p} = [R, \bar{\mathfrak{p}}]$$

is a prime ideal of R . By the second isomorphism theorem, the residue ring $R/\mathfrak{p}$ is isomorphic to a subring of $\bar{R}/\bar{\mathfrak{p}}$, and hence has no zero divisors. More precisely,

$$(R, \bar{\mathfrak{p}})/\bar{\mathfrak{p}} \simeq R/[R, \bar{\mathfrak{p}}],$$

where $(R, \bar{\mathfrak{p}})$ denotes the sum, the greatest common divisor, of the modules R and $\bar{\mathfrak{p}}$.

12.2 §2. Rings of finite rank. Representation as a direct sum

If R is a finite P -module, then R has finite rank over P , say n : there are n , but no more, linearly independent elements over P , and every system of n linearly independent elements is a module basis. If a P -module B of finite rank is divisible by another P -module A of the same finite rank, then they are equal.

From now on R is assumed to have finite rank n over P . Then every P -module in R has finite rank. In every proper divisor or multiple chain of P -modules in R , the rank of each module must therefore be greater, respectively smaller, than the rank of the preceding module; hence the chains terminate. In particular the double chain condition holds for the system of all ideals of R .

It follows that a prime ideal $\mathfrak{p}$ of R has no proper divisor other than the unit ideal. Thus the residue ring $R/\mathfrak{p}$ modulo a prime ideal different from the unit ideal is a field. If $\mathfrak{q}$ is a primary ideal belonging to $\mathfrak{p}$, say

$$\mathfrak{q} \equiv 0 \pmod{\mathfrak{p}}, \quad \mathfrak{p}^\rho \equiv 0 \pmod{\mathfrak{q}},$$

then $R/\mathfrak{q}$ is a primary ring: a power, in fact at latest the ρ -th power, of every zero divisor vanishes. Further, $R/\mathfrak{q}$ contains only zero divisors and units. This is immediate because every ideal not divisible by $\mathfrak{p}$ is relatively prime to $\mathfrak{p}^\rho$ and hence to $\mathfrak{q}$.

From the double chain condition and the existence of the unit element it follows that every ideal $\mathfrak{a}$ of R is uniquely representable as a product of finitely many pairwise relatively prime primary ideals. Because of relative primality, this product is the least common multiple, and it corresponds to a representation of the residue class ring $R/\mathfrak{a}$ as a direct sum of primary rings: $R/\mathfrak{a}$ is the greatest common divisor of these rings, and the sum representation of every element of $R/\mathfrak{a}$ is unique.

Let the decomposition of the zero ideal of R be

$$(0) = \mathfrak{q}_1 \mathfrak{q}_2 \cdots \mathfrak{q}_r = [\mathfrak{q}_1, \mathfrak{q}_2, \dots, \mathfrak{q}_r],$$

and let the corresponding direct-sum decomposition be

$$R = R_1 + \cdots + R_r, \quad R_i R_j = (0) \quad (i \neq j), \quad R_i^2 = R_i.$$

Here R_i is an ideal of R , and

$$R_i \simeq R/\mathfrak{q}_i;$$

the element $y_i \in R_i$ corresponds to the class represented by y_i in $R/\mathfrak{q}_i$.

If $\mathfrak{a}$ is any ideal of R , the distributive law gives

$$\mathfrak{a} = R_1 \mathfrak{a} + \cdots + R_r \mathfrak{a}.$$

The components

$$\mathfrak{a}_i = R_i \mathfrak{a}$$

are ideals both in R_i and in R . As ideals they are finite P -modules, and because the sum is direct their ranks add to the rank of $\mathfrak{a}$ or of R .

If the unit element has the decomposition

$$e = e_1 + \cdots + e_r,$$

where

$$e_i e_j = 0 \quad (i \neq j), \quad e_i^2 = e_i,$$

then every element y of R has the unique decomposition

$$y = y_1 + \cdots + y_r, \quad y_i = y e_i \in R_i,$$

and $R_i = R e_i$. The orthogonality relations show that the i -th component of a product is $y_i z_i$, and the i -th component of a sum is $y_i + z_i$.

The ring R_i contains a subfield

$$P_i = P e_i$$

isomorphic to P , and has finite rank over P_i ; this rank is equal to the rank of R_i as P -module. For if $c \in P$, then $ce_i = ce$ modulo $\mathfrak{q}_i$, so $ce_i \neq 0$ whenever $c \neq 0$. Hence $P \rightarrow P_i$ is an isomorphism. Linear dependence over P among elements of R_i passes, by multiplication with e_i , to linear dependence over P_i , and conversely, since $P_i = Pe_i$, every relation over P_i is already a relation over P . Thus the ranks agree.

If

$$R = R_1 + \cdots + R_r,$$

then for any extension ring $\bar{R} = R[\bar{P}]$ there is a corresponding decomposition

$$\bar{R} = \bar{R}_1 + \cdots + \bar{R}_r,$$

where $\bar{R}_i$ denotes the extension ideal of R_i in $\bar{R}$, and at the same time the ring derived from R_i in $\bar{R}$. Indeed, from $(0) = \mathfrak{q}_1 \cdots \mathfrak{q}_r$ follows by multiplication with $\bar{R}$ the product representation

$$(0) = \bar{\mathfrak{q}}_1 \cdots \bar{\mathfrak{q}}_r,$$

and the $\bar{\mathfrak{q}}_i$ remain pairwise relatively prime. The direct summands are the corresponding products of the other factors, hence the extension ideals of the R_i . But this extension ideal is, by §1, 3, just the extension ring

$$R_i[P_i]$$

with $P_i = Pe_i$. Thus in extending one may extend the individual components separately. In general $\bar{R}_i$ need no longer be a primary ring.

We shall also need a distinction between prime ideals of first and second kind. Modulo every prime ideal $\mathfrak{p}$ different from the unit ideal, the residue class ring $R/\mathfrak{p}$ is a field. It contains a subfield isomorphic to P , namely the classes represented by elements of P . This follows from the second isomorphism theorem:

$$(P, \mathfrak{p})/\mathfrak{p} \simeq P/[P, \mathfrak{p}] \simeq P,$$

because $\mathfrak{p}$ contains no non-zero element of P . The field $R/\mathfrak{p}$ has finite rank over this copy of P , and is therefore a finite algebraic extension field. According as this extension is of the first or second kind, $\mathfrak{p}$ is called a prime ideal of the first or second kind.

12.3 §3. Completely reducible rings

The ring R is called **completely reducible** if in the product decomposition of its zero ideal all primary components are prime ideals. Equivalently, R can be represented as a direct sum of finitely many fields. The completely reducible ring is called completely reducible of the first kind if all its prime ideals are of the first kind; otherwise it is completely reducible of the second kind.

If P is a perfect field, only complete reducibility of the first kind can occur; this is in particular

always the case when P is algebraically closed. In what follows $\bar{P}$ denotes the algebraic closure of P , and

$$\bar{R} = R[\bar{P}].$$

Theorem. The ring R is completely reducible of the first kind if and only if the extension ring

$$\bar{R} = R[\bar{P}]$$

with algebraically closed $\bar{P}$ is completely reducible.

The proof has three parts.

First, complete reducibility of an extension ring implies complete reducibility of R . More generally, if the zero ideal of an extension ring has a representation as a least common multiple of prime ideals,

$$(0) = [\bar{\mathfrak{p}}_1, \dots, \bar{\mathfrak{p}}_r],$$

then by intersecting with R one obtains the representation

$$(0) = [R, \bar{\mathfrak{p}}_1], \dots, [R, \bar{\mathfrak{p}}_r]$$

of the zero ideal in R ; and the contracted ideals are prime by §1, 3.

Second, complete reducibility of the first kind of R implies complete reducibility of the first kind of every extension ring, and more generally of every intermediate ring between R and the extension ring. Since by §2, 3 the direct-sum decomposition of R induces one of any extension ring, it is enough to consider one component. Thus we may assume without loss of generality that R is a field, and indeed a finite extension field of the first kind of its subfield P . Then R is obtained from P by adjoining one primitive element of the first kind. Equivalently,

$$R \simeq P[x]/(f(x)),$$

where $f(x)$ is a prime polynomial of the first kind in the polynomial ring $P[x]$. If Q is any extension field of P , then

$$Q[x]/(f(x)) \simeq R[Q],$$

for the rank of the residue ring remains equal to the degree of $f(x)$ when passing from $P[x]$ to $Q[x]$, and hence this is exactly the construction of the extension ring described in §1, 2. In $Q[x]$ the polynomial $f(x)$ decomposes into pairwise relatively prime prime functions of the first kind; each generates a prime ideal in $Q[x]$. Therefore $Q[x]/(f(x))$, and hence $R[Q]$, is completely reducible of the first kind. Taking $Q = \bar{P}$ proves the second step.

Third, if R is completely reducible of the second kind, then $\bar{R}$ is not completely reducible. As before it suffices to take R to be a field, now a finite extension field of the second kind of P . We must show that in the direct-sum decomposition of $\bar{R}$ into primary rings at least one proper primary component occurs. It is enough to exhibit a non-zero element of $\bar{R}$ whose power is zero:

then some component is non-zero but nilpotent, and the corresponding component ring is not a field.

Let γ be an element of the second kind in R , of exponent $\ell > 1$, and let

$$F(\gamma) = \gamma^{kp^\ell} + a_1\gamma^{kp^\ell-1} + \cdots + a_{kp^\ell} = 0$$

be a lowest-degree equation satisfied by γ over P , where p is the characteristic of P . The elements

$$1, \gamma, \gamma^2, \dots, \gamma^{kp^\ell-1}$$

are linearly independent over P and therefore remain linearly independent over $\bar{P}$ by the equality definition. If

$$G(\gamma) = \gamma^{kp^{\ell-1}} + \cdots$$

is chosen so that

$$G(\gamma)^p = F(\gamma),$$

then $G(\gamma)$ is non-zero in $\bar{R}$ but its p -th power vanishes. This proves the third step.

The first and third steps show that complete reducibility of R of the first kind follows from complete reducibility of $\bar{R}$; the second step gives the converse.

As a consequence, if R is completely reducible of the first kind, then $\bar{R}$ is a direct sum of rings of rank one, hence of fields isomorphic to $\bar{P}$. Such a decomposition already exists in some ring $R[Q]$, where Q is a finite extension field of P .

Indeed, by the theorem, $\bar{R}$ is a direct sum of fields which, as finite extensions of subfields isomorphic to $\bar{P}$, are identical with those subfields. Thus

$$\bar{R} = \bar{P}e_1 + \cdots + \bar{P}e_n, \quad e = e_1 + \cdots + e_n,$$

and the components e_i of the identity form a linearly independent module basis of $\bar{R}$. Expressing the e_i as linear combinations of elements of R with coefficients in $\bar{P}$, the set of all coefficients generates a finite extension field Q of P . The e_i therefore already belong to $R[Q]$, and the same decomposition exists there. In every intermediate ring between $R[Q]$ and R the indecomposable components arise as extension rings of the Qe_i .

12.4 §4. Matrix representation, traces, and discriminants in rings of finite rank

In this and the next section the hypotheses on the base ring are slightly more general than before, in order also to cover orders in number and function fields. The treatment is a general and uniform formulation of essentially known facts.

Assume that R is an extension ring of an integral domain Z , and that the unit element of R already lies in Z . For every Z -module in R , in particular for R itself, at least one finite module

basis exists which consists of elements linearly independent over Z . Two such bases $a_1, \dots, a_n$ and $b_1, \dots, b_n$ are bases of the same Z -module if and only if the transformation determinant is a unit of Z . This hypothesis is plainly fulfilled for the finite-rank rings considered above, where Z is the field P , and it is also fulfilled for orders in number and function fields, where Z is the ring of rational integers, respectively the functional domain.

Let $a_1, \dots, a_n$ be a linearly independent Z -module basis of an ideal $\mathfrak{a}$, and let y be an arbitrary element of R . Since ya_i again belongs to $\mathfrak{a}$, equations with coefficients in Z hold:

$$ya_i = c_{i1}a_1 + \dots + c_{in}a_n.$$

In matrix form, if $(a_1, \dots, a_n)$ denotes a row matrix,

$$y(a_1, \dots, a_n) = (a_1, \dots, a_n)C.$$

As y ranges over R , the matrices C form a ring $R_{\mathfrak{a}}$ homomorphic to R . More precisely,

$$R_{\mathfrak{a}} \simeq R/((0) : \mathfrak{a}),$$

where $(0) : \mathfrak{a}$ is the ideal quotient of the zero ideal by $\mathfrak{a}$. Differences and products of elements correspond to differences and products of matrices; scalar elements $c \in Z$ correspond to diagonal matrices cE . Exactly those elements y for which $y\mathfrak{a} = (0)$ correspond to the zero matrix, which proves the stated isomorphism.

A different basis of $\mathfrak{a}$ gives an equivalent matrix ring. If

$$(\bar{a}_1, \dots, \bar{a}_n) = (a_1, \dots, a_n)P$$

with determinant of P a unit of Z , then the corresponding matrices satisfy

$$\bar{C} = P^{-1}CP.$$

Two ideals $\mathfrak{a}$ and $\mathfrak{b}$ of R belong to the same ideal class if they are isomorphic as R -modules; that is, their elements correspond one-to-one in such a way that difference and multiplication by the same element of R correspond. Two matrix rings obtained as above belong to the same representation class if they are equivalent, i.e. if

$$R_2 = P^{-1}R_1P$$

for a unimodular matrix P . Ideal classes and representation classes correspond one-to-one.

Indeed, different bases of the same ideal give equivalent matrix rings, as just shown. Isomorphic ideals also give the same representation class: if $b_1, \dots, b_n$ correspond to a basis $a_1, \dots, a_n$, then ya_i and yb_i correspond and the same matrices occur. Conversely, if the matrix rings are

equivalent, replace the basis of $\mathfrak{b}$ by the transformed basis. Then the matrices coincide, and the correspondence sending

$$c_1a_1 + \cdots + c_na_n \quad \text{to} \quad c_1b_1 + \cdots + c_nb_n$$

is an R -module isomorphism. Thus the ideals are in the same class. The class of the unit ideal is called the principal class; every basis of R gives the principal representation class.

Let C be the matrix corresponding to the element y in the representation class determined by $\mathfrak{a}$. The polynomial

$$|tE - C|$$

is invariant under equivalence of representations. Hence its coefficients are invariants of the ideal class. In particular, the coefficient of $-t^{n-1}$ is called the trace of y with respect to the class $\mathfrak{a}$, written

$$S_{\mathfrak{a}}(y) = c_{11} + c_{22} + \cdots + c_{nn}.$$

The constant coefficient, up to sign, is the norm of y with respect to the class.

For a fixed class $(\mathfrak{a})$, the traces $S_{\mathfrak{a}}(y)$ form a Z -module to which R is homomorphic as a Z -module. This follows from the matrix homomorphism:

$$S_{\mathfrak{a}}(x - y) = S_{\mathfrak{a}}(x) - S_{\mathfrak{a}}(y), \quad S_{\mathfrak{a}}(cy) = cS_{\mathfrak{a}}(y).$$

If R has no zero divisors, the trace and norm of an element are independent of the class. For every non-zero ideal has the same rank as R , and a basis of an ideal is at the same time a basis of the unit ideal in the quotient field. Thus all traces and norms are generated in the quotient field by the unit ideal and are independent of the class.

Now let $\omega_1, \dots, \omega_n$ be a Z -module basis of an ideal $\mathfrak{r}$. The determinant

$$|S_{\mathfrak{a}}(\omega_i\omega_j)|$$

is well defined only up to multiplication by the square of a unit of Z , but the principal ideal it generates is an invariant of the ideal $\mathfrak{r}$ and the class $(\mathfrak{a})$. A change of basis transforms the matrix cogrediently; the determinant therefore changes by the square of the transformation determinant. This principal ideal is called the discriminant ideal of $\mathfrak{r}$ with respect to the class $(\mathfrak{a})$. If R has no zero divisors, the discriminant ideal of an ideal is independent of the chosen class, because each entry in the determinant is already independent of the class.

12.5 §5. Direct sums of classes

If the ideal $\mathfrak{a}$ is a direct sum,

$$\mathfrak{a} = \mathfrak{b} + \mathfrak{c},$$

then every ideal in the class of $\mathfrak{a}$ is likewise a direct sum, with components isomorphic to $\mathfrak{b}$ and $\mathfrak{c}$ respectively. Thus one may speak of the direct sum of the ideal class and of its component classes.

Similarly, if a ring $R_{\mathfrak{a}}$ in a representation class is a direct sum of subrings which are at the same time ideals in $R_{\mathfrak{a}}$, then the same is true for every equivalent representative, and the components are equivalent rings. Thus one may speak of the direct sum of the representation class and of its component classes. The direct sum of ideal classes corresponds to the direct sum of representation classes; in this sense we simply speak of the direct sum of a class.

Let $b_1, \dots, b_s$ be a basis of $\mathfrak{b}$ and $c_1, \dots, c_t$ a basis of $\mathfrak{c}$. Since the sum is direct, the two bases together form a basis of $\mathfrak{a}$. With respect to this basis, the matrix associated with an arbitrary element of R has block-diagonal form

$$\begin{pmatrix} D^{(b)} & 0 \\ 0 & D^{(c)} \end{pmatrix}.$$

The systems of matrices with only one block non-zero form subrings, and in fact ideals, whose direct sum is the whole representation ring. They are isomorphic to the representation rings obtained from $\mathfrak{b}$ and $\mathfrak{c}$ themselves.

The trace of an element with respect to a class is the sum of the traces with respect to the component classes. If $\beta \in \mathfrak{b}$, then the trace of β with respect to $(\mathfrak{a})$ is equal to the trace of β with respect to $(\mathfrak{b})$. Hence the discriminant ideal of the component ideal $\mathfrak{b}$, taken with respect to $(\mathfrak{a})$, is equal to its discriminant ideal taken with respect to the component class $(\mathfrak{b})$.

Finally, the discriminant ideal of $\mathfrak{a}$ with respect to $(\mathfrak{a})$ is the product of the discriminant ideals of the components with respect to their component classes. This follows by taking the basis corresponding to the direct sum: the discriminant determinant becomes block diagonal, and its determinant is the product of the component determinants.

Now let X be an extension ring without zero divisors over Z such that the intersection of R and X is Z , and let

$$\bar{R} = R[X]$$

be the extension ring of the same rank as in §1, 2. If $\mathfrak{a}$ and $\mathfrak{r}$ are ideals of R , and $\bar{\mathfrak{a}}$ and $\bar{\mathfrak{r}}$ their extensions in $\bar{R}$, then the discriminant ideal of $\bar{\mathfrak{r}}$ with respect to $(\bar{\mathfrak{a}})$ is the extension ideal in X of the discriminant ideal of $\mathfrak{r}$ with respect to $(\mathfrak{a})$. One can use the same bases in the discriminant construction. In particular, the discriminant ideal of $\bar{R}$ with respect to the principal class is the extension of the discriminant ideal of R with respect to the principal class. This ideal, in X or Z , is called simply the discriminant ideal of the ring.

12.6 §6. Discriminant criterion for complete reducibility of the first kind

From now on R is again an extension ring of a field P , and $\bar{P}$ is an extension field of P . The non-vanishing of the discriminant will be a necessary and sufficient condition for complete

reducibility of the first kind. By §5, 3, the discriminant ideal of R may be determined after passing to the extension ring $\bar{R} = R[\bar{P}]$. By §5, 2, it is enough to restrict attention to the primary rings out of which $\bar{R}$ is additively composed. We therefore begin with the discriminant ideals of primary rings.

If $\mathfrak{a}$ and $\mathfrak{r}$ are ideals of R and $\mathfrak{a}$ is divisible by $\mathfrak{r}$, then every basis of $\mathfrak{a}$ can be completed by adjoining elements to a linearly independent module basis of $\mathfrak{r}$. This is an immediate consequence of finite rank over the field P .

Assume R is a primary ring; let $\mathfrak{p}$ be the prime ideal belonging to the zero ideal, and let ρ be its exponent, so

$$\mathfrak{p}^\rho = (0), \quad \mathfrak{p}^{\rho-1} \neq (0).$$

A special P -basis of R is obtained by successively completing a basis of $\mathfrak{p}^{\rho-1}$ to a basis of $\mathfrak{p}^{\rho-2}$, then a basis of $\mathfrak{p}^{\rho-2}$ to one of $\mathfrak{p}^{\rho-3}$, and so on, counting R as $\mathfrak{p}^0$. With such a basis one proves:

If R is a primary ring, the trace, with respect to the principal class, of every element divisible by the associated prime ideal $\mathfrak{p}$ is zero.

Indeed, with the special basis the basis elements are arranged in layers, where elements in the layer $\mathfrak{p}^i$ are not all in $\mathfrak{p}^{i+1}$. Multiplication by an element of $\mathfrak{p}$ sends each layer into the next; the corresponding matrix has zeros on the diagonal. Its trace therefore vanishes.

If the zero ideal of R is already a prime ideal and P is algebraically closed, then R has rank one and is identical with P . The unit element e may be used as basis, and

$$S(e^2) = S(e) = e.$$

The discriminant ideal is therefore the unit ideal.

On the other hand, the discriminant ideal of a proper primary ring with algebraically closed subfield P is the zero ideal. Use the special basis just described, or simply complete any basis of $\mathfrak{p}$ to a basis of R . Since the residue class ring $R/\mathfrak{p}$ has rank one, the first basis element may be taken to be the unit element e . All basis products other than e^2 are then divisible by $\mathfrak{p}$, and their traces vanish. The discriminant determinant, having at least two rows because R is properly primary, therefore has all entries except possibly $S(e^2)$ equal to zero and consequently vanishes.

Theorem. A ring R of finite rank over a subfield P is completely reducible of the first kind if and only if its discriminant ideal is the unit ideal in P , equivalently if its discriminant, determined up to the square of a unit, is non-zero.

Complete reducibility of the first kind is, by §3, 2, equivalent to complete reducibility of

$$\bar{R} = R[\bar{P}]$$

with algebraically closed $\bar{P}$. By §5, 3 the discriminant ideals of R and $\bar{R}$ are simultaneously zero or unit ideals. By §5, 2 the discriminant ideal of $\bar{R}$ is the product of the discriminant ideals of its direct-sum components, each taken with respect to its component class. But those

component discriminants are exactly of the type just considered. If $\bar{R}$ is completely reducible, all components are fields isomorphic to $\bar{P}$ and all component discriminant ideals are unit ideals. Their product is a unit ideal. If $\bar{R}$ is not completely reducible, at least one component is properly primary and its discriminant ideal is zero; the product is then zero. The theorem follows.

When complete reducibility of the first kind holds, trace, norm, and discriminant admit the usual representation by conjugate elements. Let

$$R[Q] = Qe_1 + \cdots + Qe_n$$

be a decomposition as a direct sum of rank-one rings over a suitable finite extension field Q of P , chosen minimally. The components of R are represented in subfields K_i of Q , and R maps homomorphically to each K_i . With respect to the basis $e_1, \dots, e_n$, an element

$$y = c_1e_1 + \cdots + c_ne_n$$

has diagonal matrix with entries c_i . Hence

$$S(y) = c_1 + \cdots + c_n, \quad N(y) = c_1c_2 \cdots c_n.$$

If $a_1, \dots, a_n$ is a module basis of R and

$$a_j = a_{1j}e_1 + \cdots + a_{nj}e_n,$$

then the discriminant is

$$|S(a_ia_j)| = \begin{vmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{vmatrix}^2.$$

If R itself is a field, i.e. an extension field of the first kind of P , then the homomorphisms $R \rightarrow K_i$ are isomorphisms fixing P . Thus the K_i are the n conjugate fields in a common Galois extension field, each isomorphic to R . The formulas for trace, norm, and discriminant become the familiar formulas in terms of conjugates. It should again be stressed that R itself is assumed only equivalent to the K_i and need not be a subfield of the Galois extension field Q .

12.7 §7. The discriminant theorem for orders of an algebraic number or function field

The number and function fields in question - including algebraic functions of more than one indeterminate and integral algebraic functions, where one is extending a functional domain - all fit the following type of field with a prescribed notion of integral element.

Let $\mathfrak{S}$ be a principal ideal ring without zero divisors and with unit element: for example, the ring of rational integers, a polynomial ring in one indeterminate over a field, or a functional

domain. Let K be its quotient field. Let L be a finite extension of the first kind of K , and let $\mathfrak{O}$ be the ring of all elements of L integral with respect to $\mathfrak{O}$. Every subring T of $\mathfrak{O}$ which contains $\mathfrak{O}$ is called an order; $\mathfrak{O}$ itself is the principal order.

Every $\mathfrak{O}$ -module in $\mathfrak{O}$, in particular every ideal of an order and the order itself, has a linearly independent module basis over $\mathfrak{O}$. If L has degree n over K , it is enough to consider orders of rank n over $\mathfrak{O}$: an order of smaller rank would, by forming quotients, generate an intermediate field of that smaller degree, and all hypotheses would be preserved there.

Thus every order T is a ring satisfying the more general hypotheses of §§4 and 5, and is in fact a ring without zero divisors. Consequently the discriminant D_T of the order is defined uniquely up to multiplication by the square of a unit of $\mathfrak{O}$. This discriminant agrees, up to a unit of $\mathfrak{O}$, with the discriminant of the field L considered as a K -module, and is therefore non-zero because L/K is an extension of the first kind. It also admits the representation in §6, 4 as the square of a determinant of conjugate basis elements.

The ideal theory in T is given by the following theorem. Every ideal of T different from the zero ideal can be represented uniquely as a product of finitely many pairwise relatively prime primary ideals, whose associated prime ideals have no proper divisors except the unit ideal.

Let p be a prime element of $\mathfrak{O}$, so that the principal ideal $\mathfrak{O}p$ is a prime ideal different from the zero and unit ideals. Let Tp be the corresponding principal ideal of T . The residue class ring

$$R = T/Tp$$

is a ring of rank n over a subfield P isomorphic to $\mathfrak{O}/\mathfrak{O}p$. The natural homomorphism $T \rightarrow T/Tp$ is always defined. The fact that R contains a subfield isomorphic to $\mathfrak{O}/\mathfrak{O}p$ follows from the second isomorphism theorem:

$$(\mathfrak{O}, Tp)/Tp \simeq \mathfrak{O}/[\mathfrak{O}, Tp] = \mathfrak{O}/\mathfrak{O}p.$$

Moreover R has rank n over P . If

$$\alpha_1, \dots, \alpha_n$$

is a linearly independent module basis of T over $\mathfrak{O}$, and (α_i) denotes the residue class of α_i modulo Tp , then the (α_i) are linearly independent over P . For a linear relation over P would lift to a congruence

$$c_1\alpha_1 + \dots + c_n\alpha_n \equiv 0 \pmod{Tp},$$

i.e. to

$$c_1\alpha_1 + \dots + c_n\alpha_n = p\gamma,$$

with $\gamma \in T$. Expressing γ in the basis α_i and using uniqueness of the representation shows that every c_i is divisible by p , hence every class (c_i) vanishes in P .

Under the homomorphism the discriminant D_T passes to the discriminant of R , and the ideal

decomposition of Tp passes to the decomposition of the zero ideal of R . The trace used here is the one defined by matrix representation, not the representation by conjugate elements, which is valid for T but not necessarily for R . The trace and hence the discriminant are therefore built from ring operations and ring elements and pass through the homomorphism.

If

$$Tp = \mathfrak{q}_1 \cdots \mathfrak{q}_r = [\mathfrak{q}_1, \dots, \mathfrak{q}_r]$$

is the decomposition of Tp in T , then it passes to

$$(0) = [(\mathfrak{q}_1), \dots, (\mathfrak{q}_r)]$$

in R . By the first isomorphism theorem,

$$R/(\mathfrak{q}_i) \simeq T/\mathfrak{q}_i,$$

so that $\mathfrak{q}_i$ and $(\mathfrak{q}_i)$ are simultaneously proper primary ideals or prime ideals. The components remain pairwise relatively prime under the homomorphism.

Discriminant theorem. A prime element p of $\mathfrak{S}$ divides the discriminant D_T of an order T if and only if, in the decomposition of the principal ideal Tp into pairwise relatively prime primary components in T , at least one proper primary component or at least one prime ideal of the second kind occurs.

If the residue class field $\mathfrak{S}/\mathfrak{S}p$ is perfect, then precisely those prime elements p which divide D_T have at least one proper primary component. If $\mathfrak{S}/\mathfrak{S}p$ is perfect and T is the principal order, then p divides the discriminant if and only if Tp is divisible at least by the square of a prime ideal of the principal order.

Indeed, by the correspondence between the ideal decomposition of Tp in T and the zero-ideal decomposition in

$$R = T/Tp,$$

the occurrence of a proper primary component or of a prime ideal of the second kind means exactly that R is not completely reducible of the first kind. By §6, 3 this is the case if and only if the discriminant of R vanishes. By the homomorphic passage just described, this is exactly the divisibility of D_T by p .

12.8 §8. The discriminant theorem for orders of relative fields

If, instead of the principal ideal ring $\mathfrak{S}$, one starts with the principal order of a number or function field - more generally, with a multiplication ring, i.e. a ring in which the ideal theory of the principal order holds - then the discriminant D_T is replaced by the discriminant ideal of T with respect to $\mathfrak{S}$. The discriminant theorem transfers completely by the following considerations.

Let $\mathfrak{S}$ be a multiplication ring: a ring without zero divisors and with unit element, in which

every ideal different from the zero and unit ideals is uniquely representable as a product of powers of prime ideals, and where these prime ideals have no proper divisors except the unit ideal. Let $K, L, \mathfrak{S}, T$ be defined as above. Then the ideal theory stated in §7, 1 holds in T .

In particular, if a multiplication ring $\mathfrak{S}$ has only one prime ideal different from the zero and unit ideals, then $\mathfrak{S}$ is a principal ideal ring. For let this prime ideal be $\mathfrak{p}$, and choose $p \in \mathfrak{p}$ with $p \not\equiv 0 \pmod{\mathfrak{p}^2}$. Since every ideal different from zero and unit is a power of $\mathfrak{p}$, the principal ideal $\mathfrak{S}p$ must be equal to $\mathfrak{p}$. Hence every power of $\mathfrak{p}$ is principal, and the zero and unit ideals are principal as well.

The trace of an element $\gamma \in T$ is defined as the trace of γ from L , where L is considered as a module over K . Any system of n elements of L linearly independent over K may be used as a basis to generate a matrix representation defining the trace. If $\alpha_1, \dots, \alpha_n$ is such a system, then because L/K is an extension of the first kind,

$$|S(\alpha_i \alpha_j)|$$

is a non-zero element of K , namely the square of the determinant of the conjugate basis elements. From integral closedness of $\mathfrak{S}$ in K it follows that this determinant lies in $\mathfrak{S}$ whenever all α_i lie in $\mathfrak{S}$.

The discriminant ideal is defined as follows. Let $\tau_1, \dots, \tau_n$ run through all systems of n elements of T , and form the determinant

$$|S(\tau_i \tau_j)|.$$

This determinant is an element of $\mathfrak{S}$, and is non-zero when the τ_i are linearly independent. The ideal of $\mathfrak{S}$ generated by all these determinants is called the discriminant ideal of the order T with respect to $\mathfrak{S}$; it is different from the zero ideal.

If

$$\beta_1, \dots, \beta_m, \quad m \geq n,$$

is a module basis of T over $\mathfrak{S}$, then the finitely many determinants corresponding to all choices of n of the β_i form a basis of the discriminant ideal. This follows from the module homomorphism of T to the system of traces from T : the determinants $|S(\tau_i \tau_j)|$ are linear combinations, with coefficients in $\mathfrak{S}$, of the finitely many determinants $|S(\beta_i \beta_j)|$ obtained by choosing n basis elements. If the order has a linearly independent module basis $\alpha_1, \dots, \alpha_n$, then $|S(\alpha_i \alpha_j)|$ itself may be taken as a basis of the discriminant ideal.

We recall the passage to the quotient ring. If $\mathfrak{a}$ is an ideal of $\mathfrak{S}$, the quotient ring $\mathfrak{S}_{\mathfrak{a}}$ is the set of elements of the quotient field of $\mathfrak{S}$ having in the denominator an element prime to $\mathfrak{a}$. This quotient ring has no prime ideals other than zero, unit, and the extension ideals of the prime ideals belonging to $\mathfrak{a}$. There is a one-to-one correspondence, together with isomorphism of residue class rings, between the ideals of $\mathfrak{S}_{\mathfrak{a}}$ different from zero and unit and those ideals of $\mathfrak{S}$

whose associated prime ideals are among those belonging to $\mathfrak{a}$; zero and unit ideals correspond as well.

The same statements hold for a multiplication ring. If $\mathfrak{p}$ is a prime ideal, then $\mathfrak{S}_{\mathfrak{p}}$ has only one prime ideal different from zero and unit; hence, by the preceding paragraph, $\mathfrak{S}_{\mathfrak{p}}$ is a principal ideal ring. A generator p of the prime ideal derived from $\mathfrak{p}$ in $\mathfrak{S}_{\mathfrak{p}}$ is a prime element. The extension ideal of an arbitrary ideal $\mathfrak{c}$ of $\mathfrak{S}$ is generated by the primary component of $\mathfrak{c}$ belonging to $\mathfrak{p}$; hence it is $(p)^\rho$ or the unit ideal according as $\mathfrak{p}$ divides $\mathfrak{c}$ to exponent ρ or does not divide it. Therefore if two ideals have the same extensions in all quotient rings $\mathfrak{S}_{\mathfrak{p}}$, they are equal: they are divisible by the same prime ideals to the same powers.

Now let $\mathfrak{p}$ be a prime ideal of $\mathfrak{S}$, and write

$$B = T\mathfrak{p}$$

for its extension in T . The quotient ring T_B has a module basis of linearly independent elements over the principal ideal ring $\mathfrak{S}_{\mathfrak{p}}$; it is at the same time a finite $\mathfrak{S}_{\mathfrak{p}}$ -order in the extension field L of the quotient field K of $\mathfrak{S}_{\mathfrak{p}}$. Therefore the discriminant theorem of §7, 3 applies to T_B over $\mathfrak{S}_{\mathfrak{p}}$.

The discriminant D_{T_B} generates the extension in $\mathfrak{S}_{\mathfrak{p}}$ of the discriminant ideal of T over $\mathfrak{S}$. Indeed, one may choose a module basis of T_B over $\mathfrak{S}_{\mathfrak{p}}$ consisting of elements $\alpha_1, \dots, \alpha_n$ of T . Then $|S(\alpha_i \alpha_j)|$ belongs to the discriminant ideal, and every determinant $|S(\tau_i \tau_j)|$ in $\mathfrak{S}_{\mathfrak{p}}$ is divisible by it.

We finally obtain the general discriminant theorem.

General discriminant theorem. A prime ideal $\mathfrak{p}$ of a multiplication ring $\mathfrak{S}$ divides the discriminant ideal of an order T over $\mathfrak{S}$ if and only if, in the decomposition of $T\mathfrak{p}$ into pairwise relatively prime primary components in T , at least one proper primary component or at least one prime ideal of the second kind occurs.

By the preceding paragraphs, $\mathfrak{p}$ divides the discriminant ideal of T if and only if the corresponding prime element p of $\mathfrak{S}_{\mathfrak{p}}$ divides the discriminant of T_B . By §7, this is equivalent to the failure of complete reducibility of the first kind of the residue class ring

$$T_B/T_B p.$$

Since this residue class ring is isomorphic to

$$T/T\mathfrak{p},$$

the general discriminant theorem follows.

As a specialization to relative discriminants of number fields one obtains: a prime ideal $\mathfrak{p}$ of the principal order of a number field divides the relative discriminant of an extension field if and only if, in the principal order of the extension field, $\mathfrak{p}$ is divisible by the square of a prime ideal.

Göttingen, 30 March 1926.

13 With R. Brauer: On Minimal Splitting Fields of Irreducible Representations

The question of the number fields of smallest degree over a ground field P in which an irreducible representation of a finite group over P decomposes into absolutely irreducible constituents was first treated by I. Schur. Suppose, for example, that P contains the character of such a constituent. Schur showed that the degree of these fields over P - the **index** - agrees with the number of those absolutely irreducible constituents, all of which belong to the same class. Consequently the same field is already determined by the requirement that one absolutely irreducible constituent be split off. He also showed that the degree over P of every field in which such a splitting takes place is a multiple of the index.

All such fields will be called **splitting fields**, even when P does not contain the character. Schur showed by example that splitting fields of smallest degree need not be isomorphic. If P contains no associated simple character, the splitting fields must contain the field of such a character and its conjugates over P . The absolutely irreducible representations belonging to different conjugate characters are themselves conjugate, but they plainly belong to different classes. Even in that case, in order to split off a system of absolutely irreducible representations belonging to one class, it suffices to take the field of the corresponding character together with one of the corresponding splitting fields.

We now continue the study of splitting fields. Splitting fields of smallest degree can be characterized by associated division algebras; general splitting fields are characterized by two-sided simple rings invariantly connected with these division algebras. In addition, the example of the quaternion algebra shows that the degrees of minimal splitting fields are not bounded. Here a splitting field is called **minimal** if none of its proper subfields is a splitting field.

This unboundedness rests essentially on a number-theoretic existence theorem proved by H. Hasse in the immediately following note. The example is also treated elementarily and computationally. In this way, without using the general theory or the number-theoretic existence theorem, one still obtains the unboundedness of the degree numbers from a weaker but sufficient existence theorem. It should also be noted that this example again concerns splitting fields of a finite group, namely the quaternion group, which also underlies Schur's example. The representations written there are simultaneously isomorphic representations of the quaternion algebra.

13.1 1. Characterization of splitting fields

We first recall the basic concepts. Let $\mathfrak{S}$ be a hypercomplex system over a field P_0 . A representation Π of $\mathfrak{S}$ over P means a system of matrices with entries in P such that Π is a homomorphic image of $\mathfrak{S}$, and such that elements p_0 of P_0 correspond to diagonal matrices $p_0 E$.

Thus P must contain P_0 . A representation Π , together with all its transforms

$$P^{-1}\Pi P,$$

forms a representation class.

The representations of finite groups are included in this definition. The associated hypercomplex system $\mathfrak{S}$, the group ring, consists of all group numbers, that is, all linear combinations of the group elements with coefficients in P_0 , the original group elements being considered linearly independent and multiplication being defined by group multiplication and the laws of arithmetic. The terms reducible, irreducible, completely reducible, and absolutely irreducible are used in the usual sense; they will be applied also to representation classes. A system $\mathfrak{S}$ is called completely reducible over P if all its representation classes over P are completely reducible.

Assume now that $\mathfrak{S}$ is completely reducible over P_0 , and that P_0 is a perfect field. Then the same is true for every algebraic extension P of P_0 , and the representation classes remain completely reducible after every such extension. The center of $\mathfrak{S}$ is a direct sum of finitely many fields. Every extension P of P_0 isomorphic to one of these fields is the field of a simple character.

If P_0 is extended to such a field P , a two-sided simple ring splits off from $\mathfrak{S}$; to it belongs the absolutely irreducible representation class of the character. By Wedderburn's theorem this ring is the direct product of a matrix ring over P with a division algebra A , determined uniquely up to isomorphism. The center of A is P , and A has degree m^2 over P , where m is Schur's index attached to the character.

Every splitting field of $\mathfrak{S}$, with respect to the representation belonging to that character, is at the same time a splitting field of A with respect to the representations over P , and conversely. The conjugate representations of $\mathfrak{S}$ are obtained by starting with the corresponding two-sided simple ring over P_0 ; the associated division algebra is then isomorphic to A and has the conjugate field as its center. Thus the problem of splitting fields is completely reduced to the associated division algebra A and to its splittings.

In particular, the following theorem holds.

All maximal commutative subfields of A have the same degree m over P , where m is Schur's index. All these maximal commutative subfields are splitting fields of smallest degree, and every splitting field of smallest degree is contained in one of them. The degree of every splitting field is a multiple of m . The algebra A has only one absolutely irreducible representation class, and likewise only one irreducible representation class over P .

To characterize general splitting fields, let A_r denote the direct product of A with the full matrix ring of degree r over P . Then A_r is two-sided simple and has A as its associated division algebra. Conversely, every two-sided simple hypercomplex ring over P to which A is associated is obtained in this way. Equivalently, A_r may be characterized as the ring of all r by r matrices with entries in A .

Every splitting field of A of degree mr , if such a field exists, is isomorphic to a maximal

commutative subfield of A_r . Conversely, all maximal commutative subfields of A_r are splitting fields, each of degree ms , where s divides r . In the regular case all maximal commutative subfields of A_r have degree mr . The regular case means that the ground field P has the following property: for every commutative field Z over P , finite over P , there exist commutative extensions of Z of arbitrarily prescribed degree.

Whether, for $r > 1$, minimal fields occur among these splitting fields is not decided merely by the embedding in A_r . The following example shows that this can in fact happen for infinitely many r .

13.2 2. Unboundedness of the degrees of minimal splitting fields

For this unboundedness we use the quaternion algebra as example, regarded as a hypercomplex system over the rational field P . It has basis elements i, j, k , besides the unit 1, with the familiar relations

$$i^2 = j^2 = k^2 = -1, \quad ij = k, \quad jk = i, \quad ki = j,$$

with the reversed products changing sign.

We shall show that the splitting fields are exactly those algebraic number fields in which, after adjoining the field, there exists an idempotent element r , i.e.

$$r^2 = r,$$

with r neither zero nor one. These number fields can also be characterized as those in which -1 is representable as a sum of three squares, and consequently as a sum of two squares. Indeed, write

$$r = a + \frac{\beta}{2}i + \frac{\gamma}{2}j + \frac{\delta}{2}k.$$

Then

$$r^2 = r$$

is equivalent to the system

$$4a^2 - 4a - \beta^2 - \gamma^2 - \delta^2 = 0, \quad 2a\beta - \beta = 0,$$

$$2a\gamma - \gamma = 0, \quad 2a\delta - \delta = 0.$$

Since β, γ, δ are not all to vanish, this is equivalent to

$$-1 = \beta^2 + \gamma^2 + \delta^2.$$

The fact that representability of -1 as a sum of three squares implies representability as a sum

of two squares follows from the identity

$$(c^2 + d^2)(a^2 + b^2 + c^2 + d^2) = (ac + bd)^2 + (ad - bc)^2 + (c^2 + d^2)^2,$$

which is also a consequence of the product theorem for quaternion norms.

Every such idempotent element produces a splitting, and the existence of such an idempotent is necessary for splitting. This follows from standard facts about completely reducible systems. Every irreducible representation of a completely reducible system is generated by a one-sided simple ideal, more precisely by the corresponding ideal class, and every such ideal class generates the representation. These one-sided simple ideals all occur in one-sided direct decompositions, and those decompositions determine the primitive idempotent elements, the components of the identity. Conversely, every idempotent element gives a decomposition

$$\mathfrak{S} = r\mathfrak{S} + (1 - r)\mathfrak{S};$$

primitive idempotents split off simple ideals.

A division algebra, regarded as a hypercomplex system over its center and then extended to a splitting field, becomes a direct sum of m absolutely simple one-sided ideals, where m is Schur's index. For the quaternion algebra $m = 2$. Hence every idempotent different from 0 and 1 already gives the decomposition into absolutely simple ideals, and this corresponds to the splitting into absolutely irreducible representation classes. Thus the fields described above are exactly the splitting fields.

It follows further that all cyclic fields K_n of degree 2^n over the rational field in which -1 is representable as a sum of two squares are minimal splitting fields of the quaternion algebra. Such a splitting field cannot be real, because -1 is a sum of squares in it. On the other hand, the subfield of degree 2^{n-1} of K_n , and hence every proper subfield of K_n , is necessarily real: it is the intersection of K_n with the field of all real algebraic numbers. Thus K_n is Galois of degree two over this intersection, which must be the subfield of degree 2^{n-1} .

By Hasse's existence theorem there are such fields K_n for every n ; hence the degrees of the minimal splitting fields are not bounded. In fact every power of the index occurs here as the degree of a minimal splitting field.

13.3 3. Elementary treatment of the splitting fields of the quaternion algebra

Let $\mathfrak{S}$ denote the representation of the quaternion group generated by the usual two by two matrices I, J, K over a field. The group ring generated by $\mathfrak{S}$ is just a representation of the quaternion algebra as a hypercomplex system. We give an elementary proof that $\mathfrak{S}$ is representable over a field K if and only if -1 is representable in K as a sum of two squares.

Suppose first that

$$\mathfrak{S}^* = T^{-1}\mathfrak{S}T$$

is a representation rational over K . Write

$$I^* = T^{-1}IT, \quad J^* = T^{-1}JT.$$

Since J and J^* are similar matrices rational over K , there is a transformation rational over K carrying J^* to J . Replacing $\mathfrak{S}^*$ by an equivalent K -rational representation, we may therefore assume $J^* = J$. Let

$$I^* = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

with $a, b, c, d \in K$. From

$$IJ = -JI$$

one obtains $I^*J = -JI^*$, hence $a = -d$ and $b = c$. From $I^2 = -E$ follows

$$(I^*)^2 = -E,$$

which gives

$$a^2 + b^2 = -1.$$

Thus -1 is indeed a sum of two squares in K .

Conversely, suppose that in K one has

$$-1 = \alpha^2 + \beta^2.$$

Set

$$I^* = \begin{pmatrix} \alpha & \beta \\ \beta & -\alpha \end{pmatrix}, \quad J^* = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

and put $K^* = I^*J^*$. A direct calculation shows that

$$\pm E, \quad \pm I^*, \quad \pm J^*, \quad \pm K^*$$

form a group rational over K similar to the quaternion representation. This proves the characterization.

It remains to show, elementarily, that for infinitely many n there are cyclic fields of degree 2^n which are minimal splitting fields of the quaternion group. Let p be a rational prime for which, for some $r > 0$,

$$2^r + 1 \equiv 0 \pmod{p},$$

and let 2^n be the highest power of 2 dividing $p - 1$. There are primes p belonging to arbitrarily large values of n . For if k is any positive integer and p is a prime divisor of $2^{2^k} + 1$, then the congruence just required holds with $r = 2^k$. Moreover 2 modulo p has exponent 2^{k+1} , and hence the highest power 2^n dividing $p - 1$ has $n > k$.

Let ε be a primitive p -th root of unity. We show that in $P(\varepsilon)$, where P is the rational field, one can represent -1 as a sum of two squares:

$$-1 = a^2 + b^2 = (a + ib)(a - ib), \quad a, b \in P(\varepsilon).$$

This is equivalent to saying that in the field of $4p$ -th roots of unity,

$$P(\varepsilon, i) = P(\varepsilon_{4p}),$$

there is an element whose relative norm over $P(\varepsilon)$ is exactly -1 .

The element $1 + i\varepsilon^\nu$ has relative norm

$$(1 + i\varepsilon^\nu)(1 - i\varepsilon^\nu) = 1 + \varepsilon^{2\nu}.$$

Thus all numbers $1 + \varepsilon^{2\nu}$ are relative norms from $P(\varepsilon, i)$ over $P(\varepsilon)$. Hence so is their product

$$\prod_{u=0}^{r-1} (1 + \varepsilon^{2\nu}).$$

Using the congruence $2^r \equiv -1 \pmod{p}$, this product becomes -1 . Also ε itself is a relative norm. Hence -1 is a relative norm, and so -1 is representable in $P(\varepsilon)$ as a sum of two squares. Thus $P(\varepsilon)$ is a splitting field for the quaternion group.

Let K be the cyclic subfield of $P(\varepsilon)$ of degree 2^n . We claim that K is itself a splitting field. Let m be the index of the quaternion algebra with respect to K as ground field. Then $m = 1$ or $m = 2$. But $P(\varepsilon)$ is a splitting field of odd relative degree over K , so the case $m = 2$ is impossible. Therefore $m = 1$, which means that K is a splitting field.

Thus for infinitely many n there exist cyclic fields of degree 2^n which are minimal splitting fields. # Hypercomplex quantities and representation theory in arithmetical conception

13.4 Introduction

The most important general theorems on hypercomplex systems go back to Molien. Almost independently, Frobenius soon afterwards developed the theory of hypercomplex systems and their representations, especially the representation theory of finite groups, in a unified way. The basis is Dedekind's notion of the group determinant, more generally the determinant of an arbitrary hypercomplex system.

Frobenius shows that the different irreducible factors of this determinant, with the complex numbers as coefficient field, correspond to the different irreducible representation classes, and that all irreducible representation classes are exhausted in this way. Such a representation class is completely characterized by its character system. For finite groups, this character system arises by decomposing the determinant of the commutative hypercomplex system obtained from the conjugacy classes of the group. That determinant decomposes into linear factors whose

coefficients are the characters. This is a direct generalization of Dedekind's result that the group determinant of a finite abelian group decomposes into linear factors whose coefficients are the several characters of the abelian group.

Two questions are separated. The first is the reduction and decomposition of a given representation of a given ring over a given, not necessarily commutative, representation field. The second is the determination of all possible representations, or even only of all irreducible representations, of the ring in that field.

The first question is the simpler one. No special assumptions about the ring to be represented, or about the possibly non-commutative representation field, are needed. The fact that one deals with matrices of a fixed degree already supplies the necessary finiteness. The question can be treated completely by means of the theory of groups with operators.

A group M is said to have an operator domain if, together with the group, there is given a system of symbols

$$O, H, \dots$$

so that $O(a), H(a), \dots$ are defined uniquely for every $a \in M$, again as elements of M , and so that the distributive law

$$O(ab) = O(a)O(b)$$

holds. Two groups with the same operator domain are operator-isomorphic if they are isomorphic as ordinary groups and if, whenever a corresponds to $\bar{a}$, the elements $O(a)$ and $O(\bar{a})$, $H(a)$ and $H(\bar{a})$, and so on, correspond as well.

Only subgroups which themselves admit the operator domain are allowed. Thus a group is simple if it has no proper admissible normal subgroup except the identity. With this convention the Jordan-Hoelder theorem remains valid: if a group with operators has a composition series at all, the system of factor groups is uniquely determined, up to order, in the sense of operator-isomorphism. Under the same hypothesis there is also a direct-product theorem: the decomposition into directly indecomposable factors is unique in the same sense.

The connection with representation theory is that ideals and modules are examples of groups with operators. They are abelian groups with respect to addition, restricted by admitting specified multiplications by elements of a ring. Every representation is generated by a representation module, that is, by a module with respect to the ring and the representation field. More precisely, every module class, meaning a class of mutually operator-isomorphic representation modules, generates the same representation.

The uniqueness of the reduction of a representation is therefore answered by the composition-series theorem and the direct-product theorem. Applied to the representation module, a composition series gives, for every matrix, a block reduction in which the diagonal blocks corresponding to the factor groups generate irreducible representations, in the sense that they allow no further such reduction. Since the classes of those factor groups are uniquely determined, the diagonal representations are uniquely determined up to equivalence. Similarly, the direct-

product theorem gives the uniqueness of the splitting into indecomposable constituents; each constituent can then be reduced uniquely by its own composition series.

The second question is only sketched here for hypercomplex systems, and more generally for rings satisfying the double chain condition. The result is that every irreducible, or simple, module class is an ideal class. Thus composition series of one-sided ideals, through their factor groups, already supply all irreducible representations. More exactly, it is enough to take composition series in the residue class ring modulo the radical; this quotient becomes a completely reducible ring.

The representation problem is thereby reduced to the structural study of completely reducible rings. It also becomes natural to start with representations in the automorphism division rings. All simple right ideals of a two-sided simple component of such a ring belong to one class and therefore have the same automorphism ring. Because the ideals are simple, this automorphism ring is a division ring, and it is at the same time the automorphism ring of the simple left ideals. The simple ring itself becomes isomorphic to the full matrix ring over this automorphism division ring.

All representations over a subfield, and in particular over the coefficient domain of the hypercomplex system, arise from this matrix representation by replacing the elements of the automorphism division ring by the corresponding matrices. In this way Wedderburn's structure theorem for hypercomplex systems is reinterpreted through the general group-theoretic notion of the automorphism division ring, and is at the same time recognized as the source of the representation theory of hypercomplex systems.

The new questions which appear are those of a Galois theory for non-commutative division rings, closely connected with the question of splitting fields. At the same time one obtains a prospect of structure theorems for more general, not completely reducible, rings through the automorphism rings of indecomposable ideals. Group theory, extended to the most general groups with operators, again appears as the organizing principle.

14 Hypercomplex quantities and representation theory

14.1 Introduction

The general theory just indicated is now developed in a strictly arithmetical form. Hypercomplex quantities and representation theory are not treated as separate subjects, but as parts of one theory: the theory of module classes and ideal classes over non-commutative rings satisfying suitable finiteness conditions.

The central assertion is that the irreducible module classes are already exhausted by the corresponding ideal classes. In particular, for rings without radical, every module class decomposes into irreducible classes. This is the arithmetical counterpart of Frobenius's theorem that the irreducible factors of the regular group determinant, or more generally of the system determinant, already give all irreducible representation classes, and that for systems without radical complete

reducibility holds.

The determinant and its factorization may be regarded as a passage to the norm. Here, instead, the direct-sum decomposition of ideals and their composition series are treated directly. Representation theory becomes a theory of module classes by means of the representation module.

The introduction of module and ideal classes is group-theoretic. Modules and ideals are abelian groups under addition, with additional operators supplied by multiplication by ring elements. They are therefore groups with operators. For such groups the ordinary isomorphism is replaced by operator-isomorphism; groups which are operator-isomorphic are collected into a class. In the case of ideals of a number field this class notion coincides with the usual one.

The theory of groups with operators supplies the group-theoretic foundation. Composition series, direct products or direct sums, and completely reducible groups all retain their usual uniqueness properties, interpreted in the sense of operator-isomorphism. These results will be used first to obtain the general uniqueness theorem for the irreducible diagonal constituents of arbitrary representations. Since representation classes correspond one-to-one to the classes of representation modules, and these are classes of groups with operators, the assertion follows formally from the group theory.

The further question, the determination of all representations, requires a closer analysis of the structure of the rings to be represented. In the case of hypercomplex systems this means an arithmetical reconstruction and extension of the Wedderburn results, guided by the study of one-sided ideals. It turns out that the multiple-chain condition for right ideals, or equivalently the minimal condition for right ideals, is already sufficient as the finiteness hypothesis.

Rings without radical satisfying this minimal condition are exactly the completely reducible right rings with identity. In such rings all simple right ideals of a two-sided indecomposable ring belong to one class; their automorphism ring is a division ring; and the ring itself is a full matrix ring over that division ring. From this, the representation theory and the questions about splitting fields follow in the same conceptual framework.

14.2 Chapter I. Group-theoretic foundations

14.3 §1. Groups with operators

Let G be a group, finite or infinite. An operator domain Ω for G is a set of new symbols

$$O, H, \dots$$

such that for each $a \in G$ and each $O \in \Omega$ an element $Oa \in G$ is uniquely defined and the distributive law

$$O(ab) = Oa \cdot Ob$$

is satisfied. Thus every operator defines a homomorphism of the group into itself, possibly onto a proper subgroup. The identity is carried to the identity, and inverses to inverses.

An operator domain is called absolute if distinct operators define distinct homomorphisms. From any operator domain one obtains an absolute one by identifying all operators which define the same homomorphism. An absolute operator domain is then in one-to-one correspondence with a subset of the set of all endomorphisms of the group.

A subgroup A of G is admissible, with respect to the fixed operator domain Ω , if it is stable under the operators: for every $a \in A$ and every $O \in \Omega$ one has $Oa \in A$.

As examples, take first the inner automorphisms

$$a \mapsto c^{-1}ac.$$

The admissible subgroups are precisely the normal subgroups. If all automorphisms are taken as operators, then the admissible subgroups are the characteristic subgroups, invariant under every automorphism.

A ring is also an example. A ring is here an abelian group under addition equipped with multiplication satisfying

$$r(a + b) = ra + rb, \quad r(ab) = (ra)b, \quad (a + b)r = ar + br, \quad (ab)r = a(br).$$

Each element r gives two operators, left multiplication $x \mapsto rx$ and right multiplication $x \mapsto xr$. The admissible subgroups are then the ideals: left ideals if they are stable under all left multiplications, right ideals if they are stable under all right multiplications, and two-sided ideals if they are stable under both.

The ideals become only the trivial ideals when the ring is a division ring, that is, when the non-zero elements form a group under multiplication. No commutativity of multiplication is assumed here, either for rings or for division rings.

Let o be a ring, and let M be an abelian group, written additively. Suppose that a multiplication of elements of o with elements of M is given on the left, and that

$$r(a + b) = ra + rb, \quad (rs)a = r(sa), \quad (r + s)a = ra + sa$$

for $r, s \in o$ and $a, b \in M$. Then M is a left o -module. The ring o is at the same time an operator domain. If different elements of o induce different operations, the multiplier domain is absolute. As before, one passes from an arbitrary multiplier domain to the absolute one by identifying elements which induce the same operation; the absolute multiplier domain is again a ring. The admissible subgroups of a module are its submodules. Taking $M = o$ gives the left ideals.

Right modules are defined analogously by the rules

$$(a + b)r = ar + br, \quad a(rs) = (ar)s, \quad a(r + s) = ar + as.$$

If M is simultaneously a left module over o and a right module over o' , and if

$$r(ar') = (ra)r'$$

for $r \in o$, $a \in M$, $r' \in o'$, then M is a bimodule.

Finally, for an abelian group M the endomorphisms of M form a ring. Addition and multiplication are defined by

$$(H + O)a = Ha + Oa, \quad (HO)a = H(Oa).$$

With respect to this ring, the abelian group itself is a module. From now on, when subgroups are mentioned in the context of groups with operators, only admissible subgroups are meant. The intersection of two admissible subgroups is admissible. The product of two such subgroups is admissible whenever at least one of them is normal.

14.4 §2. The isomorphism theorems

Let two groups have the same operator domain. A map from G onto $\bar{G}$ is an operator-homomorphism if it is a homomorphism in the ordinary sense and, moreover, if a maps to $\bar{a}$ then Oa maps to $O\bar{a}$ for every operator O . If the correspondence is one-to-one, it is an operator-isomorphism.

The homomorphism theorem has the same form as for ordinary groups. If $\bar{G}$ is an operator-homomorphic image of G , then $\bar{G}$ is operator-isomorphic to a factor group G/N , where N is the admissible normal subgroup consisting of the elements of G which map to the identity of $\bar{G}$. Conversely, every admissible normal subgroup N defines a factor group G/N which admits the same operator domain and is an operator-homomorphic image of G .

A group is simple if it has no normal subgroups except itself and the identity group. Every homomorphism of a simple group is either an isomorphism or sends every element to the identity.

The first isomorphism theorem says: if $\bar{G}$ is a homomorphic image of G , if A is a normal subgroup of $\bar{G}$, and if B is the set of elements of G which map into A , then B is a normal subgroup and

$$\bar{G}/A \simeq G/B.$$

In the notation of the homomorphism theorem, if $\bar{G} = G/N$, then A necessarily contains the image of N , and one recovers A from its inverse image. The formula can be written as

$$(G/N)/(B/N) \simeq G/B.$$

The second isomorphism theorem says: if A is a subgroup and B is a normal subgroup of G , then $A \cap B$ is normal in A , and

$$AB/B \simeq A/(A \cap B).$$

Indeed, under the homomorphism $G \rightarrow G/B$, the elements of A go to the cosets making up AB/B , and the kernel is $A \cap B$.

An operator-homomorphic map of a group into itself, or into a subgroup of itself, will be called an endomorphism of the group. If the group is abelian and written additively, the operator-endomorphisms form a ring under the addition and multiplication just described; this ring is the automorphism ring in Noether's terminology.

The automorphism ring of a simple abelian group with operators is a division ring. For every endomorphism either maps the group onto itself or maps it to the zero group. The non-zero endomorphisms are isomorphisms, and the isomorphisms of the group onto itself form a group under multiplication. Hence the ring with the zero operator omitted is a group, which is exactly the assertion that the ring is a division ring.

If G is a left o -module and the operator-endomorphisms are written as right operators, then G becomes a bimodule over o on the left and over its automorphism ring on the right. The identities

$$(a + b)T = aT + bT, \quad (ra)T = r(aT)$$

state precisely that T is a homomorphism with respect to the left operators. Conversely, in any bimodule the same formulas say that each right multiplier induces an operator-endomorphism relative to the left multipliers.

14.5 §3. Composition series

A finite chain of admissible subgroups

$$G = A_0 \supset A_1 \supset \cdots \supset A_r = E$$

is a composition series if each A_i is normal in the preceding subgroup and no further subgroup with the same property can be inserted between two consecutive terms. The factor groups

$$A_0/A_1, A_1/A_2, \dots, A_{r-1}/A_r$$

are the composition factors. They are simple groups.

If all inner automorphisms are included among the operators, such a series consists entirely of normal subgroups of G and is called a principal series. If all automorphisms are included, it is called a characteristic series. In the examples above one obtains composition series of ideals, modules, or bimodules.

The Jordan-Hoelder theorem remains valid in this setting. If a group with operators has two composition series, then the two series have the same length, and their composition factors are pairwise operator-isomorphic after a suitable rearrangement. Moreover, if a composition series exists, then one can draw such a series through every admissible normal subgroup.

The existence of a composition series is automatic for finite groups, and more generally for

groups satisfying both the maximal and minimal conditions. The maximal condition says that every set of admissible subgroups contains a maximal member, equivalently that every ascending chain

$$A_1 \subset A_2 \subset A_3 \subset \cdots$$

terminates. The minimal condition says that every set of admissible subgroups contains a minimal member, equivalently that every descending chain terminates.

When only normal subgroups are admissible, as for abelian groups, the existence of a composition series conversely implies both chain conditions. Each normal subgroup has a composition series, and its length is the length of the normal subgroup. If $A \subset B$, the length of A is smaller than the length of B , since a composition series for B can be drawn through A . Hence a subgroup of shortest length is minimal, and one of greatest length is maximal.

14.6 §4. Direct products and direct sums

A group G is the direct product of two factors, written

$$G = A \times B,$$

if A and B are normal subgroups, $AB = G$, and $A \cap B = E$. Equivalently, every element $g \in G$ has a unique expression

$$g = ab, \quad a \in A, \quad b \in B,$$

and the elements of A commute with the elements of B .

For finitely many factors one writes

$$G = A_1 \times \cdots \times A_n.$$

This means that, for each i , G is the direct product of A_i and the product of all remaining factors. Equivalently, every element has a unique representation

$$g = a_1 a_2 \cdots a_n, \quad a_i \in A_i,$$

and the elements of different factors commute with one another.

A group is directly indecomposable if it cannot be written as a direct product of two proper factors. Every group satisfying the minimal condition is a direct product of finitely many directly indecomposable groups. In additive notation, products become sums and direct products become direct sums.

The direct-product theorem used later is a uniqueness theorem in the sense of operator-isomorphism. Under the chain hypotheses stated above, the decomposition into directly indecomposable factors is unique up to order and operator-isomorphism. This is the additive form that will be applied to ideals and modules.

15 34. Hypercomplex Quantities and Representation Theory

15.1 Continuation of §4. Direct products and direct intersections

A group D is called the **direct intersection** of two groups R and S with respect to G if

- a) R and S are normal subgroups of G ,
- b) $R \cap S = D$,
- c) $RS = G$.

Similarly, D is called the direct intersection of n groups $S_1, \dots, S_n$ if, after putting

$$R_i = S_1 \cap \dots \cap S_{i-1} \cap S_{i+1} \cap \dots \cap S_n,$$

one has

$$D = R_i \cap S_i$$

directly for each i .

It follows at once from the definition that D must be a normal subgroup of G .

If

$$D = S_1 \cap \dots \cap S_n$$

is direct and, under the homomorphism $G \rightarrow G/D$, the groups G, S_i, D go over into $\overline{G}, \overline{S}_i, E$, then

$$E = \overline{S}_1 \cap \dots \cap \overline{S}_n$$

is direct. Conversely, from every such representation

$$E = \overline{S}_1 \cap \dots \cap \overline{S}_n$$

one returns to a direct representation

$$D = S_1 \cap \dots \cap S_n.$$

This correspondence reduces theorems on direct intersections with arbitrary D to the special case $D = E$.

For $D = E$ and two factors, the conditions for direct product and direct intersection coincide. For n factors there is the following one-to-one relation between direct product and direct intersection.

I. If

$$G = A_1 \times \dots \times A_n = A_i \times B_i,$$

then

$$E = B_1 \cap \dots \cap B_n = B_i \cap C_i$$

directly, and $C_i = A_i$.

II. If

$$E = S_1 \cap \dots \cap S_n = R_i \cap C_i$$

directly, then

$$G = R_1 \times \cdots \times R_n = R_i \times T_i,$$

and $T_i = S_i$.

For I, write

$$B = B_1 \cap \cdots \cap B_n = B_i \cap C_i.$$

Then $C_i \supseteq A_i$; we show that $C_i \subseteq A_i$. If, for example, $c \in C_1$, then c belongs to $B_2, \dots, B_n$, and its component representation with respect to the factors A_i gives

$$c = a_1 a_2 \cdots a_n = a_1 e a_3 \cdots a_n = a_1 a_2 e \cdots a_n = \cdots = a_1 \cdots a_{n-1} e.$$

Since the product of all the A_i is direct, these representations agree. At every position except the first there appears once the identity element. Hence $c = a_1$, and so $C_1 \subseteq A_1$. Thus $C_i = A_i$ for all i . It follows that

$$B = B_i \cap A_i = E, \quad B_i A_i = G,$$

so that the intersection is direct.

For II, write

$$H = R_1 \cdots R_n = R_i T_i.$$

Then $T_i \subseteq S_i$; we show that $S_i \subseteq T_i$. Let, for instance, $s \in S_1$. Since $G = S_1 \times R_1$, one may write

$$s = s_1 e = s_2 r_2 = \cdots = s_n r_n.$$

Put

$$t_1 = e r_2 \cdots r_n = t_i r_i.$$

Using the elementwise commutativity of R_i and S_i one obtains

$$s t_1^{-1} = s_i t_i^{-1},$$

an element of all the S_i , hence the identity. Thus $S_1 \subseteq T_1$, and so $T_1 = S_1$; similarly $T_i = S_i$ for every i . Consequently

$$H = R_i T_i = R_i S_i = G, \quad R_i \cap T_i = R_i \cap S_i = E,$$

and therefore G is the direct product of the R_i .

15.2 §5. Completely reducible groups

A group is called **completely reducible** if it is the direct product of finitely many simple groups:

$$G = A_1 \times \cdots \times A_n.$$

In this case the groups

$$G, A_2 \times \cdots \times A_n, A_3 \times \cdots \times A_n, \dots, A_n, E$$

form a composition series. Its length is n , and its composition factors are

$$A_1, A_2, \dots, A_n.$$

If G is completely reducible, then every normal subgroup is a direct factor. The complementary factor can be chosen as the product of simple groups occurring in a prescribed product decomposition of G . The normal subgroup itself is completely reducible.

Indeed, suppose

$$G = HA_1A_2 \cdots A_n.$$

Put $H_0 = H$ and $H_i = H_{i-1}A_i$. Either $A_i \subseteq H_{i-1}$, in which case the factor A_i may be omitted, or $H_{i-1} \cap A_i$ is a proper normal subgroup of A_i and hence equals E ; in that case

$$H_i = H_{i-1} \times A_i.$$

After omitting the superfluous factors the decomposition becomes

$$G = H \times A_{i_1} \times \cdots \times A_{i_s},$$

so H is a direct factor. Moreover

$$H \simeq G/(A_{i_1} \times \cdots \times A_{i_s}),$$

which exhibits H as completely reducible.

It follows that every decomposition of a completely reducible group into directly indecomposable factors is already a decomposition into simple factors. It therefore gives a composition series, and the factors are uniquely determined, up to isomorphism and order.

15.3 §6. Modules relative to a field. Hypercomplex systems

An almost trivial example of completely reducible groups with operators is furnished by modules with finite basis relative to a field, not necessarily commutative.

Let G be a right K -module, and assume that the unit element e of K is at the same time the identity operator: $ae = a$ for $a \in G$. Every submodule aK generated by a non-zero element a is simple; indeed it is equal to the module generated by any one of its non-zero elements. Thus, if H is the submodule generated by certain elements and if $a \notin H$, then

$$H \cap aK = 0,$$

and $H + aK$ is direct. Starting with a basis element a_1 one may adjoin further basis elements $a_2, a_3, \dots$, obtaining

$$G = a_1K + a_2K + \cdots + a_nK$$

as a direct sum. Hence G is completely reducible.

The number n , the length of the composition series, is called the **rank** of G relative to K . Because the representation of the elements is unique, and because e is assumed to be the identity operator, the basis $a_1, \dots, a_n$ is linearly independent.

Every submodule H is a direct summand:

$$G = H + R = (a_1K + \cdots + a_rK) + (a_{r+1}K + \cdots + a_nK).$$

Equivalently, every linearly independent basis of H can be extended to a linearly independent basis of G . The complementary basis elements may even be chosen from the original basis elements.

If $(a_1, \dots, a_n)$ and $(c_1, \dots, c_n)$ are two linearly independent bases, then

$$c_j = \sum_i a_i p_{ij},$$

or, in matrix notation,

$$(c_1, \dots, c_n) = (a_1, \dots, a_n)P.$$

Conversely,

$$(a_1, \dots, a_n) = (c_1, \dots, c_n)Q.$$

It follows that

$$(c_1, \dots, c_n) = (c_1, \dots, c_n)QP,$$

and, since the c_i are linearly independent,

$$QP = E.$$

The corresponding relation gives $PQ = E$ as well.

Special K -modules which are at the same time rings are the **hypercomplex systems**.

A ring $\mathfrak{o}$ which is also a right module relative to a commutative field K is called a hypercomplex system relative to K (also called, in the literature, an algebra over K) if the following hold.

1. The rank is finite; let $u_1, \dots, u_n$ be a linearly independent basis.
2. Elements of K commute with multiplication in $\mathfrak{o}$:

$$(ab)x = a(bx) = (ax)b.$$

One says that K is connected commutatively with $\mathfrak{o}$.

3. The unit element e of K is the identity operator:

$$ae = a \quad (a \in \mathfrak{o}).$$

Because

$$\left(\sum_i u_i \alpha_i \right) \left(\sum_j u_j \beta_j \right) = \sum_{i,j} (u_i u_j) (\alpha_i \beta_j),$$

the system is completely determined, besides K , by the multiplication table

$$u_i u_j = \sum_k u_k \gamma_{ij}^k.$$

The coefficients γ_{ij}^k have to satisfy the familiar relations imposed by associativity.

If, as will often be assumed, $\mathfrak{o}$ has a unit element 1 with $1a = a1 = a$ for all a , then the elements $x \in K$ may be identified with $1x$. Thus K may be regarded as a subfield of $\mathfrak{o}$. In that case the hypercomplex system may be described as a ring of finite rank over a field lying in the center.

One example is the group ring of a finite group: take the group elements as basis elements u_i , and define multiplication by the group multiplication. The field K is arbitrary.

15.4 §7. Matrices

The square matrices (x_{ij}) with entries in a ring $\mathfrak{o}$ form a ring under the usual matrix multiplication and addition.

A matrix with entries in a field K is called **regular** if it has both a right inverse and a left inverse.

In §6 we saw that the transition matrix between two linearly independent bases of a K -module is regular. For regularity it is enough to have a right inverse. To see this, form a right K -module with linearly independent basis $(a_1, \dots, a_n)$, and put

$$c_j = \sum_i a_i p_{ij}.$$

If P has a right inverse, then the a_i may be expressed through the c_j ; hence the c_j are again a basis, and in particular linearly independent. Thus P is regular. The right inverse is then also the left inverse. The same argument applies if a left inverse is given first. If P has a left inverse, P is not a left zero divisor: from $PA = 0$ one obtains $A = P^{-1}PA = 0$.

Let P_{ij} denote, as usual, the matrix whose (i, j) entry is x , all other entries being zero; and let E_{ij} denote the matrix which has the unit element at position (i, j) and zeros elsewhere. Then

$$(x_{ij}) = \sum E_{ij} x_{ij}.$$

Thus the matrices over K form a K -module of rank n^2 . The matrix units satisfy

$$E_{ij}E_{jk} = E_{ik}, \quad E_{ij}E_{hk} = 0 \quad (j \neq h).$$

Consequently the matrices over K form a finite K -module and, when K is commutative, a hypercomplex system.

15.5 Chapter II. Non-commutative ideal theory

15.6 §8. The homomorphism theorem for rings

If $\mathfrak{a}$ is a two-sided ideal of $\mathfrak{o}$, then the residue class domain $\mathfrak{o}/\mathfrak{a}$ is not only an $\mathfrak{o}$ -module but also a ring. Every homomorphic image of $\mathfrak{o}$ is again a ring and is isomorphic to a residue class ring $\mathfrak{o}/\mathfrak{a}$, where $\mathfrak{a}$ is a two-sided ideal. The proof is the same as for groups.

In particular, if $\mathfrak{o}$ is a division ring, then there are no ideals other than (0) and $\mathfrak{o}$; hence a homomorphism is either an isomorphism or sends every element to zero.

If the ring $\mathfrak{o}$ is a right K -module and K is a ring which commutes elementwise with $\mathfrak{o}$,

$$(ab)x = a(bx) = (ax)b,$$

for example if $\mathfrak{o}$ is a hypercomplex system relative to a commutative field K , then only those right, left, or two-sided ideals are admitted which are at the same time K -modules. Besides ring homomorphisms, one demands operator homomorphisms with respect to multiplication by K . The admissible ideals are bimodules relative to $\mathfrak{o}$ and K . In hypercomplex systems, the word “ideal” by itself always means such an ideal, admissible relative to the underlying coefficient field K . With this enlargement, the homomorphism theorem just stated remains valid.

The passage from a multiplier domain $\mathfrak{o}$ to the absolute multiplier domain (§1, Example 4) is an example of ring homomorphism. Thus the absolute multiplier domain is isomorphic to a residue class ring $\mathfrak{o}/\mathfrak{d}$, where $\mathfrak{d}$ is the two-sided ideal of all elements of $\mathfrak{o}$ which annihilate the module M .

From the homomorphism theorem one obtains, as in §2, the first and second isomorphism theorems for ring isomorphism and operator-isomorphism.

All products AB of subsets of $\mathfrak{o}$ are to be understood as module products: AB is the set of all sums of products ab with $a \in A$, $b \in B$; and aB is the set of all sums ab with $b \in B$. If B is a right module with respect to any ring as operator domain, then AB , respectively aB , is again a right module. In particular it may be a right ideal, an admissible ideal relative to a coefficient domain, and so on. If (A, B) denotes the sum of the modules A and B , the calculation rules are

$$A(B, C) = (AB, AC), \quad (A, B)C = (AC, BC).$$

The direct sum is denoted by $A + B$.

In what follows we often consider rings satisfying a maximal or a minimal condition for right ideals. These include, in particular, the hypercomplex systems, since by convention only K -modules are admitted as ideals, and their ranks are therefore bounded.

15.7 §9. Idempotent elements. Direct-sum decomposition into right ideals

Let $\mathfrak{o}$ be a ring with unit element.

If $\mathfrak{r}$ is a right ideal, then $\mathfrak{r}\mathfrak{o} \subseteq \mathfrak{r}$; because there is a unit element, in fact $\mathfrak{r}\mathfrak{o} = \mathfrak{r}$. Similarly for left ideals, $\mathfrak{o}\mathfrak{l} = \mathfrak{l}$.

An element e is called **idempotent** if $e^2 = e$.

Let

$$\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_s$$

be a decomposition into right ideals, and let

$$1 = e_1 + \cdots + e_s, \quad e_i \in \mathfrak{r}_i.$$

Then

$$e_i e_k = 0 \quad (i \neq k), \quad \mathfrak{r}_i = e_i \mathfrak{o}.$$

These are the orthogonality relations.

For if $r \in \mathfrak{r}_i$, then

$$r = re = re_1 + \cdots + re_s,$$

while also

$$r = 0 + \cdots + 0 + r + 0 + \cdots + 0.$$

Hence $re_k = 0$ for $k \neq i$, and $re_i = r$. The first formula gives $\mathfrak{r}_i \subseteq e_i\mathfrak{o}$, while the reverse inclusion is clear, so $\mathfrak{r}_i = e_i\mathfrak{o}$. Taking $r = e_i$ gives

$$e_i^2 = e_i, \quad e_ie_k = 0 \quad (i \neq k).$$

Conversely, if

$$1 = e_1 + \cdots + e_s, \quad e_i^2 = e_i, \quad e_ie_k = 0 \quad (i \neq k),$$

and if $\mathfrak{r}_i = e_i\mathfrak{o}$, then

$$\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_s$$

directly. Every element r of $\mathfrak{o}$ is

$$r = e_1r + \cdots + e_sr.$$

The representation is unique, since from

$$0 = e_1a_1 + \cdots + e_sa_s$$

multiplication by e_i gives $e_ia_i = 0$.

From a right decomposition

$$\mathfrak{o} = e_1\mathfrak{o} + \cdots + e_s\mathfrak{o}$$

one obtains a left decomposition

$$\mathfrak{o} = \mathfrak{o}e_1 + \cdots + \mathfrak{o}e_s.$$

If the right ideals $\mathfrak{r}_i$ are directly indecomposable, then so are the corresponding left ideals $\mathfrak{l}_i = \mathfrak{o}e_i$; otherwise a decomposition of some $\mathfrak{l}_i$ would give a corresponding decomposition of $\mathfrak{r}_i$ by the same orthogonal idempotent calculation.

There is also the following converse form. If $e_ie_k = 0$ for $i \neq k$, if $e_i^2 = e_i$, and if $e = e_1 + \cdots + e_s$, then e is a left identity for the ring

$$e_1\mathfrak{o} + \cdots + e_s\mathfrak{o}.$$

The directness of the sum is proved as above.

15.8 §10. Decomposition into two-sided ideals

Assume again that $\mathfrak{o}$ is a ring with unit element, and that

$$\mathfrak{o} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_s$$

is a decomposition of $\mathfrak{o}$ into two-sided ideals. Then the orthogonality relations hold also for the ideals:

$$\mathfrak{a}_i\mathfrak{a}_k = 0 \quad (i \neq k), \quad \mathfrak{a}_i^2 = \mathfrak{a}_i.$$

Indeed, if $\mathfrak{b}_i$ is the sum of all $\mathfrak{a}_j$ with $j \neq i$, then $\mathfrak{o} = \mathfrak{a}_i + \mathfrak{b}_i$ and

$$\mathfrak{a}_i \mathfrak{b}_i \subseteq \mathfrak{a}_i \cap \mathfrak{b}_i = 0.$$

A fortiori $\mathfrak{a}_i \mathfrak{a}_k = 0$ for $i \neq k$, and

$$\mathfrak{a}_i \mathfrak{o} = \mathfrak{a}_i (\mathfrak{a}_i + \mathfrak{b}_i) = \mathfrak{a}_i^2.$$

Since $\mathfrak{a}_i \mathfrak{o} = \mathfrak{a}_i$, this gives $\mathfrak{a}_i^2 = \mathfrak{a}_i$.

Conversely, if a direct sum decomposition satisfies these orthogonality relations, then each summand $\mathfrak{a}_i$ is a two-sided ideal. Right ideals in the ring $\mathfrak{a}_i$ are right ideals in $\mathfrak{o}$.

If $\mathfrak{r}$ is a right ideal of $\mathfrak{o}$, then $\mathfrak{r}$ is the direct sum of the right ideals

$$\mathfrak{r}\mathfrak{a}_1, \dots, \mathfrak{r}\mathfrak{a}_s.$$

For

$$\mathfrak{r} = \mathfrak{r}\mathfrak{o} = (\mathfrak{r}\mathfrak{a}_1, \dots, \mathfrak{r}\mathfrak{a}_s),$$

and the orthogonality relations make the sum direct. If $\mathfrak{r}$ is directly indecomposable, it can have only one component, and therefore must be contained in one of the $\mathfrak{a}_i$.

These facts reduce the ideal theory of $\mathfrak{o}$ to the ideal theory of the individual $\mathfrak{a}_i$. For this reason we shall usually restrict ourselves to rings which are two-sided indecomposable.

Two right ideals contained in different $\mathfrak{a}_i$ can never be operator-isomorphic. An ideal contained in $\mathfrak{a}_i$ is annihilated by all the other $\mathfrak{a}_j$, but not by $\mathfrak{a}_i$; the opposite holds for an ideal contained in $\mathfrak{a}_j$.

If a ring can be decomposed as

$$\mathfrak{o} = \mathfrak{a}_1 + \dots + \mathfrak{a}_s$$

with the $\mathfrak{a}_i$ two-sided indecomposable, and if each $\mathfrak{a}_i$ is decomposed into one-sided indecomposable right ideals, then operator-isomorphic right ideals are to be looked for only among those belonging to the same $\mathfrak{a}_i$. It may nevertheless happen that there are several classes of non-isomorphic right ideals within the same $\mathfrak{a}_i$.

For example, let K be the field of rational numbers and let

$$\mathfrak{o} = e_1 K + e_2 K + u K$$

be a hypercomplex system in which e_1, e_2, u commute with the elements of K , and multiplication is given by

	e_1	e_2	u
e_1	e_1	0	u
e_2	0	e_2	0
u	u	0	0

The unit element is $e = e_1 + e_2$. The e_i satisfy the orthogonality relations. Thus the right decomposition is

$$\mathfrak{o} = e_1 \mathfrak{o} + e_2 \mathfrak{o} = (e_1, u) + (e_2),$$

and the left decomposition is

$$\mathfrak{o} = \mathfrak{o}e_1 + \mathfrak{o}e_2 = (e_1) + (e_2, u).$$

Since $e_1\mathfrak{o}$ and $\mathfrak{o}e_1$ are visibly indecomposable, the corresponding opposite one-sided ideals are indecomposable as well. A two-sided decomposition is impossible, since one component would have to contain $e_1\mathfrak{o}$ and the other $e_2\mathfrak{o}$; then the first would contain u , but the second would contain $ue_1 = u$ as well. Yet $e_1\mathfrak{o}$ and $e_2\mathfrak{o}$ are not operator-isomorphic, since they have different ranks over K .

Now suppose

$$\mathfrak{o} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_s$$

with the $\mathfrak{a}_i$ two-sided indecomposable. Then the $\mathfrak{a}_i$ are uniquely determined. Indeed, suppose also

$$\mathfrak{o} = \mathfrak{c}_1 + \cdots + \mathfrak{c}_t.$$

Then

$$\mathfrak{a}_i = \mathfrak{a}_i\mathfrak{o} = (\mathfrak{a}_i\mathfrak{c}_1, \dots, \mathfrak{a}_i\mathfrak{c}_t),$$

and this sum is direct, since $\mathfrak{a}_i\mathfrak{c}_j \subseteq \mathfrak{c}_j$. Since $\mathfrak{a}_i$ is indecomposable, all these summands vanish except one. Thus $\mathfrak{a}_i = \mathfrak{a}_i\mathfrak{c}_j$. Symmetrically each $\mathfrak{c}_j$ contains precisely one of the $\mathfrak{a}_i$, and the two decompositions coincide up to order.

15.9 §11. The center

The **center** Z of a ring $\mathfrak{o}$ is the set of all elements z commuting with every ring element:

$$za = az \quad (a \in \mathfrak{o}).$$

It is a commutative ring.

Every one-sided ideal $\mathfrak{a}$ of $\mathfrak{o}$ has a contracted ideal $\mathfrak{a} \cap Z$ in Z , since both $\mathfrak{a}$ and Z are Z -modules.

Every ideal A in Z has an extended ideal: the ideal generated by A in $\mathfrak{o}$,

$$\mathfrak{a} = (A, \mathfrak{o}A).$$

This ideal is two-sided, since

$$\mathfrak{a}\mathfrak{o} = (\mathfrak{o}A, A\mathfrak{o})\mathfrak{o} = (\mathfrak{o}A\mathfrak{o}, A\mathfrak{o}^2) \subseteq (\mathfrak{o}A, A\mathfrak{o}) = \mathfrak{a}.$$

When $\mathfrak{o}$ has a unit element one simply writes $\mathfrak{a} = \mathfrak{o}A$.

In that case the following theorem holds. Every two-sided decomposition of $\mathfrak{o}$ corresponds one-to-one to a decomposition of the center. From

$$\mathfrak{o} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_s, \quad A_i = \mathfrak{a}_i \cap Z$$

one obtains

$$Z = A_1 + \cdots + A_s, \quad \mathfrak{a}_i = \mathfrak{o}A_i,$$

and conversely.

For let $z \in Z$ and write its decomposition with respect to the two-sided decomposition of $\mathfrak{o}$:

$$z = z_1 + \cdots + z_s.$$

Then the z_i are again central. Indeed, for all a ,

$$za = z_1a + \cdots + z_sa = az = az_1 + \cdots + az_s,$$

and comparison of components gives $z_ia = az_i$, since the $\mathfrak{a}_i$ are two-sided. Hence $z_i \in \mathfrak{a}_i \cap Z = A_i$, and the above is the asserted direct-sum decomposition of Z . In particular, if

$$1 = e_1 + \cdots + e_s$$

is the decomposition of the unit, then the e_i are central elements. Finally

$$\mathfrak{o}A_i = \mathfrak{o}Ze_i = \mathfrak{o}e_i = \mathfrak{a}_i.$$

Conversely, let

$$Z = A_1 + \cdots + A_s$$

be a decomposition of the center. By §9 the components of the identity satisfy

$$1 = e_1 + \cdots + e_s, \quad e_i^2 = e_i, \quad e_ie_j = 0 \ (i \neq j).$$

The orthogonality relations give

$$\mathfrak{o} = \mathfrak{o}e_1 + \cdots + \mathfrak{o}e_s.$$

Putting $\mathfrak{a}_i = \mathfrak{o}A_i$ gives, by the first part of the proof,

$$A_i = \mathfrak{a}_i \cap Z.$$

It follows that $\mathfrak{a}_i = \mathfrak{o}e_i$, and the decomposition is the desired two-sided decomposition.

Consequently $\mathfrak{o}$ and Z are at the same time two-sided directly decomposable, or not.

15.10 §12. Nilpotent ideals

An ideal $\mathfrak{c}$ is called **nilpotent** if $\mathfrak{c}^r = 0$ for some r .

The ideal (u) in the example of §10 is nilpotent.

If $\mathfrak{o}$ contains a nilpotent right ideal $\mathfrak{r}$, then it contains a nilpotent left ideal, indeed a two-sided nilpotent ideal. If $\mathfrak{r}^r = 0$, then $\mathfrak{o}\mathfrak{r}$ is a left ideal; if $\mathfrak{o}\mathfrak{r}$ could be zero because $\mathfrak{o}$ has no unit element, take instead $(\mathfrak{o}\mathfrak{r}, \mathfrak{r})$. This ideal is nilpotent.

The sum of two nilpotent right ideals is again nilpotent. If $\mathfrak{c}^r = 0$ and $\mathfrak{d}^s = 0$, then in

$$(\mathfrak{c}, \mathfrak{d})^{r+s-1}$$

every term contains either at least r factors from $\mathfrak{c}$ or at least s factors from $\mathfrak{d}$, and so vanishes. Hence the whole power is zero.

If the maximal condition for right ideals holds in $\mathfrak{o}$, there is therefore a maximal nilpotent right ideal. It contains all other nilpotent right ideals, for otherwise one could form the sum with such an ideal. Call it $\mathfrak{c}$. Since $\mathfrak{o}\mathfrak{c}$ is again a nilpotent right ideal, $\mathfrak{o}\mathfrak{c} \subseteq \mathfrak{c}$, and therefore $\mathfrak{c}$ is a left ideal as well. Every nilpotent left ideal $\mathfrak{d}$ is contained in $\mathfrak{c}$, because $(\mathfrak{d}, \mathfrak{d}\mathfrak{o})$ is a nilpotent right ideal. Thus there is a maximal two-sided nilpotent ideal $\mathfrak{c}$, the **radical**, which contains all nilpotent right and left ideals.

In the example of §10, (u) is the maximal nilpotent ideal.

If the radical is the zero ideal, one speaks of a **ring without radical**: equivalently, a semisimple ring, or a Dedekind system. The residue class ring modulo the radical is always a ring without radical.

Let Z be the center of $\mathfrak{o}$ and $\mathfrak{c}$ the radical of $\mathfrak{o}$. Then

$$Z \cap \mathfrak{c}$$

is the radical of Z . It is plainly nilpotent. If there were a larger nilpotent ideal C in Z , it would extend to a nilpotent ideal of $\mathfrak{o}$ not wholly contained in $\mathfrak{c}$:

$$(C\mathfrak{o})^r = C^r \mathfrak{o}^r = 0.$$

This is impossible by maximality of $\mathfrak{c}$.

It follows that if $\mathfrak{o}$ is without radical, then so is its center. The converse is false, as the example in §10 shows. The center of $\mathfrak{o}/\mathfrak{c}$ contains a ring isomorphic to $Z/(Z \cap \mathfrak{c})$, but this ring may be a proper subring.

15.11 §13. Completely reducible rings

By §5, a ring is called **right completely reducible** if it is the direct sum of finitely many simple right ideals.

A right completely reducible ring with unit element has no non-zero nilpotent ideal, hence no radical. Indeed every right ideal $\mathfrak{r}$ is a direct summand:

$$\mathfrak{o} = \mathfrak{r} + \mathfrak{s} = e_1\mathfrak{o} + e_2\mathfrak{o}.$$

Since $e_1^2 = e_1$, one also has $e_1 = e_1^r$. If $\mathfrak{r}$ were nilpotent, $e_1^r = 0$ and so $e_1 = 0$, whence $\mathfrak{r} = 0$.

We now prove the two converses.

First: A ring without radical satisfying the minimal condition for right ideals is completely reducible relative to right ideals.

Let $\mathfrak{t} \neq 0$ be a minimal right ideal. We first show that $\mathfrak{t}$ contains an idempotent. Since $\mathfrak{t}^2 \subseteq \mathfrak{t}$ and $\mathfrak{t}^2 \neq 0$, we have $\mathfrak{t}^2 = \mathfrak{t}$. Thus there is an $a \in \mathfrak{t}$ with $a\mathfrak{t} \neq 0$, and therefore $a\mathfrak{t} = \mathfrak{t}$. The set of all $b \in \mathfrak{t}$ annihilated by a , so that $ab = 0$, is a right ideal. It is not all of $\mathfrak{t}$, hence it is zero. Therefore $ab = 0$ implies $b = 0$.

Because $\mathfrak{t} = a\mathfrak{t}$, one may write $a = ac$ with $c \neq 0$. It follows that

$$a(c^2 - c) = 0,$$

and therefore $c^2 = c$.

We next show that $\mathfrak{t}$ is a direct summand. The right ideal $c\mathfrak{o}$ is contained in $\mathfrak{t}$ and is non-zero because $c^2 = c$ lies in it; hence $c\mathfrak{o} = \mathfrak{t}$. Every element $r \in \mathfrak{o}$ has the Peirce decomposition

$$r = cr + (r - cr).$$

The elements cr form $\mathfrak{t}$. The elements $r - cr$ form another right ideal, annihilated by c :

$$c(r - cr) = 0.$$

Thus

$$\mathfrak{o} = \mathfrak{t} + \mathfrak{r}.$$

Because the elements of $\mathfrak{t}$ are not annihilated by c , the sum is direct.

Now decompose $\mathfrak{o}$, using the minimal condition, as a direct sum of directly indecomposable ideals,

$$\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_s.$$

The $\mathfrak{r}_i$ must be simple, equivalently minimal. If, for instance, $\mathfrak{r}_1$ were not simple and $\mathfrak{t}$ were a minimal ideal in $\mathfrak{r}_1$, the direct-summand result above would give a decomposition of $\mathfrak{r}_1$, contradiction. Thus $\mathfrak{o}$ is completely reducible.

Second: A ring without radical satisfying the minimal condition has a unit element.

To establish first the existence of a left unit, it suffices to prove the following. If a non-zero right ideal $\mathfrak{a} = e_1\mathfrak{o}$ has a left unit e_1 , then there is a right ideal $\mathfrak{b}$ of larger length which also has a left unit e . Since complete reducibility, hence boundedness of all lengths, has already been proved, finitely many repetitions yield a left unit for $\mathfrak{o}$ itself.

Assume $\mathfrak{a} = e_1\mathfrak{o}$ and $e_1^2 = e_1$. The formula

$$r = e_1r + (r - e_1r)$$

gives a Peirce decomposition $\mathfrak{o} = e_1\mathfrak{o} + \mathfrak{r}$. Let e_2 be an idempotent in $\mathfrak{r}$ as above. Then $e_1e_2 = 0$, because e_1 annihilates all elements of $\mathfrak{r}$. Put

$$e'_2 = e_2 - e_2e_1.$$

Then

$$(e'_2)^2 = e'_2, \quad e'_2e_1 = 0, \quad e_1e'_2 = 0,$$

so $e'_2\mathfrak{o}$ is non-zero; and $e = e_1 + e'_2$ is a left unit for the ring $e_1\mathfrak{o} + e'_2\mathfrak{o}$, which has larger right-ideal length than $\mathfrak{a}$.

To show that the left unit thus constructed is also a right unit, make the Peirce decomposition into left ideals:

$$\mathfrak{o} = \mathfrak{o}e + \mathfrak{l}.$$

We have $\mathfrak{l}e = 0$ and $\mathfrak{l} = e\mathfrak{l}$, so

$$\mathfrak{l}^2 = \mathfrak{l}e\mathfrak{l} = 0.$$

Since by hypothesis no nilpotent left ideal can exist, $\mathfrak{l} = 0$. Thus e is a right unit as well, and hence a unit element.

Theorem. From right complete reducibility and the existence of a unit element follows two-sided complete reducibility:

$$\mathfrak{o} = \mathfrak{d}_1 + \cdots + \mathfrak{d}_s,$$

where the $\mathfrak{d}_i$ are two-sided simple.

As before, it is enough to show that every minimal two-sided ideal $\mathfrak{a}$ is a direct summand. As a right ideal,

$$\mathfrak{o} = \mathfrak{a} + \mathfrak{r} = e_1\mathfrak{o} + e_2\mathfrak{o}.$$

The corresponding left decomposition is

$$\mathfrak{o} = \mathfrak{o}e_1 + \mathfrak{o}e_2 = \mathfrak{a} + \mathfrak{l}.$$

One obtains

$$\mathfrak{a}\mathfrak{l} = 0, \quad (\mathfrak{a}\mathfrak{l})^2 = 0,$$

and hence $\mathfrak{a}\mathfrak{l} = 0$; similarly $\mathfrak{l}\mathfrak{a} = 0$. It follows that $\mathfrak{r} = \mathfrak{l}$ and is two-sided, so $\mathfrak{a}$ is a two-sided direct summand. The rest of the proof is the same as for one-sided complete reducibility.

The $\mathfrak{d}_i$ are two-sided simple also as rings, because all ideals in $\mathfrak{d}_i$ are ideals in $\mathfrak{o}$. We now examine their structure.

15.12 §14. Two-sided simple completely reducible rings with unit element

Let $\mathfrak{o}$ be such a ring, and let

$$\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_n = e_1\mathfrak{o} + \cdots + e_n\mathfrak{o}$$

be a decomposition into simple right ideals. Then

$$\mathfrak{o} = \mathfrak{l}_1 + \cdots + \mathfrak{l}_n = \mathfrak{o}e_1 + \cdots + \mathfrak{o}e_n,$$

where the $\mathfrak{l}_i$ are directly indecomposable left ideals.

The product $\mathfrak{l}_i\mathfrak{r}_j$ is a non-zero two-sided ideal whenever i, j are fixed and the product is non-zero; since $\mathfrak{o}$ is two-sided simple, one has, in fact, enough non-zero components to recover all of $\mathfrak{o}$. Put

$$A_{ij} = \mathfrak{l}_i\mathfrak{r}_j.$$

Then

$$\mathfrak{o} = \sum_{i,j} A_{ij}$$

directly: from

$$\sum a_{ij} = 0, \quad a_{ij} \in A_{ij},$$

one first uses the directness of the right decomposition and then the directness of the left decomposition to conclude each $a_{ij} = 0$.

Moreover

$$A_{ij}\mathfrak{r}_j = A_{ij} \neq 0,$$

and for any non-zero $a_{ij} \in A_{ij}$ one has

$$a_{ij}\mathfrak{r}_j \subseteq \mathfrak{r}_i, \quad a_{ij}\mathfrak{r}_j \neq 0.$$

Since $\mathfrak{r}_j$ is simple, multiplication by a_{ij} gives an operator homomorphism

$$\mathfrak{r}_j \rightarrow \mathfrak{r}_i.$$

The kernel is an ideal in $\mathfrak{r}_j$, hence zero. Thus it is an isomorphism. Therefore any two simple right ideals occurring in a decomposition of $\mathfrak{o}$ are operator-isomorphic; the isomorphisms are mediated by elements of A_{ij} . Since any two simple right ideals can be made to occur together in some decomposition of $\mathfrak{o}$, all simple right ideals are mutually isomorphic.

All homomorphisms from $\mathfrak{r}_j$ to $\mathfrak{r}_i$ are mediated by elements of A_{ij} . In particular, endomorphisms of $\mathfrak{r}_i$ are mediated by elements of A_{ii} . The product and sum of elements of A_{ii} correspond to product and sum of endomorphisms. Distinct elements give distinct endomorphisms, since e_i goes to $a_{ii}e_i = a_{ii}$. Hence the ring A_{ii} is isomorphic to the endomorphism ring of $\mathfrak{r}_i$.

But the endomorphism ring of a simple ideal is a division ring. Therefore A_{ii} is a division ring. Since all $\mathfrak{r}_i$ are operator-isomorphic, their endomorphism rings are ring-isomorphic. Thus A_{ii} is determined by $\mathfrak{o}$ uniquely up to ring isomorphism. The various possible A_{ii} are merely different concrete realizations of the abstract endomorphism division ring of the simple right ideals.

Main theorem. Every completely reducible two-sided simple ring $\mathfrak{o}$ with unit element is isomorphic to the ring of all n by n matrices over a division ring K . The division ring K is isomorphic to the automorphism division ring of the right ideals of $\mathfrak{o}$.

We construct matrix units. Let I_i be the identity automorphism of $\mathfrak{r}_i$, and let I_{ij} be an arbitrary isomorphism $\mathfrak{r}_j \rightarrow \mathfrak{r}_i$. Put

$$I_{ik} = I_{ij}I_{jk}.$$

If I_{ij} is mediated by an element $c_{ij} \in A_{ij}$, then

$$c_{ij} = e_i c_{ij} = c_{ij} e_j,$$

and

$$c_{ij}c_{jk} = c_{ik}, \quad c_{ij}c_{hk} = 0 \quad (j \neq h).$$

The c_{ij} are therefore matrix units.

Now define K . To an element $a_{11} \in A_{11}$ assign its conjugates

$$a_{ii} = c_{i1}a_{11}c_{1i} \in A_{ii},$$

and set

$$\alpha = a_{11} + a_{22} + \cdots + a_{nn}.$$

The correspondence $a_{11} \mapsto \alpha$ is a ring isomorphism from A_{11} onto the set of all such α ; call this

set K . Every $\alpha \in K$ commutes with all matrix units c_{ij} . Moreover

$$\mathfrak{o} = \sum_{i,j} c_{ij} K.$$

Thus every element of $\mathfrak{o}$ can be written uniquely in the form

$$\sum_{i,j} c_{ij} \alpha_{ij},$$

and assigning to it the matrix (α_{ij}) gives an isomorphism of $\mathfrak{o}$ onto the full matrix ring over K . This proves the theorem.

Conversely, the ring of all n by n matrices over a division ring A is two-sided simple and completely reducible; A is isomorphic to the automorphism ring of the right ideals, and the construction above recovers the subring $K = A$.

Let c_{ij} be the usual matrix units, and put $\mathfrak{r}_i = c_{ii} \mathfrak{o}$. Then

$$\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_n.$$

The $\mathfrak{r}_i$ are simple right ideals, since every non-zero element $\sum_j c_{ij} \alpha_j$ generates the whole $\mathfrak{r}_i$. Indeed, if $\alpha_k \neq 0$, then multiplication by $\alpha_k^{-1} c_{ki}$ produces c_{ii} , and $c_{ii} \beta$ runs through all elements of $\mathfrak{r}_i$.

The right ideals $\mathfrak{r}_i$ are operator-isomorphic by $\mathfrak{r}_i \mapsto c_{ji} \mathfrak{r}_i$. It follows that the ring is two-sided indecomposable; by the theorem of §13, since it is already right completely reducible and has a unit, it is two-sided simple. Equivalently, every non-zero element generates the full two-sided ideal.

The center of $\mathfrak{o}$ is the center of K , and hence is a field. Indeed, a central element in the matrix ring must commute with all matrix units, so it has the form

$$\alpha c_{11} + \cdots + \alpha c_{nn},$$

where α lies in the center of K . Conversely, every such scalar diagonal matrix is central.

Since a matrix ring is also left completely reducible, and every right completely reducible ring with unit is a direct sum of matrix rings, it follows that every right completely reducible ring with unit is also left completely reducible, and conversely. The center of a completely reducible ring with unit is a direct sum of commutative fields corresponding to the two-sided simple matrix-ring components.

We shall also use this fact: any two decompositions

$$\mathfrak{o} = \sum c_{ij} K, \quad \mathfrak{o} = \sum c'_{ij} K'$$

are carried into each other by an inner automorphism,

$$x \mapsto y^{-1} x y, \quad c'_{ij} = y^{-1} c_{ij} y.$$

The proof is by choosing elements mediating reciprocal isomorphisms between corresponding right ideals and forming the usual conjugating element.

15.13 Chapter III. Module and representation theory

15.14 §15. Representations and representation modules

Let $\mathfrak{o}$ be a ring and K a ring with unit element. In the later applications K will always be a field.

A representation of degree n of $\mathfrak{o}$ in K is a homomorphism

$$\mathfrak{o} \rightarrow D,$$

where D is a ring of n by n matrices with entries in K .

A **representation module** of $\mathfrak{o}$ relative to K is a bimodule M which is a left $\mathfrak{o}$ -module and a right K -module:

$$\mathfrak{o}M \subseteq M, \quad MK \subseteq M,$$

which is a direct sum of finitely many one-generator K -modules,

$$M = x_1K + \cdots + x_nK,$$

and in which the unit element of K is the identity operator.

Every representation module gives a representation. If $c \in \mathfrak{o}$ and

$$cx_i = \sum_j x_j \gamma_{ji},$$

then, in matrix notation,

$$c(x_1, x_n) = (x_1, \dots, x_n)C, \quad C = (\gamma_{ji}).$$

The matrices C form a representation, for

$$(b+c)x_i = bx_i + cx_i, \quad (bc)x_i = b(cx_i),$$

and these are exactly the matrix rules $B+C$ and BC .

Conversely, every representation D of $\mathfrak{o}$ belongs to a representation module, and indeed to a particular basis of that module. Let M be the set of formal linear forms

$$y = x_1\alpha_1 + \cdots + x_n\alpha_n, \quad \alpha_i \in K.$$

Then M is a right K -module. If the element c is represented by the matrix $C = (\gamma_{ji})$, define

$$cx_i = \sum_j x_j \gamma_{ji},$$

and extend by linearity. The homomorphism relations for the matrices give

$$(c+d)y = cy + dy, \quad (cd)y = c(dy),$$

and the remaining bimodule laws

$$c(y + z) = cy + cz, \quad c(y\alpha) = (cy)\alpha$$

are immediate. Thus M is a representation module, and, relative to the chosen basis, it gives exactly the representation $\mathfrak{o} \rightarrow D$.

If $(y_1, \dots, y_n)$ is another basis of the same representation module and

$$(y_1, \dots, y_n) = (x_1, \dots, x_n)P,$$

then the matrices obtained from this new basis are

$$P^{-1}CP.$$

Representations related by

$$C \mapsto P^{-1}CP$$

are called equivalent and are counted in the same representation class. Since every regular matrix P transforms one basis of M into another, every representation module determines uniquely a representation class.

Equivalently, one may formulate the correspondence invariantly, without a special choice of basis. A representation module assigns to the multiplier domain $\mathfrak{o}$ a homomorphic system of linear transformations of the module. Choosing a basis only writes these transformations as matrices. Conversely, any homomorphic system of linear transformations of a finite linear-form module gives a representation module. Thus the relation between modules and representation classes is independent of coordinates.

It is clear that operator-isomorphic representation modules determine the same representation class. Conversely, if two representation modules with bases (x_i) and (z_i) produce the same representation, then

$$x_i \mapsto z_i$$

extends to an operator-isomorphism. The multiplication law is completely determined by the matrices C .

In particular, if two bases of the same module produce the same representation, then they differ by an operator-isomorphism of the module onto itself. Equivalently, if

$$C = P^{-1}CP$$

for every C and one fixed P , then the transformation of the module induced by P is an automorphism. Conversely, every automorphism of the bimodule is mediated by a matrix commuting with all matrices of the representation.

The notion of representation can also be extended to rings which are commutatively connected with a commutative multiplier domain P . In that case the representing rings K must contain P in their center, and a representation is required to be not only a ring homomorphism but also

an operator homomorphism: from $a \mapsto A$ one must have

$$a\rho \mapsto A\rho \quad (\rho \in P).$$

For representation modules this gives the additional rule

$$am \cdot \rho = a(m\rho) = a\rho \cdot m.$$

One says that the module and the ring are commutatively connected with P .

15.15 §16. Reducible representations

Let A be a submodule of the representation module M , and suppose it is possible to choose a basis for M consisting of a basis $z_1, \dots, z_t$ of A , completed by elements $y_1, \dots, y_r$:

$$M = y_1K + \dots + y_rK + z_1K + \dots + z_tK.$$

Then the representing matrices have block triangular form

$$C = \begin{pmatrix} R & S \\ 0 & T \end{pmatrix}.$$

The matrices T alone form a representation of degree t , mediated by A , and the matrices R form a representation of degree r , mediated by M/A .

For the products cz_i are again elements of A , so they are expressible in terms of the z_i alone; hence the lower-left block is zero in the chosen order. The remaining blocks give the two induced representations. Conversely, if a representation is given in such a block form, then the subspace generated by the last t basis elements is a submodule.

Let

$$M = A_0 \supset A_1 \supset \dots \supset A_s = 0$$

be a composition series for M , and suppose that each A_i is a direct summand of the preceding module as a K -module. Then one can choose a basis adapted to the series, and the matrices have the form

$$\begin{pmatrix} R_{11} & * & * & \cdots & * \\ 0 & R_{22} & * & \cdots & * \\ 0 & 0 & R_{33} & \cdots & * \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & R_{ss} \end{pmatrix}.$$

The diagonal blocks R_{ii} are representations mediated by the composition factors

$$A_{i-1}/A_i.$$

These representations are irreducible, since the composition factors are simple.

If K is a field, every submodule is a direct summand, and conversely every representation reduced as far as possible in the above sense gives a composition series. By the Jordan-Hölder

theorem, the composition factors are uniquely determined up to operator-isomorphism; hence the diagonal representations R_{ii} are uniquely determined up to equivalence and order.

If

$$M = A + B$$

is a direct sum of representation modules, then with a basis adapted to the two summands the representation has block diagonal form

$$\begin{pmatrix} R & 0 \\ 0 & S \end{pmatrix},$$

where R is mediated by A and S by B . Conversely, such a block diagonal form corresponds to a direct-sum decomposition of the representation module.

If one first decomposes M as a direct sum of directly indecomposable constituents and then draws composition series in these constituents, the representation can be put into a block form in which the large diagonal boxes correspond to the directly indecomposable summands, and inside each box the diagonal pieces correspond to the composition factors. If K is a field, every representation reduced as far as possible in this way conversely gives such a module decomposition. By the theorem of Krull and Otto Schmidt, the classes of the directly indecomposable constituents are uniquely determined up to order.

If the module M is completely reducible, all boxes consist of single irreducible representations. The corresponding representation is called **completely reducible**.

15.16 §17. Direct-sum decomposition of representation modules for rings with unit element

Let M be an $\mathfrak{o}$ -module, with $\mathfrak{o}$ a ring with unit element. Then

$$M = M_e + M_0,$$

where the unit is the identity operator on M_e and the zero operator on M_0 . If M is also a right K -module, then M_e and M_0 are right K -modules as well.

Every $m \in M$ is decomposed as

$$m = em + (m - em).$$

Let M_e be the set of all em , and M_0 the set of all $m - em$. These are additive groups, and right K -modules if M is one. Moreover

$$r(em) = erm,$$

so M_e is an $\mathfrak{o}$ -module; also

$$r(m - em) = rm - erm = rm - rm = 0,$$

so M_0 is an $\mathfrak{o}$ -module. The unit e acts as the identity on the elements em and as zero on the elements $m - em$. Hence only the zero element belongs to both M_e and M_0 , and the sum is direct.

For representation modules M_0 gives the trivial representation, in which every element is assigned the zero matrix. After splitting off M_0 there remains the representation mediated by M_e , in which the unit element is represented by the identity matrix. From now on we may and shall restrict ourselves to such representations.

Let

$$\mathfrak{o} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_s$$

be a direct sum of two-sided ideals, let

$$e = e_1 + \cdots + e_s$$

be the corresponding decomposition of the unit, and let M be an $\mathfrak{o}$ -module on which e is the identity operator. Then

$$M = \mathfrak{a}_1 M + \cdots + \mathfrak{a}_s M$$

directly.

Indeed $M = \mathfrak{o}M = (\mathfrak{a}_1 M, \dots, \mathfrak{a}_s M)$. If $\mathfrak{b}_i$ denotes the sum of all components except $\mathfrak{a}_i$, then

$$M = (\mathfrak{a}_i M, \mathfrak{b}_i M),$$

and this sum is direct because e_i acts as identity on $\mathfrak{a}_i M$ and as zero on $\mathfrak{b}_i M$.

We shall therefore usually restrict ourselves to representations of two-sided indecomposable rings.

15.17 §18. Module and representation theory of completely reducible rings

Let $\mathfrak{o}$ be completely reducible and two-sided simple, and let M be a finite $\mathfrak{o}$ -module on which the unit of $\mathfrak{o}$ is the identity operator. Then M is completely reducible, and its simple constituents are operator-isomorphic to the simple left ideals of $\mathfrak{o}$.

For let

$$\mathfrak{o} = \mathfrak{l}_1 + \cdots + \mathfrak{l}_n$$

be the decomposition into simple left ideals, and suppose

$$M = (\mathfrak{o}m_1, \dots, \mathfrak{o}m_r).$$

Then

$$\mathfrak{o}m_j = (\mathfrak{l}_1 m_j, \dots, \mathfrak{l}_n m_j).$$

The non-zero modules $\mathfrak{l}_i m_j$ are isomorphic to $\mathfrak{l}_i$, since the map $a \mapsto am_j$ is an operator homomorphism whose kernel is a left ideal. Omitting those summands already contained in previous ones, one obtains a direct sum of simple modules. Thus M is completely reducible. If M is directly indecomposable, it is simple and isomorphic to one of the $\mathfrak{l}_i$.

Let A be the automorphism division ring of such a simple module M , and let A_0 be that of $\mathfrak{l}_1$. Since M and $\mathfrak{l}_1$ are operator-isomorphic, the two division rings are ring-isomorphic. By §1, both M and $\mathfrak{l}_1$ may be regarded as bimodules with these automorphism division rings on the right. They are representation modules: $\mathfrak{l}_1$, and therefore M , has finite rank over its automorphism

division ring. The representations mediated by them correspond under the ring isomorphism $A \simeq A_0$. Thus the study of this representation class may be reduced to the representation mediated by $\mathfrak{l}_1$ in its own automorphism division ring.

Choose for $\mathfrak{l}_1$ a basis consisting of matrix units

$$\mathfrak{l}_1 = (c_{11}, \dots, c_{n1})K,$$

where K denotes the concrete realization, from §14, of the automorphism division ring. Write every element $a \in \mathfrak{o}$ as

$$a = \sum_{i,j} c_{ij} \alpha_{ij}.$$

Then the representation mediated by $\mathfrak{l}_1$ over K is

$$a \mapsto (\alpha_{ij}).$$

Indeed, multiplication of a on the basis column $(c_{11}, \dots, c_{n1})$ gives the corresponding matrix of coefficients.

Now let Γ be a subfield of K such that K has finite rank over Γ :

$$K = x_1\Gamma + \dots + x_t\Gamma.$$

Then K itself is a representation module over Γ . Each element $\beta \in K$ is represented over Γ by a matrix $B = (\beta_{ij})$ determined by

$$\beta x_i = \sum_j x_j \beta_{ji}.$$

Regarding $\mathfrak{l}_1$ as representation module relative to Γ , the representation of $\mathfrak{o}$ over Γ is obtained from

$$a \mapsto (\alpha_{ij})$$

by replacing each entry α_{ij} by the matrix A_{ij} which represents that element of K over Γ .

The representations just studied - those in the automorphism division ring of the left ideals and in its finite subfields - are, up to isomorphism, the only ones that can be mediated by simple $\mathfrak{o}$ -modules, or by left ideals. For if one regards an $\mathfrak{o}$ -module M as a bimodule relative to some field K , then the elements of K induce module homomorphisms. Hence K must be mapped isomorphically into a subfield of the automorphism division ring of M . The full automorphism division ring must have finite degree over that subfield, otherwise the representation module would have infinite rank.

These irreducible representations are not yet all irreducible representations in general. There may be representation modules which are reducible as $\mathfrak{o}$ -modules but irreducible as bimodules, and which therefore still yield irreducible representations.

15.18 §19. The simple composition factors in modules and representation modules

Let $\mathfrak{o}$ be a ring satisfying the maximal and minimal conditions for left ideals, and let $\mathfrak{c}$ be its maximal nilpotent ideal. Then every simple $\mathfrak{o}$ -module M is either annihilated by $\mathfrak{o}$, or else it is

isomorphic to a simple left ideal $\mathfrak{l}$ of the radical-free ring $\mathfrak{o}/\mathfrak{c}$. In the latter case the module is annihilated by precisely those two-sided ideals of $\mathfrak{o}/\mathfrak{c}$ which do not contain $\mathfrak{l}$, and its absolute multiplier domain is isomorphic to the two-sided simple ring containing $\mathfrak{l}$.

If $\mathfrak{c}^r = 0$, then necessarily $\mathfrak{c}M = 0$; otherwise

$$M = \mathfrak{c}M = \mathfrak{c}^2M = \cdots = 0,$$

a contradiction. Thus M may be regarded as an $\mathfrak{o}/\mathfrak{c}$ -module. The ring $\mathfrak{o}/\mathfrak{c}$ has a unit element and no radical. Hence M decomposes into one part annihilated by $\mathfrak{o}/\mathfrak{c}$ and one part on which the unit is the identity; since M is simple, only one part can occur. If the unit is the identity, the argument of §18 gives an isomorphism to a simple left ideal. The remaining assertions are clear for such left ideals and pass immediately to all modules isomorphic to them.

Consequently, if M is an $\mathfrak{o}$ -module with a composition series, its simple composition factors are either annihilated by $\mathfrak{o}$ or are isomorphic to simple left ideals of $\mathfrak{o}/\mathfrak{c}$.

If P is a ring commutatively connected with $\mathfrak{o}$, all results remain valid after replacing modules by bimodules relative to P on the right and $\mathfrak{o}$ on the left, subject to

$$am \cdot \rho = a(m\rho) = a\rho \cdot m.$$

The unit element of P is to be the identity operator. Submodules are always required to admit multiplication by P . The submodules constructed in the proofs, such as $\mathfrak{c}M, \mathfrak{c}^2M, \dots$, have this property.

In particular, if P is a field and M is a finite-rank P -module, then M is a representation module. The representation mediated by M is a ring homomorphism and also an operator homomorphism with respect to P . Conversely, all such representations over P arise from such modules. Irreducible representations correspond to simple modules, and conversely. Therefore the preceding theorems may be read as theorems about representations of $\mathfrak{o}$ over P : apart from the zero representations, all irreducible representations are mediated by the simple left ideals of $\mathfrak{o}/\mathfrak{c}$. If an arbitrary representation is reduced by means of a composition series as in §16, then the only diagonal representations which occur, besides zeros, are equivalent to those mediated by simple left ideals of $\mathfrak{o}/\mathfrak{c}$.

For these statements, no finiteness condition on $\mathfrak{o}$ is needed beyond the hypotheses already placed on the representation over P . Every representation is an isomorphic representation of its absolute multiplier domain. This absolute multiplier domain is, as the inverse image of a finite matrix ring, itself a finite-rank P -module. Since the admissible ideals are by definition P -modules, the maximal and minimal conditions hold for it.

If P is a commutative field, this absolute multiplier domain is a hypercomplex system over P . This assumption is automatically fulfilled whenever non-zero representations occur on the diagonal: in that case P is isomorphic to a subfield of the automorphism division ring of a simple left ideal, and, in the realization by the subfield K of $\mathfrak{o}/\mathfrak{c}$, the commutative connection forces it to lie in the center of K .

15.19 Chapter IV. Representations of groups and hypercomplex systems

15.20 §20. Inclusion of hypercomplex systems

We consider hypercomplex systems $\mathfrak{o}$ relative to a commutative field P . By convention, only P -modules are counted as ideals in $\mathfrak{o}$. Hence the maximal and minimal conditions hold for left and right ideals, and the entire ring theory of Chapter II applies.

In what follows, a representation of the hypercomplex system $\mathfrak{o}$ always means a representation in the field P , and more precisely one satisfying: from $a \mapsto A$ follows

$$a\rho \mapsto A\rho \quad (\rho \in P),$$

that is, module homomorphy relative to P . For representation modules this says

$$a\rho \cdot m = a(m\rho).$$

The results of §19 therefore apply. All irreducible representations are mediated by simple left ideals of

$$\mathfrak{o}_0 = \mathfrak{o}/\mathfrak{c},$$

where $\mathfrak{c}$ is the radical. Thus there are as many inequivalent irreducible representations as there are two-sided simple summands $\mathfrak{a}$ in a decomposition

$$\mathfrak{o}_0 = \mathfrak{a}_1 + \cdots + \mathfrak{a}_s.$$

To determine explicitly the representation defined by such a left ideal, first pass from the elements of $\mathfrak{o}$ to their residue classes in $\mathfrak{o}_0$. Multiplication by an element of P is an isomorphism for all ideals of $\mathfrak{o}_0$; hence P is isomorphic to a subfield $e^{(\nu)}P$ of the automorphism division ring K_ν of every simple left ideal. The representation over P mediated by a simple left ideal $\mathfrak{l}_\nu$ is obtained by first studying the representation over this subfield $e^{(\nu)}P$, and then transporting it to P through the isomorphism $e^{(\nu)}P \simeq P$. The division ring K_ν has finite rank over P ; thus we are dealing with representation in a finite-degree subfield of the automorphism division ring of $\mathfrak{l}_\nu$.

To find the representation, write each element c of $\mathfrak{o}_0$ according to its two-sided components,

$$c = c_1 + \cdots + c_s.$$

Only the matrix of c_ν is then relevant, since the other c_i annihilate $\mathfrak{l}_\nu$ and are represented by zero matrices. Write c_ν with respect to the matrix units $c_{ij}^{(\nu)}$ of $\mathfrak{a}_\nu$:

$$c_\nu = \sum_{i,j} c_{ij}^{(\nu)} \alpha_{ij}^{(\nu)}.$$

Then replace each entry $\alpha_{ij}^{(\nu)}$ by the matrix which represents it in the representation of K_ν over P .

The division ring K_ν has finite rank over P . Every element $x \in K_\nu$ satisfies an equation $f(x) = 0$ with coefficients in $e^{(\nu)}P$, since its powers are linearly dependent. If P is algebraically closed, this equation splits into linear factors; because K_ν is a division ring, x is already a root

of one of those linear factors, and so $x \in e^{(\nu)}P$. Thus $K_\nu = e^{(\nu)}P$. In this case the matrices $(\alpha_{ij}^{(\nu)})$ themselves are the irreducible representations.

Combining this observation with the end of §19 gives Burnside's theorem in the following form.

Let P be algebraically closed and commutatively connected with a ring $\mathfrak{o}$. Then every irreducible representation of degree n over P contains exactly n^2 linearly independent matrices.

This immediately implies the usual formulation: a system of n by n matrices over P , closed under multiplication, is irreducible if and only if there is no non-trivial homogeneous linear relation

$$\sum \alpha_{ij} x_{ij} = 0$$

with coefficients in P satisfied by the entries x_{ij} of every matrix in the system. For by adjoining all linear combinations one obtains a ring and a P -module, and may regard this ring as its own representation.

Indeed, the absolute multiplier domain is isomorphic to a two-sided simple completely reducible ring with unit, hence to a full matrix ring over the automorphism division ring of its left ideals. Algebraic closedness makes that division ring equal to P . An irreducible representation is generated by a simple left ideal. If this left ideal has rank m , and the representation has degree n , then $m = n$, and there are exactly n^2 linearly independent elements in the full matrix ring.

Similarly one obtains the generalized Burnside theorem. Under the same hypotheses, if D is a completely reducible representation which decomposes into pairwise inequivalent constituents of degrees

$$n_1, n_2, \dots, n_s,$$

then D has rank

$$n_1^2 + n_2^2 + \dots + n_s^2$$

over P .

If $\mathfrak{o}$ is a hypercomplex system without radical over an arbitrary field, then $\mathfrak{o}$ is completely reducible and has a unit element. It follows from §§17 and 18 that every representation of $\mathfrak{o}$ is completely reducible.

Conversely, every completely reducible representation of a ring $\mathfrak{o}$ commutatively connected with P has as absolute multiplier domain a hypercomplex system without radical. Indeed, the absolute multiplier domain is isomorphic to a ring and P -module of matrices over P , hence is a hypercomplex system. The elements of its radical act as zero in every irreducible constituent and hence in the whole representation; therefore the radical is zero.

The **regular representation** of a hypercomplex system $\mathfrak{o}$ is the representation mediated by the unit ideal $\mathfrak{o}$ itself. For a group ring one chooses specifically the basis consisting of the group elements. Since all left ideals of $\mathfrak{o}/\mathfrak{c}$ occur as composition factors in $\mathfrak{o}$, all irreducible representations already occur on the diagonal of the regular representation, after reducing the regular representation by a composition series as in §16.

A representation is known once the representing matrices of the basis elements $u_1, \dots, u_h$ are known. If the system is a group ring, so that the u_i are group elements, every homomorphic matrix representation of the group induces a representation of the group ring. Thus the representation

problem for groups is a special case of the representation problem for hypercomplex systems.

15.21 §21. Extension of the ground field. Representations of the center

Let

$$\mathfrak{o} = a_1P + \cdots + a_hP$$

be a hypercomplex system, and let Ω be an extension field of P . One forms

$$\mathfrak{o}\Omega = a_1\Omega + \cdots + a_h\Omega$$

with the old multiplication rules for the basis elements. This is again hypercomplex over Ω .

Every representation of $\mathfrak{o}$ gives a representation of $\mathfrak{o}\Omega$, since the representation of all elements is known as soon as the representations of the basis elements are known.

An ideal, a representation module, or the corresponding representation is called **absolutely irreducible** if it remains irreducible after passage to the algebraic closure.

If $\mathfrak{o}\Omega$ has no radical, then neither has $\mathfrak{o}$: a nilpotent ideal of $\mathfrak{o}$ would give a nilpotent extended ideal in $\mathfrak{o}\Omega$. The converse is false, as will appear below.

The center of $\mathfrak{o}\Omega$ is the extended center $Z\Omega$. If $\mathfrak{o}$ is completely reducible, then

$$Z = Z_1 + \cdots + Z_s$$

is a direct sum of fields. If, after passage to Ω , the algebra $\mathfrak{o}\Omega$ remains completely reducible, so that no nilpotent ideal appears, then the new center

$$Z\Omega = Z_1\Omega + \cdots + Z_s\Omega$$

is again a direct sum of fields. If Ω is algebraically closed, these fields have rank one over Ω .

For the representations of the center the following statements hold.

Every irreducible representation of a commutative hypercomplex system Z over an algebraically closed field Ω has degree one. Equivalently, irreducible representations are the same as homomorphisms

$$Z \rightarrow \Omega.$$

Indeed, every irreducible representation of Z gives one of $Z\Omega$, and hence of $Z\Omega/\mathfrak{c}$, where $\mathfrak{c}$ is the radical. This quotient is commutative and without radical, hence a direct sum of fields. Over an algebraically closed field these fields are of first degree, and they generate all irreducible representations. Since equivalent one-dimensional representations are necessarily equal, the number of distinct homomorphisms $Z \rightarrow \Omega$ is the rank of $Z\Omega/\mathfrak{c}$.

For the special case in which Z is a field over P , this says: Z is completely reducible, i.e. has no radical after scalar extension, if and only if Z is an extension of the first kind over P . For a field the homomorphisms are isomorphisms. Their number is the rank of $Z\Omega/\mathfrak{c}$, and it equals the field degree precisely when $\mathfrak{c} = 0$.

If

$$Z = Z_1 + \cdots + Z_s$$

is completely reducible, then under every irreducible representation exactly one of the fields Z_i

is mapped isomorphically, and the others are represented by zero.

Applying these statements to hypercomplex systems gives first, for systems without radical: if $\mathfrak{o}\Omega$, and therefore $\mathfrak{o}$, is without radical, then in every absolutely irreducible representation of $\mathfrak{o}$ the central elements z are represented by scalar diagonal matrices

$$\xi E,$$

and $z \mapsto \xi$ is a one-dimensional representation of the center. The correspondence between absolutely irreducible representation classes of $\mathfrak{o}$ and those of Z is one-to-one. Hence the number of such classes equals the rank of the center.

For the decomposition into two-sided indecomposable ideals

$$\mathfrak{o}\Omega = \mathfrak{a}_1 + \cdots + \mathfrak{a}_s$$

corresponds to a decomposition of the center into fields

$$Z\Omega = Z_1 + \cdots + Z_s.$$

In an irreducible representation one $\mathfrak{a}_i$ is represented isomorphically, the others by zero; this two-sided summand determines the representation class. The summand $\mathfrak{a}_i$ is represented by a full matrix ring, and the center of such a ring consists of scalar diagonal matrices. Thus the representation of the center is a homomorphism onto the corresponding scalar field, and this proves the assertion.

The following consequence is useful. If the center has a component Z_i which is an extension of the second kind over P , then $\mathfrak{o}\Omega$ certainly has a radical, since $Z_i\Omega$ already has one.

15.22 §22. Application to abelian groups

Let

$$G = G_1 \times G_2 \times \cdots \times G_s$$

be a finite abelian group, decomposed into cyclic groups G_i of orders h_i . Let

$$G_i = \{1, a_i, a_i^2, \dots, a_i^{h_i-1}\}.$$

Let P be a field whose characteristic does not divide the group order

$$h = h_1 h_2 \cdots h_s.$$

The group ring Z consists of all sums

$$\sum c_{i_1 \dots i_s} a_1^{i_1} \cdots a_s^{i_s}, \quad c_{i_1 \dots i_s} \in P.$$

It is a homomorphic image of the polynomial ring $P[x_1, \dots, x_s]$ by the substitution $x_i \mapsto a_i$. Hence

$$Z \simeq P[x_1, \dots, x_s]/\mathfrak{m},$$

where

$$\mathfrak{m} = (x_1^{h_1} - 1, \dots, x_s^{h_s} - 1).$$

In an extension field Ω in which all these polynomials split, each $x_i^{h_i} - 1$ decomposes into distinct linear factors, since the characteristic does not divide h_i . Thus $\mathfrak{m}$ decomposes as the intersection of distinct prime ideals

$$(x_1 - \xi_1, \dots, x_s - \xi_s),$$

where each ξ_i runs through the h_i -th roots of unity. The direct-sum decomposition corresponding to this intersection gives one field for each system $(\xi_1, \dots, \xi_s)$. The associated representations are

$$a_i \mapsto \xi_i, \quad a_1^{i_1} \cdots a_s^{i_s} \mapsto \xi_1^{i_1} \cdots \xi_s^{i_s}.$$

These are the characters. Their number is

$$h_1 h_2 \cdots h_s = h,$$

as the general theory requires.

15.23 §23. Determinant of a hypercomplex system

Let

$$\mathfrak{o} = a_1 P + \cdots + a_h P$$

be a hypercomplex system. Adjoin to P independent variables $x_1, \dots, x_h$ and form

$$\mathfrak{o}^* = a_1 P(x_1, \dots, x_h) + \cdots + a_h P(x_1, \dots, x_h)$$

with the old multiplication rules; the variables commute with the a_i . Thus the rules in $\mathfrak{o}^*$ are already determined.

In $\mathfrak{o}^*$ lies the **general element** of $\mathfrak{o}$,

$$w = a_1 x_1 + \cdots + a_h x_h.$$

If $a_i \mapsto A_i$ in a representation, then w is represented by

$$W = A_1 x_1 + \cdots + A_h x_h.$$

The matrix W is the system matrix attached to the representation; when the a_i are group elements and $\mathfrak{o}$ is the group ring, it is the group matrix. For the regular representation one has the regular system matrix.

The entries of W are linear forms in the x_i . The determinant $|W|$ is therefore a homogeneous form of degree n when the representation has degree n . In particular, the regular system determinant has degree h .

The system determinant does not change under passage to an equivalent representation, because

$$|PWP^{-1}| = |P| |W| |P^{-1}| = |W|.$$

If one changes from the basis $(a_1, \dots, a_h)$ to a new basis $(b_1, \dots, b_h)$ and writes

$$w = \sum_i b_i y_i,$$

then the new determinant is obtained from the old one by an invertible linear substitution

$$x_i = \sum_j \gamma_{ij} y_j.$$

If a composition series of the representation module is given, then, for a suitable choice of basis, the matrix W is upper triangular in blocks:

$$\begin{pmatrix} W_1 & * & \cdots & * \\ 0 & W_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & W_r \end{pmatrix}.$$

The determinants $|W_i|$ which occur in an arbitrary representation are among those occurring in the regular representation of $\mathfrak{o}$, indeed already in that of $\mathfrak{o}/\mathfrak{c}$, where $\mathfrak{c}$ is the radical.

Assume now that P is algebraically closed. Then the determinant belonging to an irreducible representation is an irreducible polynomial in the x_i , and inequivalent representations give different irreducible factors.

It is enough to work in the radical-free ring $\mathfrak{o}/\mathfrak{c}$. There we use matrix units $c_{ij}^{(\nu)}$ as a basis; the general element has the form

$$w = \sum_{\nu, i, j} c_{ij}^{(\nu)} x_{ij}^{(\nu)}.$$

The matrices of the irreducible representations are precisely

$$W_\nu = (x_{ij}^{(\nu)}).$$

The determinants

$$|W_\nu| = |x_{ij}^{(\nu)}|$$

are the familiar irreducible determinant forms and are plainly distinct.

To compute the $|W_\nu|$, one may factor the regular system matrix of $\mathfrak{o}/\mathfrak{c}$. Each irreducible factor appears with multiplicity equal to the degree of the corresponding irreducible representation, since the corresponding left ideal occurs that many times in a composition series.

One may also start from the regular system matrix of $\mathfrak{o}$ itself. Then each irreducible factor appears with a higher multiplicity, namely as often as the corresponding simple left ideal occurs as composition factor in the left ideals of $\mathfrak{o}$. If $\mathfrak{o}$ is instead regarded as right ideal, one gets a second regular system matrix, Frobenius's antistrophic matrix. It contains the same irreducible factors - the system determinants of all irreducible representations - possibly with different exponents.

The system determinant of a commutative system splits into linear factors, because all irreducible representations have degree one. These linear factors are themselves the irreducible representations and, for abelian groups, the characters. This was Dedekind's starting point in

the study of the group determinant of non-abelian groups.

15.24 §24. Traces and characters

If, in a representation of a hypercomplex system $\mathfrak{o}$, the element a is assigned the matrix A , set

$$\mathrm{Sp}(a) = \mathrm{Sp} A.$$

Traces are linear functions:

$$\mathrm{Sp}(c + d) = \mathrm{Sp}(c) + \mathrm{Sp}(d), \quad \mathrm{Sp}(\alpha c) = \alpha \mathrm{Sp}(c).$$

Equivalent representations have the same traces.

The trace in a reducible representation is the sum of the traces in the representations mediated by the composition factors, because the representing matrix is block triangular and its trace is the sum of the traces of the diagonal blocks.

The **principal trace** is the trace in the regular representation. The **reduced trace** is the sum of the traces in the distinct irreducible representations.

If c belongs to the maximal nilpotent ideal $\mathfrak{c}$, then

$$\mathrm{Sp}(c) = 0$$

for every representation. It suffices to consider the representations mediated by the simple left ideals of $\mathfrak{o}/\mathfrak{c}$, since in every other representation only these composition factors occur and trace is additive. But $\mathfrak{c}$ annihilates all elements of $\mathfrak{o}/\mathfrak{c}$; hence every $c \in \mathfrak{c}$ is represented by the zero matrix.

Let $\mathfrak{o}$ be two-sided simple and let P be algebraically closed; thus $\mathfrak{o}$ is a matrix ring

$$\mathfrak{o} = \sum c_{ij} P.$$

In the irreducible representation, the element

$$a = \sum c_{ij} \alpha_{ij}$$

is assigned the matrix (α_{ij}) . Hence

$$\mathrm{Sp}_{\mathrm{red}}(a) = \sum_i \alpha_{ii},$$

while the principal trace is

$$\mathrm{Sp}_{\mathrm{pr}}(a) = n \sum_i \alpha_{ii}.$$

In particular,

$$\mathrm{Sp}_{\mathrm{red}}(c_{ij}) = 0 \quad (i \neq j), \quad \mathrm{Sp}_{\mathrm{red}}(c_{ii}) = 1,$$

and

$$\mathrm{Sp}_{\mathrm{pr}}(c_{ii}) = n.$$

Passing from P to an algebraically closed field Ω does not change the principal trace, since it may be computed from the same basis. This is not true for the reduced trace.

The traces of the elements a of the radical-free system $\mathfrak{o}$ in the absolutely irreducible representations are called **characters** and are denoted by

$$\chi(a),$$

or, if the representation is to be specified, by

$$\chi^{(\nu)}(a).$$

In an irreducible representation of degree n_ν , the central elements, by §21, are represented by scalar matrices

$$\xi^{(\nu)} E,$$

where $z \mapsto \xi^{(\nu)}$ is a one-dimensional representation of the center, i.e. a homomorphism of the center into Ω . Hence the trace has value $n_\nu \xi^{(\nu)}$. Thus the homomorphisms Θ of the center and the characters χ are connected by

$$\chi(z) = n_\nu \Theta(z).$$

In the commutative case $n_\nu = 1$, and the characters themselves are the homomorphisms.

If the algebraically closed field Ω has characteristic zero, as will be assumed below, one may divide by n_ν :

$$\Theta(z) = \frac{\chi(z)}{n_\nu}.$$

The homomorphism property of Θ becomes

$$\frac{\chi(z)}{n_\nu} + \frac{\chi(z')}{n_\nu} = \frac{\chi(z+z')}{n_\nu},$$

and

$$\frac{\chi(z)}{n_\nu} \frac{\chi(z')}{n_\nu} = \frac{\chi(zz')}{n_\nu}.$$

A representation class is already uniquely determined by the traces of its matrices. To know all traces it suffices, of course, to know the traces of the basis elements in the given representation. If the representation module is R , it is determined by the multiplicities with which the irreducible modules M_ν occur as direct summands. If this multiplicity is p_ν , then the trace of the idempotent $e^{(\nu)}$ in the representation equals

$$p_\nu n_\nu.$$

Thus

$$p_\nu = \frac{\text{Sp}(e^{(\nu)})}{n_\nu}.$$

15.25 §25. Discriminants

Let

$$\mathfrak{o} = a_1 P + \cdots + a_h P$$

be a hypercomplex system.

The **discriminant matrix** is the matrix whose entries are the principal traces

$$\mathrm{Sp}_{\mathrm{pr}}(a_i a_j).$$

The **reduced discriminant matrix** is the analogous matrix formed from reduced traces after passing to the algebraically closed field. The determinants of these matrices are the discriminant and reduced discriminant.

The discriminant remains the same under extension of the ground field. Under passage to another basis it is multiplied by the square of the transformation determinant.

Indeed, suppose

$$b_j = \sum_i a_i p_{ij}.$$

Then

$$(\mathrm{Sp}(a_i a_j)) = (\mathrm{Sp}(a_i b_j)) P,$$

and similarly, using the transposed matrix,

$$(\mathrm{Sp}(a_i b_j)) = P^t (\mathrm{Sp}(b_i b_j)).$$

Thus

$$(\mathrm{Sp}(a_i a_j)) = P^t (\mathrm{Sp}(b_i b_j)) P,$$

and taking determinants proves the assertion. The determinant is therefore determined only up to a square factor from P , but its vanishing or non-vanishing is invariant. The same calculation also shows that, up to a non-zero factor, one may compute the discriminant from two different bases as

$$|\mathrm{Sp}(a_i b_j)|.$$

The discriminant vanishes if $\mathfrak{o}$ has a nilpotent ideal $\mathfrak{c}$. Choose a basis

$$c_1, \dots, c_r, d_1, \dots, d_s$$

so that $c_1, \dots, c_r$ is a basis of $\mathfrak{c}$. Since products $c_i c_j$ are nilpotent, and the trace of nilpotent elements is zero in every representation, the corresponding block of the discriminant matrix is zero, forcing the determinant to vanish. The same holds for the reduced discriminant.

Consequently, the discriminant also vanishes if a nilpotent ideal appears only after passing to the algebraic closure.

If

$$\mathfrak{o} = \mathfrak{a}_1 + \dots + \mathfrak{a}_s$$

is a direct two-sided decomposition, and if $M, M_1, \dots, M_s$ are the discriminant matrices of $\mathfrak{o}, \mathfrak{a}_1, \dots, \mathfrak{a}_s$, then M is block diagonal with blocks M_i , and hence

$$|M| = |M_1| \cdots |M_s|.$$

Indeed, choose a basis of $\mathfrak{o}$ composed of bases of the $\mathfrak{a}_i$. If $a \in \mathfrak{a}_i$, then its principal trace

computed in $\mathfrak{o}$ is the same as its principal trace computed in $\mathfrak{a}_i$. Products from different components vanish, so the off-diagonal trace blocks are zero. The same reasoning applies to the reduced discriminant.

For the full matrix ring

$$\sum c_{ij}P$$

one can compute the discriminant by taking two bases consisting of the matrix units, once ordered by first indices and once by second indices. The product table shows that

$$\mathrm{Sp}(c_{ij}c_{kl})$$

vanishes except in the matching cases. The resulting determinant is

$$D_{\mathrm{red}} = 1$$

for reduced traces, and

$$D = n^{n^2}$$

for principal traces.

Thus, if $\mathfrak{o}$ decomposes over the algebraic closure into matrix rings of degrees

$$n_1, \dots, n_s,$$

then

$$D = (n_1^{n_1^2} \cdots n_s^{n_s^2}) e$$

up to a non-zero square factor, whereas the reduced discriminant is non-zero. In particular, the reduced discriminant is non-zero for radical-free systems, and the principal discriminant is non-zero precisely when none of the n_i is divisible by the characteristic of the field.

Hence non-vanishing of the reduced discriminant is necessary and sufficient for the absence of radical after algebraic closure of the field P . In characteristic zero, non-vanishing of the principal discriminant is likewise necessary and sufficient for the absence of radical after algebraic closure.

15.26 §26. The group ring

Let $a_1, \dots, a_h$ be the elements of a finite group, and form the group ring over a field whose characteristic does not divide h . Then the discriminant D is non-zero; hence the group ring is radical-free.

For the principal trace in the regular representation satisfies

$$\mathrm{Sp}(1) = h,$$

and

$$\mathrm{Sp}(a_i) = 0$$

for $a_i \neq 1$. In the regular representation the identity is represented by a diagonal matrix with h diagonal entries equal to 1. If $a_i \neq 1$, then multiplication by a_i permutes the group elements without fixed points; the corresponding matrix therefore has zeros on the main diagonal, so its

trace is zero.

Take the two bases

$$a_1, \dots, a_h$$

and

$$a_1^{-1}, \dots, a_h^{-1}.$$

The matrix

$$(\text{Sp}(a_i a_j^{-1}))$$

is diagonal with diagonal entries h . Therefore

$$D = h^h \neq 0.$$

The **class** of a group element is the sum, formed in the group ring,

$$K_a = \sum_s s^{-1} a s,$$

where the sum ranges over the distinct conjugates. The class sums K_a are central, since they commute with all group elements. They generate the center: if an element $\sum \alpha_i a_i$ of the group ring commutes with all s , then

$$\sum_i \alpha_i a_i = s^{-1} \left(\sum_i \alpha_i a_i \right) s$$

for every s , so the coefficients are constant on conjugacy classes.

Thus the rank of the center is the number of conjugacy classes. Consequently the number of absolutely irreducible representations is equal to the number of classes of conjugate group elements.

The matrices A and $S^{-1}AS$ have the same trace; hence conjugate group elements have the same character. Taking the trace of the class sum gives

$$\chi(K_a) = h_a \chi(a),$$

where h_a is the number of elements in the conjugacy class of a . In words: the character of a group element is the character of its class sum divided by the number of elements of that class.

16 35. On Maximal Domains of Integral Functions

The following investigations are closely connected with Hilbert's problem on relatively integral functions. They go back to an occasional question of E. Hecke, who needed a special case as an auxiliary lemma and handled it by direct calculation.

The question is this. By a maximal domain of integral functions in the variables $x_1, \dots, x_n$ - polynomials, power series, or quotients of such - inside a given domain $\mathfrak{B}$, I mean a domain which can no longer be enlarged by adjoining integral denominators. Thus, whenever $a g(x)$ belongs to it, with $g(x) \in \mathfrak{B}$ and a a non-zero rational integer, then $g(x)$ itself must already belong to it.

Every integral domain $\mathfrak{A}$ in $\mathfrak{B}$ generates uniquely such a maximal domain $M(\mathfrak{A})$. It consists of all functions $g(x)$ in $\mathfrak{B}$ for which there is at least one non-zero integer a such that

$$ag(x) \in \mathfrak{A}.$$

The question is what can be said about the denominators a of this maximal domain when the original integral domain $\mathfrak{A}$ was finite, that is, when it consists of all integral polynomials in finitely many functions

$$f_1(x), \dots, f_r(x)$$

from $\mathfrak{B}$.

For $\mathfrak{B}$ one may take all integral polynomials or power series in $x_1, \dots, x_n$, or else those quotients which contain no denominators except those already occurring among the $f_i(x)$, and their powers. I show - under the additional hypothesis, in the case of power series or quotients of power series, that only algebraic relations occur among the $f_i(x)$ - that the integers a may always be chosen so that only finitely many different prime numbers occur in them. The powers of those primes, however, need not be bounded.

This last point is immediate. If $ag(x)$ belongs to $\mathfrak{A}$ but $a^\rho g(x)$ does not belong to $\mathfrak{A}$ for any fixed bound on ρ , then all powers of a may occur. Thus the reasonable boundedness question is different: can one give, in general, an infinite system of generators of $M(\mathfrak{A})$ in which the exponents of the primes in the corresponding denominators remain bounded? Since only finitely many primes are involved, this is equivalent, by known arguments, to the finiteness of $M(\mathfrak{A})$ itself. It is therefore Hilbert's problem on relatively integral functions in its integral form. The methods used here do not decide that question.

The proof that only finitely many different primes p occur in the denominators rests on the following idea. To every such prime there corresponds at least one relation modulo p among the functions $f_i(x)$ generating $\mathfrak{A}$, a relation with integral coefficients which is not identically satisfied in the variables x before reduction. In another language, let $\mathfrak{o}$ be the polynomial ring over the integers in indeterminates $u_1, \dots, u_r$. The assignment

$$u_i \longmapsto f_i(x)$$

makes $\mathfrak{A}$ a homomorphic image of $\mathfrak{o}$; the homomorphism is defined by a prime ideal $\mathfrak{a}$ of $\mathfrak{o}$, so that

$$\mathfrak{A} \simeq \mathfrak{o}/\mathfrak{a}.$$

After reduction modulo p , $\mathfrak{B}_p$ is likewise a homomorphic image of the reduced polynomial ring $\mathfrak{o}_p$ except for the finitely many primes already appearing in the denominators of the $f_i(x)$. The new homomorphism is defined by a prime ideal $\mathfrak{c}_p$ of $\mathfrak{o}_p$ containing $\mathfrak{a}_p$.

A prime p occurs in the denominator of $M(\mathfrak{A})$ if and only if $\mathfrak{c}_p$ is a proper over-ideal of $\mathfrak{a}_p$. This can happen only if either the transcendence degree of $\mathfrak{B}_p$ drops in comparison with that of $\mathfrak{A}$, or if $\mathfrak{a}_p$ ceases to be a prime ideal. Both events occur only for finitely many primes. The first is controlled by functional determinants modulo p ; the second by the fact that $\mathfrak{a}$ is an absolute prime ideal and can lose primality only modulo finitely many primes.

All arguments remain valid, with minor modifications, if the rational integers are replaced

by the integers of a finite algebraic number field and the rational primes by the prime ideals of that field.

16.1 §1. Functional determinants over coefficient fields of characteristic p

Let

$$f_1(x_1, \dots, x_n), \dots, f_r(x_1, \dots, x_n)$$

be rational functions, or more generally formally defined power series or quotients of such, with coefficients in a field P of characteristic p . Assume that the derivatives $\partial f_i / \partial x_j$ are formally defined and satisfy the usual rules of calculation. Let Ω be an algebraically closed extension field of P .

Then the following statement holds as in characteristic zero. If there is an algebraic dependence among the $f_i(x)$ with coefficients in Ω , then the rank of the functional matrix

$$\left(\frac{\partial f_i}{\partial x_j} \right)$$

is smaller than the number of algebraically independent functions under consideration, unless the dependence is purely inseparable. More precisely, if the functions are algebraically dependent and $G(u_1, \dots, u_r)$ is a non-zero polynomial relation of least degree, then

$$G(f_1, \dots, f_r) = 0.$$

Differentiation gives

$$\sum_i \frac{\partial G}{\partial u_i}(f) \frac{\partial f_i}{\partial x_j} = 0$$

for every j . If the functional matrix had full rank, all partial derivatives $\partial G / \partial u_i$ would vanish. In characteristic p , this means that G is a polynomial in the p -th powers,

$$G(u) = H(u_1^p, \dots, u_r^p).$$

Since Ω is algebraically closed, this contradicts the choice of G as an irreducible polynomial of least degree. Thus the rank must drop.

The consequence needed below is the following. Let

$$F_1(x), \dots, F_t(x)$$

be integral functions - polynomials, power series, or quotients of such - whose functional matrix has rank exactly t . Choose a prime p which occurs neither in the denominators of the $F_i(x)$ nor in all t -rowed determinants of the functional matrix. Then the reduced functions

$$f_1(x), \dots, f_t(x)$$

modulo p remain algebraically independent.

Indeed, their functional matrix is obtained by reducing the entries of the original functional matrix modulo p . By the choice of p , the rank remains t . Algebraic dependence would force the

rank to be smaller, by the preceding argument.

16.2 §2. Formulation of the theorem

Let $\mathfrak{A}$ be an integral domain of integral functions in $x_1, \dots, x_n$: polynomials, power series, or quotients of such, with rational integral coefficients. In the case of power series one may understand formal power series or convergent power series; only the fact that the functions form an integral domain is used.

Let $\mathfrak{Q}$ be the quotient field of $\mathfrak{A}$, and let $\mathfrak{B}$ be an integral domain between $\mathfrak{A}$ and $\mathfrak{Q}$. The domain $\mathfrak{M}$ is called maximal in $\mathfrak{B}$ if the following condition holds: whenever $a \neq 0$ is an integer, $g(x) \in \mathfrak{B}$, and $ag(x) \in \mathfrak{M}$, then $g(x) \in \mathfrak{M}$.

The maximal domain generated by $\mathfrak{A}$ in $\mathfrak{B}$ exists uniquely. It is the intersection of all maximal domains in $\mathfrak{B}$ containing $\mathfrak{A}$, and it consists exactly of those $g(x) \in \mathfrak{B}$ for which $ag(x) \in \mathfrak{A}$ for at least one non-zero integer a .

Assume now that $\mathfrak{A}$ is finite and has a unit element. Let

$$f_1(x), \dots, f_r(x)$$

be an integral basis, so that $\mathfrak{A}$ consists of all integral polynomials in these functions. Let $\mathfrak{B}$ consist of the allowed polynomials, power series, or quotients, with no denominators except those appearing among the finitely many $f_i(x)$.

In the case of power series or quotients one adds the following condition. If t is the rank of the functional matrix of the $f_i(x)$, and if, after renumbering, $f_1, \dots, f_t$ already have a functional matrix of rank t , then the quotient field $\mathfrak{Q}$ is to be a finite algebraic extension of

$$\mathbb{Q}(f_1, \dots, f_t).$$

Equivalently, the remaining $f_{t+1}, \dots, f_r$ are algebraic over the field generated by $f_1, \dots, f_t$.

Let $\mathfrak{o} = \mathbb{Z}[u_1, \dots, u_r]$. Then $\mathfrak{A}$ is a homomorphic image of $\mathfrak{o}$ under $u_i \mapsto f_i(x)$, hence

$$\mathfrak{A} \simeq \mathfrak{o}/\mathfrak{a}$$

for a prime ideal $\mathfrak{a}$. The theorem says that, except for finitely many rational primes p , reduction modulo p gives no new denominator obstruction: if a is not divisible by any of those exceptional primes and $ag(x)$ lies in $\mathfrak{A}$, then $g(x)$ already lies in $\mathfrak{A}$.

Equivalently, only finitely many primes can occur in the denominators required to pass from $\mathfrak{A}$ to its maximal domain.

16.3 §3. Proof of the theorem

The first auxiliary assertion is that the prime ideal $\mathfrak{a}$ which defines the homomorphism from $\mathfrak{o}$ to $\mathfrak{A}$ is absolutely prime, under the stated hypotheses. In other words, after extending the coefficient field to an algebraic closure, the corresponding ideal remains prime. This is a standard consequence of the irreducibility of the system $f_1, \dots, f_t$ and of the algebraic dependence of the remaining functions over it.

Therefore $\mathfrak{a}$ can cease to be prime after reduction modulo p for at most finitely many primes. This is one part of the exceptional set. A second finite exceptional set is formed by the primes in the denominators of the $f_i(x)$. A third is formed by the primes at which all suitable t -rowed functional determinants vanish modulo p . Outside these finitely many primes, three facts hold simultaneously:

1. the reduction of the functions $f_i(x)$ is defined;
2. the reduced ideal $\mathfrak{a}_p$ is still prime;
3. the reduced functions $f_1, \dots, f_t$ remain algebraically independent.

Choose p outside the exceptional set. The homomorphism from the reduced polynomial ring $\mathfrak{o}_p$ to $\mathfrak{B}_p$ is defined by a prime ideal $\mathfrak{c}_p$ containing $\mathfrak{a}_p$. It remains to show that in fact

$$\mathfrak{c}_p = \mathfrak{a}_p.$$

For this it is enough to compare transcendence degrees. The residue field of $\mathfrak{c}_p$ is the quotient field of the reduced image domain, so its transcendence degree is at least t . The residue field of $\mathfrak{a}_p$ also has transcendence degree exactly t , because the classes of $u_1, \dots, u_t$ remain algebraically independent and the remaining classes are algebraic over them.

To see the latter point explicitly, use the algebraic relations in characteristic zero. The ideal $\mathfrak{a}$ contains primitive integral polynomials

$$G_{t+1}(u_1, \dots, u_t, u_{t+1}), \dots, G_r(u_1, \dots, u_t, u_r)$$

expressing the algebraic dependence of the remaining generators. For all but finitely many primes these polynomials do not vanish after reduction and still involve the corresponding variable u_j genuinely. Hence the classes of $u_{t+1}, \dots, u_r$ are algebraic over the classes of $u_1, \dots, u_t$ modulo $\mathfrak{a}_p$.

Thus $\mathfrak{a}_p$ and $\mathfrak{c}_p$ are prime ideals of the same transcendence degree, and $\mathfrak{c}_p$ contains $\mathfrak{a}_p$. A prime ideal has no proper prime over-ideal of the same transcendence degree. Therefore $\mathfrak{c}_p = \mathfrak{a}_p$.

This proves that no such prime p can occur in a denominator. The theorem follows.

16.4 §4. Algebraic integer coefficient domains

The same reasoning applies if the rational integers are replaced by the ring of integers of a finite algebraic number field K , and rational primes by prime ideals of K .

The determinant argument and the formulation of the theorem are unchanged. In the proof, however, one must add to the exceptional prime ideals those which divide the finitely many algebraic integers occurring as contents of the polynomials expressing the algebraic dependences; these polynomials can no longer all be assumed primitive over $\mathbb{Z}$ in the same naive sense.

The final formulation is as follows. Let a be an integer of K not divisible by any of the finitely many exceptional prime ideals. If $ag(x)$ lies in the original domain $\mathfrak{A}$ and $g(x) \in \mathfrak{B}$, then $g(x)$ lies in the maximal domain generated by $\mathfrak{A}$ without needing any of the non-exceptional prime ideals as denominators.

Equivalently, after localizing away from the exceptional prime ideals, the same argument as above shows that the relevant reduced homomorphisms have the same prime ideal kernel. If

$$(a) = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_s^{e_s}$$

is the prime-ideal factorization and none of the $\mathfrak{p}_i$ is exceptional, the localization makes these prime ideals principal and leaves the ideal decomposition unchanged. The preceding proof then applies word for word.

If the exceptional prime ideals are generated in the localized ring by elements

$$\lambda_1, \dots, \lambda_q,$$

then every needed denominator may be chosen as a product of powers of these finitely many λ_i . Returning from the localized ring to the original ring of integers gives the asserted finite-prime statement.

17 36. Ideal Differentiation and the Different

It is shown how the different of an algebraic number field may be understood as a differential quotient of a defining ideal. The agreement of this definition with the usual one follows from structural theorems which are interesting in themselves and which, in an analogous sense, generalize Lagrange's interpolation formula. A full account is to appear in the *Mathematische Annalen*.

18 37. Normal Basis for Fields Without Higher Ramification

By a normal basis of a Galois number field K/k one means a basis of the maximal order $\mathfrak{O}$ of K over the maximal order $\mathfrak{o}$ of k which consists of conjugate elements. It is known that a necessary condition for the existence of such a basis is the absence of higher ramification groups in K/k . The same remains true locally: after passing to the $\mathfrak{p}$ -adic extension $k_{\mathfrak{p}}$ of k and the corresponding extension $K_{\mathfrak{p}}$ of K .

I prove the converse in the following form. For every prime $\mathfrak{p}$ whose residue characteristic does not divide the degree of K/k , there exists a normal basis. In particular, if K/k has no higher ramification, then at every ramified prime there is a normal basis; the discriminant at such a prime becomes the square of a group determinant.

This last assertion must be supplemented. Contrary to a conjecture of E. Artin, the passage from this factorization to Artin conductors still requires further ideal-theoretic considerations. The decomposition of the group determinant according to irreducible characters gives generalized root numbers. A suitable collection of these gives a factorization in the ground field enlarged by the character values. In the simplest cases these new factors can be identified with Artin's conductors by means of the ideal-theoretic factorization of the root numbers.

Even in the general case, where no normal basis exists, a decomposition into generalized root numbers is always possible by means of a composition series as Galois modules. This again gives a factorization in the enlarged ground field, and again ideal-theoretic questions enter.

The proof of the existence of the normal basis under the hypotheses stated above rests on this point: the module $\mathfrak{D}/\mathfrak{o}$, regarded as a Galois module - that is, as an $\mathfrak{o}$ -module with the substitutions of the Galois group as operator domain - becomes operator-isomorphic to a one-sided ideal of the integral group ring enlarged by $\mathfrak{o}$. At all primes with only ordinary ramification this group ring is a maximal order. Ideals in maximal orders of p -adic semisimple systems are principal. Hence the corresponding Galois module is generated by one element under the substitutions of the group, which is exactly the assertion that there is a normal basis.

At primes with higher ramification the integral group ring passes to a non-maximal order, and the ideal attached to $\mathfrak{D}/\mathfrak{o}$ passes to a non-principal ideal. The same reasoning applies to function fields of one variable, provided that the characteristic of the constant field does not divide the degree of K/k .

18.1 §1. The p -adically extended integral group ring

Let $\mathfrak{o}$ and $\mathfrak{o}_{\mathfrak{p}}$ denote the maximal orders of an algebraic number field k and of its $\mathfrak{p}$ -adic extension $k_{\mathfrak{p}}$, respectively. Let G be a finite group of order n . Write (G) for the rational group ring and $[G]$ for the integral group ring. Then $(G)_k$ and $(G)_{k_{\mathfrak{p}}}$ denote the corresponding group rings with coefficients in k and $k_{\mathfrak{p}}$; similarly $[G]_{\mathfrak{o}}$ and $[G]_{\mathfrak{o}_{\mathfrak{p}}}$ denote the integral group rings after extension of the coefficient ring.

Thus $(G)_k$ and $(G)_{k_{\mathfrak{p}}}$ are semisimple systems, and $[G]_{\mathfrak{o}}$ and $[G]_{\mathfrak{o}_{\mathfrak{p}}}$ are orders in them.

Theorem 1. If the prime ideal $\mathfrak{p}$ does not divide the group order n , then $[G]_{\mathfrak{o}_{\mathfrak{p}}}$ is a maximal order in $(G)_{k_{\mathfrak{p}}}$. If $\mathfrak{p}$ divides n , it is not maximal.

The discriminant of the integral group ring, and hence of the extended group rings, is a power of the group order n . If $\mathfrak{p}$ does not divide n , this discriminant is a unit in $\mathfrak{o}_{\mathfrak{p}}$. Thus $[G]_{\mathfrak{o}_{\mathfrak{p}}}$ cannot be enlarged to a strictly larger order while keeping the coefficient ring fixed: such an enlargement would split off a non-unit square factor from the discriminant. Since $\mathfrak{o}_{\mathfrak{p}}$ is already the maximal order of the local field, this proves the first part of the theorem.

If $\mathfrak{p}$ divides n , the direct summand corresponding to the trivial representation gives a genuine enlargement. The idempotent attached to the trivial character has denominator n , and the corresponding order is strictly larger than the integral group ring. Thus the group ring is then not maximal.

Theorem 2. If $\mathfrak{p}$ does not divide n , then every integral or fractional ideal with respect to $[G]_{\mathfrak{o}_{\mathfrak{p}}}$ is principal.

This follows because $[G]_{\mathfrak{o}_{\mathfrak{p}}}$ is then a maximal order in a p -adic semisimple system, and ideals in such maximal orders are principal. The assertion is first proved for simple components and then follows for semisimple systems by decomposing into components.

18.2 §2. Galois modules, operator-isomorphism, normal bases

Let K/k be Galois with group G . If $S \in G$, write z^S for the conjugate of $z \in K$ under S . The substitutions of G act as operator automorphisms on K/k considered as a k -module. Hence K/k becomes a module over the rational group ring $(G)_k$ by putting

$$z \left(\sum_i c_i S_i \right) = \sum_i c_i z^{S_i}.$$

A module over $(G)_k$ contained in K/k will be called a rational Galois module. Similarly, $[G]_{\mathfrak{o}}$ -modules contained in K/k are called integral Galois modules; in particular $\mathfrak{D}/\mathfrak{o}$ is an integral Galois module.

Theorem 3. As a rational Galois module, K/k is operator-isomorphic to the group ring $(G)_k$.

This is the usual existence of a field normal basis: there is an element $z \in K$ whose conjugates z^S form a k -basis of K . The map

$$\sum_i c_i S_i \mapsto \sum_i c_i z^{S_i}$$

is an operator homomorphism, and because the two sides have the same rank over k , it is an isomorphism.

Theorem 4. As an integral Galois module, $\mathfrak{D}/\mathfrak{o}$ is operator-isomorphic to a $[G]_{\mathfrak{o}}$ -module of highest rank n inside $(G)_k$. Locally, $\mathfrak{D}_{\mathfrak{p}}/\mathfrak{o}_{\mathfrak{p}}$ is operator-isomorphic to a $[G]_{\mathfrak{o}_{\mathfrak{p}}}$ -module of rank n inside $(G)_{k_{\mathfrak{p}}}$.

Indeed, the rational isomorphism of Theorem 3 carries the integral module $\mathfrak{D}/\mathfrak{o}$ to a module over the integral group ring. Extending the coefficients to $k_{\mathfrak{p}}$ gives the corresponding local statement.

Theorem 5. At every prime $\mathfrak{p}$ not dividing $n = [K : k]$, the local module $\mathfrak{D}_{\mathfrak{p}}/\mathfrak{o}_{\mathfrak{p}}$ has a normal basis.

By Theorem 1, $[G]_{\mathfrak{o}_{\mathfrak{p}}}$ is a maximal order at such a prime. By Theorem 2, the module corresponding to $\mathfrak{D}_{\mathfrak{p}}/\mathfrak{o}_{\mathfrak{p}}$ is a principal ideal:

$$W[G]_{\mathfrak{o}_{\mathfrak{p}}}.$$

Its basis is WS , where S runs through G . Passing back through the operator-isomorphism, if w is the element corresponding to W , then the conjugates w^S form an $\mathfrak{o}_{\mathfrak{p}}$ -basis of $\mathfrak{D}_{\mathfrak{p}}$. This is a normal basis.

If $\mathfrak{p}$ divides n , the associated module cannot be principal when no normal basis exists. This is precisely what happens at primes with higher ramification.

The operator-isomorphism of Theorem 3 also gives a concise form of Speiser's results on Klein's problem of forms. Every representation of the Galois group irreducible over k is generated by an irreducible rational Galois module contained in K/k ; every absolutely irreducible one is generated after passing to a splitting field Z , using the extended system K_Z/Z . This is exactly the formulation in which a basis $v_1, \dots, v_m$ of such a Galois module satisfies

$$v_i^S = \sum_j s_{ji} v_j,$$

where $S \mapsto (s_{ji})$ is the corresponding representation.

18.3 §3. The discriminant as a group determinant

To decompose the discriminant it is enough to work at the individual ramified primes. In fields without higher ramification, these are precisely the primes at which the normal basis just

constructed is available.

Theorem 6. If K/k is a Galois field without higher ramification, then at every ramified prime its discriminant is the square of the group determinant of the Galois group, with the indeterminates replaced by the elements of a normal basis. Corresponding to the irreducible representations, the group determinant decomposes into generalized root numbers; the discriminant decomposes into ideals in the ground field enlarged by the character values, with conjugate-complex characters giving the same ideals.

Let w^S be a normal basis. The discriminant is the square of

$$D = |w^{ST}|_{S,T \in G},$$

which is the group determinant with the w^S in place of the indeterminates. The decomposition of the group determinant according to the absolutely irreducible representations gives factors $D_1, \dots, D_r$, the generalized root numbers. In the cyclic case these are simply the Lagrange resolvents

$$\sum_S w^S \chi(S).$$

In general, D_i lies in the coefficient field obtained by adjoining the values of the corresponding character. Pairing an irreducible representation with its adjoint gives real factors

$$A_i = D_i \overline{D_i}$$

in the real subfield of the character field. The discriminant ideal is thereby expressed as a product of such factors.

If one can show that the ideals generated by the A_i actually descend to the ground field and are equal for conjugate characters, then they agree with Artin's conductors, provided compound characters are assigned the corresponding products. The functional equation and the behavior for trivial representations of subgroups give the same result for rationally irreducible characters.

Theorem 7. Let $k = \mathbb{Q}$, and let K/k be cyclic of prime degree ℓ , with ℓ prime to the discriminant. Then for the non-trivial representations the ideals generated by the factors A_i are all equal to one another and agree with the conductor of K/k .

This follows from the classical factorization of Lagrange root numbers. At a ramified prime p , after adjoining the ℓ -th roots of unity, the Lagrange resolvents factor in such a way that

$$(A_i) = (D_i \overline{D_i})$$

becomes the same prime-ideal factor for every non-trivial character. Taking norms down to the rational field gives exactly the conductor. The existence of a normal basis, already supplied above, replaces the usual appeal to the realization of absolutely cyclic fields by cyclotomic fields.

19 38. With R. Brauer and H. Hasse: Proof of a Main Theorem in the Theory of Algebras

We have finally succeeded, by joint effort, in proving the following theorem, which is fundamental for the structure theory of algebras over algebraic number fields and beyond.

Main theorem. Every normal division algebra over an algebraic number field is cyclic, or, in the usual terminology, of Dickson type.

It is a special pleasure to present this result, essentially an achievement of the p -adic method, to Kurt Hensel, the founder of that method, on his seventieth birthday.

The proof consists of three reductions, contributed by the three authors.

19.1 1. Hasse's first reduction

The first reduction is due to H. Hasse and is based on his theory of cyclic algebras over algebraic number fields.

Reduction 1. The main theorem follows if the following assertion is proved:

I. Every algebra over Q which is split everywhere is already split over Q .

Here, as in what follows, an “algebra over Q ” means a normal simple algebra over the algebraic number field Q , that is, a simple hypercomplex system with center Q . We write

$$A \sim Q$$

when A is a full matrix algebra over Q , i.e. belongs to the special similarity class determined by Q itself as division algebra. An algebra is said to be split everywhere if, for every prime place $\mathfrak{p}$ of Q , the $\mathfrak{p}$ -adic extension

$$A_{\mathfrak{p}} = A \otimes_Q Q_{\mathfrak{p}}$$

is split over $Q_{\mathfrak{p}}$.

Let D be a division algebra over Q . Choose a cyclic field Z/Q such that for every prime place $\mathfrak{p}$ of Q the local degree of Z at $\mathfrak{p}$ is a multiple of the local index of D at $\mathfrak{p}$. Then at every prime of Z over $\mathfrak{p}$, the local field $Z_{\mathfrak{P}}$ is a splitting field for $D_{\mathfrak{p}}$. Hence the algebra D_Z obtained by extending the center from Q to Z is split everywhere. If assertion I is known for every ground field, then

$$D_Z \sim Z.$$

Thus D has a cyclic splitting field. Hasse's theory then gives a cyclic splitting field whose degree is the degree of D itself; hence D is cyclic. Therefore the main theorem follows from I.

19.2 2. Brauer's second reduction

R. Brauer supplied the decisive idea for the second reduction. He had observed, in the related question of determining the exponent of a division algebra, that Sylow's theorem reduces the problem to the case of a solvable splitting field. The same idea applies to cyclicity.

Reduction 2. Assertion I follows if the following assertion is proved:

II. Every algebra over Q which is split everywhere and has a solvable splitting field is already split over Q .

Let A be split everywhere over Q , and let K/Q be a Galois splitting field for A . Fix a rational prime p , and let L be the fixed field of a Sylow p -subgroup of the Galois group. Then A_L is again split everywhere and has a solvable splitting field, namely K/L . If II is known, then

$$A_L \sim L.$$

Thus A has a splitting field of degree prime to p . The index of A , which divides the degree of any splitting field, is therefore prime to p . Since this holds for every prime p , the index is 1, and A is split. Hence I follows from II.

19.3 3. Noether's third reduction

The third reduction, which together with the first two leads to the proof, was found by E. Noether after a communication from Hasse. Hasse had proved the main theorem in the special case of an abelian splitting field by using Reduction 1 and a formula reduction of the corresponding factor system; Noether extracted the essential reduction from that argument. Independently, Brauer had also considered this reduction.

Reduction 3. Assertion II follows if the following assertion is proved:

III. Every algebra over Q which is split everywhere and has a cyclic splitting field of prime degree is already split over Q .

Let A be split everywhere and have a solvable splitting field K/Q . Choose a chain of fields

$$K = A_0 \supset A_1 \supset \cdots \supset A_r = Q$$

such that each A_i/A_{i+1} is cyclic of prime degree. Then A_{A_1} is split everywhere over A_1 and has the cyclic splitting field $K = A_0$ of prime degree. If III is known, A_{A_1} is split. Hence A_1 is a splitting field for A . Repeating the argument down the chain gives finally $A_Q \sim Q$. Thus II follows from III.

But III is known from Hasse's results. Therefore, proceeding backwards through Reductions 3, 2, and 1, the main theorem is proved.

Noether adds that the truth of III, more generally for cyclic splitting fields of arbitrary degree, is equivalent to Hasse's norm theorem. For Reduction 3 only the prime-degree case is required, hence only the original Hilbert-Furtwängler norm theorem. Thus the reduction gives a new simple proof of Hasse's norm theorem.

Indeed, let Z/Q be cyclic and let $\alpha \in Q^*$ have trivial norm residue symbol at every prime place of Q . Then the cyclic algebra

$$A = (\alpha, Z)$$

splits everywhere. By Reduction 3, using only the Hilbert-Furtwängler norm theorem, A is split over Q . But this is equivalent to saying that α is the norm of an element of Z .

The proper generalization of the norm theorem to higher, even non-abelian, cases is therefore not the literal norm statement but assertion I: an algebra which splits everywhere splits globally.

19.4 Consequences

The immediate consequence of the main theorem is that Hasse's theory of cyclic algebras over algebraic number fields becomes a general structure and invariant theory of normal simple algebras, in particular normal division algebras, over algebraic number fields.

Theorem 1. The exponent of a normal simple algebra over an algebraic number field is equal to its index.

Theorem 2. The ground ideal of a genuine normal division algebra over Q is divisible by at least one prime place of Q , and in fact by at least two.

Here the ground ideal may be defined as the reduced norm of the reduced different; an infinite prime is included exactly when the algebra reduces there to the quaternion algebra rather than to the real or complex field. Hasse's local theory says that a prime place appears in the ground ideal exactly when the local index is not 1. If all local indices were 1, the algebra would be split everywhere and hence would not be a genuine division algebra. A single exceptional prime is also impossible, because the product formula for the norm residue symbols forces the product of all local invariants to be trivial.

The theorem also gives the arithmetic characterization of splitting fields.

Theorem 3. For an algebra A over Q , an algebraic number field K/Q is a splitting field for A if and only if, for every prime place $\mathfrak{p}$ of Q , the local degree of K at $\mathfrak{p}$ is divisible by the $\mathfrak{p}$ -index of A .

Necessity is local. For sufficiency, extend A to K . At each prime of K , the local algebra is split by the divisibility condition. Hence A_K is split everywhere, and by assertion I it is split globally.

Further, the main theorem yields a decomposition theorem for prime divisors in splitting fields and an ordering theorem for splitting fields. If K and K' are splitting fields and $K \subset K'$, then the local degree conditions for K imply those for K' ; conversely the partial order among splitting fields is controlled by divisibility of the corresponding local degrees. In particular, minimal splitting fields may be studied arithmetically through local indices.

Finally, the main theorem applies to representations of finite groups. All absolutely irreducible representations of a finite group are realizable over algebraic number fields with the expected splitting behavior, because the simple components of the rational group algebra are normal simple algebras over number fields and hence cyclic algebras. Schur's representation-theoretic indices are therefore exactly the indices of cyclic algebras and are determined by the local invariants.

20 39. Hypercomplex Systems in Their Relations to Commutative Algebra and Number Theory

The theory of hypercomplex systems, or algebras, has developed rapidly in recent years. Only very recently, however, has the significance of this theory for commutative questions become clear. I will describe that significance through two classical problems going back to Gauss: the principal genus theorem and the closely related norm theorem.

These problems have repeatedly changed their form. In Gauss they appear at the conclusion of his theory of binary quadratic forms. Later they play an essential role in characterizing

relatively cyclic and abelian number fields through class field theory. Finally, they can be stated as theorems about automorphisms and splitting of algebras. This last formulation at the same time extends the theorems to arbitrary relatively Galois number fields.

The guiding principle is the use of non-commutative theory for commutative problems. One seeks, through the theory of algebras, invariant and simple formulations of known facts, and then those formulations lead to generalizations.

20.1 1. Crossed products

Let K/k be a Galois extension of degree n with group G . A crossed product is a simultaneous embedding of K and the group G into an algebra A in such a way that the automorphisms of K become inner automorphisms in A .

Let u_S be symbols corresponding to the group elements $S \in G$. First put

$$A = \sum_{S \in G} u_S K,$$

so that A is a module of rank n over K . The condition that u_S implement the automorphism S is

$$zu_S = u_S z^S$$

for every $z \in K$. Multiplication among the u_S is prescribed by

$$u_S u_T = u_{ST} a_{S,T}, \quad a_{S,T} \in K^*.$$

Associativity is equivalent to the factor-system law

$$a_{S,T} a_{ST,U} = a_{S,TU}^U a_{T,U}.$$

The resulting algebra is the crossed product of K with G for the factor system $a_{S,T}$.

Changing u_S to

$$v_S = u_S c_S, \quad c_S \in K^*,$$

changes the factor system to an associated one,

$$a'_{S,T} = a_{S,T} c_S^T c_{ST}^{-1}.$$

Associated factor systems form one class. Likewise, similar algebras are collected into one algebra class. The classes of crossed products with fixed splitting field K form an abelian group under direct product; this is Brauer's group of algebra classes, expressed in the language of factor systems.

20.2 2. Cyclic fields and norms

If K/k is cyclic and S is a generator of its group, the crossed product becomes particularly simple. One may choose one element u corresponding to S , and then

$$uz = z^S u, \quad u^n = a \in k^*.$$

The factor system is determined by the single element a , and changing u to uc replaces a by

$$a N_{K/k}(c).$$

Thus cyclic algebra classes with splitting field K are described by the quotient group

$$k^*/N_{K/k}(K^*).$$

A cyclic algebra (a, K) splits if and only if a is the norm of an element of K .

This is the bridge from the norm theorem to the theorem on split algebras. The norm theorem for cyclic fields says, in one formulation, that if a is a local norm at every finite and infinite place, then a is a global norm. In algebraic language, the cyclic algebra (a, K) splits at every place and therefore splits globally.

The general theorem on algebras says: if an algebra over an algebraic number field splits at every place, then it splits globally. The cyclic norm theorem is exactly its cyclic special case. The proof of the general theorem from the cyclic case proceeds by purely algebraic-arithmetic reductions.

One important consequence, drawn by Hasse, is that every normal simple algebra over an algebraic number field is cyclic. This long-conjectured statement was the problem whose solution led to the general formulation.

20.3 3. Principal genus theorem

A second consequence of the theorem on split algebras is the principal genus theorem. Its invariant formulation rests on the fact that the defining relations of the crossed product are purely multiplicative. They continue to make sense if K^* is replaced by an abelian group I on which the Galois group operates.

Taking I to be the group of ideals of K , one obtains factor systems of ideals. A class division of ideals induces a class division of the factor systems, usually finer than the original ideal class division. The identity class of factor systems is defined by those $a_{S,T}$ which generate split algebras at all ramified finite and infinite places.

Let G^* denote the extension of the Galois group obtained in this way. The invariant form of the principal genus theorem is this:

If, under the class division just defined, the substitution

$$v_S = u_S(c_S)$$

induces an automorphism of G^* , then this automorphism is inner and is generated by an ideal class (b) .

Equivalently, if the transformation factors

$$(c_S)^T(c_T)(c_{ST})^{-1}$$

belong to the identity class of factor systems, then there is an ideal class (b) such that

$$(c_S) = \frac{(b)}{(b)^S}$$

for all $S \in G$. In the cyclic case this says that if the norm of an ideal class lies in the identity class of factor systems, then the class is a symbolic $(1 - S)$ -th power.

20.4 4. Relation with the classical form

If one restricts to ideals prime to the ramified places, the content remains the same. For quadratic fields the principal genus just defined agrees with Gauss's principal genus. Ideal classes correspond to quadratic forms, and norms of ideal classes correspond to integers represented by those forms. The condition that the identity class generate split algebras at the ramified places says exactly that those represented integers are quadratic residues at the ramified places. Thus the forms have the total character of the principal form.

For cyclic fields the theorem becomes the usual statement: the principal genus consists of those ideal classes whose norms are norm residues at the finite and infinite ramified places. This is equivalent to being a norm residue modulo the conductor.

In the general Galois case one can likewise introduce a conductor built only from the ramified primes, normalized so that the factor systems at each ramified place contain only elements of the identity class. The relation with Artin's conductors, which involve the same prime ideals, leads back to the theory of Galois modules and hence to the second hypercomplex method. How far these two methods reach remains a question for future work.

21 Non-commutative Algebras

It is well known that the main theorems of commutative algebra are contained in Galois theory, preceded by the theory of detachment fields and splitting fields: fields large enough, respectively, to split off one linear factor of a given polynomial or to decompose it completely into linear factors.

The corresponding parts of algebra are developed here in the non-commutative setting, and especially for hypercomplex systems. The method is deliberately non-commutative: it works with representations in non-commutative division rings. At the end I show how the corresponding theorems of commutative algebra may be proved in a completely parallel manner by representation in commutative fields.

The representation theory used here is a further development of the theory based on representation modules. It is preceded by a short automorphism theory. Alongside direct representations I consider reciprocal representations. Each can be reduced to the other, but the reciprocal representation is generated by the reciprocal representation module. The advantage is that this reciprocal representation module is a bimodule and may also be regarded as a one-sided module over an extension ring. Thus representation in non-commutative division rings is reduced to ideal theory in the extension ring: the irreducible representation classes correspond to the irreducible ideal classes of that extension ring, in exact analogy with the facts for representations of hypercomplex systems over their commutative coefficient domain.

From this one obtains structure theorems for matrix rings over non-commutative division rings. The basic observation is that every subring defines a representation by the division ring, or reciprocally by the oppositely isomorphic division ring. This oppositely isomorphic division ring is the first analogue of a minimal detachment or splitting field in the commutative case: it mediates all reciprocal representations of first degree and, when the division ring has finite rank over its center, also gives a complete splitting into direct summands of rank one.

This yields the Galois theory for division rings and at the same time expresses the fact that reciprocally isomorphic division algebras generate inverse classes in Brauer's group of algebra classes. A second proof of the Galois theory, applying more generally to simple systems, is almost an immediate consequence of a theorem on mutually commuting subrings. That theorem itself is tied directly to the preliminary automorphism considerations.

Up to this point the arguments are purely non-commutative. But the problem of commutative splitting fields is also handled non-commutatively, by representing the splitting field in the division algebra. The final sections transfer the same proof method to the commutative case and develop splitting fields for arbitrary systems, thereby giving the connection with the ordinary commutative representation theory.

21.1 §1. Automorphisms, modules, and bimodules

The representation theory based on representation modules rests on the theory of the automorphism ring of abelian groups with or without operators. The elementary arguments concerning mappings and laws of calculation will be collected here to avoid repetition.

21.1.1 1. Multiplicative mapping and associativity

Let G first be a group without operators, and let A be its absolute endomorphism domain, the set of all homomorphisms of G into itself. This domain is multiplicatively closed. If $s, t \in A$, the product st is defined by

$$g(st) = (gs)t,$$

and associativity is immediate.

Now let G be a group with operators. Thus there is a set B of symbols $O, H, \dots$ such that $gO, gH, \dots$ are uniquely defined elements of G and each symbol induces an endomorphism of G . Hence there is a definite, though not necessarily one-to-one, map from B into the absolute endomorphism domain A .

If the operator domain B is itself multiplicatively closed, then the map from B to A is multiplicatively homomorphic if and only if the corresponding associativity relation

$$g(OH) = (gO)H$$

holds for all $g \in G$ and $O, H \in B$. For left operators the same assertion gives a reciprocal homomorphism, since left multiplication must be read in the opposite order whereas operators are read from left to right.

21.1.2 2. Operator-homomorphic mapping

If G is a group with operator domain B , a mapping $G \rightarrow G'$ is operator-homomorphic when it is a group homomorphism and respects the operators. When the operator domains themselves are also mapped, this includes ring homomorphisms as a special case.

The essential principle is this. Whenever a group is made into a group with operators by a multiplicatively closed operator domain, the associativity law is exactly the statement that the operator domain maps homomorphically into the endomorphism ring. Conversely, a homomorphic image of an operator domain in endomorphisms supplies the required associative action.

21.1.3 3. Modules and bimodules over rings

Let R be a ring and M an abelian group. A right R -module structure on M is an operator structure satisfying

$$(m_1 + m_2)r = m_1r + m_2r,$$

$$m(r_1 + r_2) = mr_1 + mr_2,$$

$$m(r_1r_2) = (mr_1)r_2.$$

Left modules are defined similarly, with the order in the last formula reversed. If M is a right module over R and a left module over S , and

$$s(mr) = (sm)r,$$

then M is a bimodule.

The automorphism ring of a module is the ring of all module endomorphisms. If these endomorphisms are written on the side opposite the original ring action, a module becomes a bimodule over the original ring and its automorphism ring. Conversely, every bimodule determines homomorphisms from one of its multiplier rings into the automorphism ring relative to the other.

21.1.4 4. Passage from a right bimodule to a one-sided module

Suppose M is a right module over a ring R , and suppose that R contains two subrings which commute elementwise. Then M can be regarded as a one-sided module over the extension ring generated by those two subrings. Conversely, a one-sided module over such an extension ring gives a bimodule by restriction to the two commuting subrings.

This elementary passage is the key technical device. It permits one to replace reciprocal representation modules, which are naturally bimodules, by one-sided modules over an extension ring. Ideal theory for the extension ring can then be applied.

21.2 §2. Representation modules

21.2.1 1. Modules of linear forms

The transition from the preceding automorphism theory to representation theory proceeds through modules of linear forms. Let a hypercomplex system S act on a module M by right multiplication. Choosing a basis of M over the coefficient division ring gives matrices for the action of every element of S . The entries of those matrices depend on the chosen basis, but the module class does not.

The automorphism ring of a module of linear forms commutes with the representing matrices. Conversely, matrices commuting with a given representation arise as endomorphisms of the corresponding representation module. This is the module-theoretic form of Schur's lemma and its generalizations.

Theorem. If M is a module of linear forms for a system S , then its S -automorphism ring is the complete commutant of the matrices representing S on M .

In particular, when M is irreducible, this automorphism ring is a division ring.

21.2.2 2. Representation and representation module

A direct representation of a system S assigns to every element $s \in S$ a matrix in such a way that sums and products are respected. The representation module is the module whose basis vectors are the rows or columns on which these matrices act. Equivalent representations yield operator-isomorphic modules, and operator-isomorphic representation modules yield equivalent representations.

A reciprocal representation is defined by reversing the multiplication order. It is generated by the reciprocal representation module. The reciprocal module is naturally a bimodule: the represented system operates on one side, and the coefficient division ring operates on the other.

Theorem. Every reciprocal or direct representation module generates a representation, and equivalent modules generate equivalent representations. Conversely, a representation determines its module class.

The usefulness of the reciprocal module is that it may be viewed as a one-sided module over a suitable extension ring. Thus questions about irreducible reciprocal representations become questions about simple one-sided ideals in that extension ring.

21.2.3 3. Passage from reciprocal modules to ordinary modules

Let A be the coefficient division ring and let S be represented reciprocally in A_n , the full matrix ring over A . The reciprocal representation module is a right module over an extension ring generated by S and A . By the preceding passage, it may be treated as a one-sided module.

The consequence is the following fundamental connection. The classes of irreducible reciprocal representations of S over A correspond one-to-one to the irreducible ideal classes of the extension ring. Under complete reducibility, direct-sum decompositions of the module correspond to decompositions of representations into irreducible components.

This is the non-commutative analogue of the familiar fact that, for a hypercomplex system represented over its own commutative coefficient field, irreducible representations are obtained from one-sided ideals.

21.2.4 4. The theorem on commuting matrices

The module formulation gives the theorem on commuting matrices. If a system of matrices over a division ring is irreducible, its commutant is a division ring. If the representation is completely reducible, the commutant is a full matrix ring over the division ring attached to the irreducible class, with one full matrix factor for each multiplicity.

Conversely, if two systems of matrices are full mutual commutants in a full matrix ring, each is completely reducible relative to the other. This double-centralizer idea is the mechanism behind the later theorem on mutually commuting subrings.

21.3 §3. Extension modules

21.3.1 1. Normal bases of submodules

Let A be a division ring and let P be a subfield of its center or, more generally, a central subfield. If M is a module over A , a P -submodule generated by a basis may be extended by multiplication with A to the whole module. Linear independence over P and over A are related exactly as in the commutative case once left and right conventions are fixed.

This gives a normal-basis type statement for modules. A suitable basis of a submodule may be completed to a basis of the larger module after scalar extension. Such bases are used to compare ranks in the double-centralizer calculations.

21.3.2 2. Extension modules

Let A be a division ring and P a subfield. An A -module M of rank n is called an extension module of P -rank r when it is generated over A by a P -module of the corresponding rank and when the two scalar structures are compatible. The extension module records precisely how a representation over a smaller field becomes a representation over the larger division ring.

If $z_1, \dots, z_m$ are linearly independent over P , then their images remain independent under suitable scalar extension. Every P -module

$$T = z_1 P + \dots + z_m P$$

inside M generates an extension module by multiplication with A .

21.3.3 3. Invariant modules

Let $M = M_1 + \dots + M_r$ be a direct decomposition into simple components. A submodule invariant under a second commuting system corresponds to a sum of certain components. The precise theorem is that extension modules of invariant submodules and invariant submodules of extension modules are in one-to-one correspondence.

The proof is a straightforward use of the uniqueness of direct-sum decompositions into simple modules. If a submodule is invariant, its projection onto each simple summand is either zero or the full summand. Conversely, any such sum is invariant.

This will be used to identify two-sided ideals with invariant modules.

21.4 §4. Simple systems

21.4.1 1. Simple systems and modules

Let S be a simple hypercomplex system over P . A representation over a division ring A is irreducible if its representation module is simple. The corresponding extension ring is then a simple ring. Conversely, a simple ideal class of the extension ring gives an irreducible representation.

If S_A denotes the extension of S to A , then a decomposition of S_A as a module reflects the decomposition of the representation after extension. When A is large enough to split the system, the simple components have rank one in the relevant sense, and the representation is absolutely irreducible.

21.4.2 2. Representation classes

A representation class is a class of operator-isomorphic representation modules. For a simple system with center P , representation classes over a division ring A correspond to simple ideal classes in the extension ring. The number of irreducible classes, their dimensions, and their automorphism division rings are obtained from the decomposition of this extension ring.

Theorem on representation classes. Let S be hypercomplex over P and let A be a division ring containing P in its center in the appropriate sense. The irreducible reciprocal representation classes of S in A are in one-to-one correspondence with the irreducible right ideal classes of the extended system. The direct-sum decomposition of the extended system gives the decomposition of any completely reducible representation.

21.4.3 3. Rank relation

When S is simple, the rank of an irreducible representation, the rank of the corresponding ideal, and the rank of the associated division algebra satisfy the usual product formula. If a simple algebra S is embedded irreducibly in a full matrix ring over A , and if B is the commuting division ring, then the product of the ranks of S and B equals the rank of the full matrix ring. This is the numerical form of the double-centralizer theorem.

21.4.4 4. Sharpened rank relation

If A has finite rank over its center, the rank relation can be sharpened. The division ring reciprocal to A appears as the full commutant of an irreducible embedding. The two central simple algebras obtained in this way have reciprocal classes in the Brauer group. Their indices multiply in the expected manner, and splitting of one corresponds to complete reduction of the other.

21.5 §5. Commuting subrings

21.5.1 1. Reducible and irreducible embeddings

Let A_n be a full matrix ring over a division ring A , and let S be a subring. An embedding of S is irreducible if the natural module has no proper invariant submodule. It is completely reducible if the module is a direct sum of irreducible invariant submodules.

If the embedding is reducible, the matrices of S may be simultaneously put into block triangular form. If it is completely reducible, they may be put into block diagonal form. These statements are merely the module-theoretic reduction theorems translated into matrices.

21.5.2 2. Inner automorphisms

If two simple subrings of A_n both contain the center P and are isomorphic over P , then under suitable irreducibility hypotheses an isomorphism between them is induced by an inner automorphism of A_n . That is, there exists an invertible element $u \in A_n$ such that

$$\varphi(s) = u^{-1}su.$$

This is the non-commutative analogue of the extension of field isomorphisms.

The proof uses the representation modules attached to the two embeddings. Since the embeddings are irreducible and isomorphic, the modules belong to the same class. An operator-isomorphism of the modules is given by multiplication with an invertible matrix, and this matrix implements the desired inner automorphism.

21.5.3 3. The theorem on mutually commuting subrings

The central theorem is the following.

Theorem. Let S and T be simple subrings of a full matrix ring over a division ring, each containing the center P , and suppose that every element of S commutes with every element of T . Then the full commutant of S is generated by T together with the center in the appropriate way precisely when the corresponding representation is irreducible; in general the two full commutants are obtained by passing to the completely reducible decomposition.

In the finite-rank case this becomes the usual double-centralizer theorem: if S is a central simple subalgebra of a central simple algebra A , then the centralizer $C_A(S)$ is central simple over the same center, and

$$[S : P][C_A(S) : P] = [A : P].$$

Moreover, S and its centralizer are mutual centralizers.

21.5.4 4. Sharpened commutation theorem

If S is a simple subring and A is a division algebra finite over its center, then the centralizer of S in a full matrix algebra over A has class inverse to the class represented by the embedding of S . In particular, if one passes from S to its reciprocal system, the associated algebra class is inverted in the Brauer group.

This sharpened theorem is the bridge to Galois theory. It says that intermediate division rings and closed subgroups of the inner automorphism group determine one another through centralizers.

21.6 §6. Galois theory of simple systems

The sharpened commutation theorem expresses the Galois theory of simple systems relative to the center. We first consider division rings, then simple systems in general.

21.6.1 1. Galois theory of division rings finite over the center

Let A be a division ring with center P . Its Galois group is the group of automorphisms of A which extend the identity on P . By the Skolem-Noether theorem in this setting, these are the inner automorphisms. Thus the group is isomorphic to

$$A^*/P^*,$$

where A^* and P^* denote the multiplicative groups of non-zero elements.

Subgroups of this group correspond to subgroups H^* of A^* containing P^* . A subgroup is called closed if, after adjoining zero, H^* becomes a division subring H of A .

The statement that a division subring C is fixed by a subgroup is equivalent to saying that C commutes elementwise with the division ring H attached to the subgroup. Therefore the commutation theorem gives the main theorem of non-commutative Galois theory:

Main theorem of the Galois theory of division rings. The division rings C between P and A and the closed subgroups H of the Galois group correspond one-to-one, in such a way that C is the full invariant division ring of H and H is the full invariant group of C .

21.6.2 2. Galois theory of simple systems

Let A now be a simple system with center P . Its Galois group is again the group of inner automorphisms, isomorphic to A^*/P^* , where now A^* denotes the group of regular elements of A .

A subgroup is called simply closed if the subring generated by the corresponding regular elements is a simple system and if the group consists of all regular elements of that simple system, modulo P^* . The key lemma is that every simple system containing P is generated by its regular elements. This is clear when P is infinite, since a general element with indeterminate coefficients is regular and may be specialized to regular basis elements. Shoda proved it in general.

Using this lemma, commutativity with a simple system is equivalent to invariance under the automorphisms induced by its regular elements. Hence the commutation theorem gives:

Main theorem of the Galois theory of simple systems. The simple systems S in A containing P and the simply closed subgroups of the Galois group correspond one-to-one, in such a way that S is the full invariant domain of the subgroup and the subgroup is the full invariant group of S .

21.6.3 3. Proof by the extension principle

For division rings there is another proof of the main theorem, based on extending isomorphisms, i.e. representations of first degree. It avoids rank counting and therefore requires fewer finiteness hypotheses. It also shows in what direction an extension to division rings of infinite rank might be possible.

Assume that a simple system S over P admits a reciprocal representation of first degree in A , so that S is itself a division ring. Let

$$S_A = e_1 S_A + \cdots + e_n S_A$$

be a decomposition into simple right ideals. The representation generated by e_i is defined by

$e_i w = e_i w$ in A .

Lemma 1. The n reciprocal representations generated by $e_1, \dots, e_n$ are distinct. Thus S has at least as many distinct representations as its rank.

If two of them were the same, then $e_i w = e_j w$ for every $w \in S_A$. But this is impossible because $e_i e_j = 0$ for $i \neq j$.

Lemma 2. Let T be an intermediate division ring between P and S , and let $s = [S : T]$. Then every reciprocal isomorphism of T into A has at least s distinct extensions to S .

Indeed, decompose the representation of T by idempotents. Multiplication by S splits each component into exactly s components, and Lemma 1 gives their distinctness.

Define the class division of the reciprocal isomorphisms of S induced by T : two isomorphisms are equivalent if they induce the same isomorphism on T . Then:

Main theorem in this form. If $\mathfrak{I}_T$ is the class division induced by T , then T is a maximal intermediate division ring with respect to this class division.

Passing from the set of isomorphisms to the automorphism group by multiplication with the inverse of one fixed isomorphism gives the usual subgroup formulation. The assertion that closed subgroups are full invariant groups is included by taking the division ring H attached to such a subgroup as T .

21.7 §7. Similarity classes of algebras and splitting fields

From now on A , B , and their matrix rings are assumed finite over their centers P . Thus they are normal simple algebras over P ; A and B may be division algebras.

All algebras A_n with the same associated division algebra A are collected into one similarity class. Isomorphic division algebras are identified.

21.7.1 1. The group of algebra classes

If A and B are algebras over P , then their direct product over P is again an algebra over P . At the level of associated division algebras this product determines a class, independent of matrix factors. Thus the classes form a multiplicatively closed commutative system.

Brauer's theorem states that these similarity classes form an abelian group under direct product. The identity is the class of full matrix algebras over P . The inverse of the class of A is represented by the algebra $\bar{A}$ reciprocally isomorphic to A . The relation

$$A \otimes_P \bar{A} \sim P$$

follows from the sharpened commutation theorem.

This is the Brauer group of P .

21.7.2 2. Splitting fields

An extension field Z/P is a splitting field of A if

$$A_Z \sim Z.$$

Equivalently, Z contains a maximal commutative subfield of the division algebra associated with A , or at least is large enough to make the associated division algebra into a matrix ring.

A field is a detachment field if it splits off one simple component of the representation; in the central simple case detachment and splitting agree for maximal commutative subfields. In general, every maximal commutative subfield of a division algebra is a splitting field, and every splitting field has degree divisible by the index.

21.7.3 3. Rank relations and index

Let A be central simple over P , and let Z be a maximal commutative subfield. If $[A : P] = m^2$, then $[Z : P] = m$. The integer m is the Schur index. It is the degree of any maximal commutative subfield embedded in the division algebra, and it is also the minimal possible degree of a splitting field in the regular case.

For a general splitting field of degree n , one has

$$n = mr$$

for some integer r . This follows either from rank comparison or from the fact that the index divides every splitting-field degree.

If A is embedded in a larger division algebra D and L is the centralizer, then the refined relation reads

$$[A : P][L : P] = [D : P]$$

with the appropriate square-root conventions for indices. These are exactly the double-centralizer rank formulas.

21.7.4 4. Existence of separable splitting fields

Every algebra class has separable splitting fields, even such as can be embedded in the algebra itself. Let the ground field have characteristic p , and write the index as

$$m = sp^r,$$

with s prime to p . Suppose first that a splitting field has inseparable part of degree p^r . One shows that this inseparable part can be reduced. The reduced discriminant of the division algebra after extension to an algebraic closure is non-zero, so there exists an element with non-zero reduced trace. Such an element cannot lie in the purely inseparable ground part, since traces there vanish by multiplication by p . Adjoining it gives a proper separable extension and reduces the remaining index. Repetition yields a separable splitting field of degree equal to the index.

21.8 §8. Splitting fields, detachment fields, and Galois theory in the commutative case

We now pass from splitting fields of central simple systems to those of arbitrary simple systems by developing the splitting theory of the center. Everything in this paragraph is commutative.

21.8.1 1. Commutative simple systems

A commutative simple system Z over P is a field extension. Let Ω be an algebraically closed extension field of P . Then Z_Ω has no radical if and only if Z/P is separable. Equivalently, Z has as many distinct embeddings into Ω as its degree.

An extension field A of P is called a splitting field of Z if Z_A decomposes into composition factors of first degree. It is called a detachment field if at least one composition factor of first degree splits off. Thus A is a detachment field if it contains a subfield isomorphic to Z , and is a splitting field if it contains a field isomorphic to the Galois closure of Z .

A minimal detachment field is isomorphic to Z . It is also a splitting field if and only if Z/P is normal.

21.8.2 2. Galois theory for separable fields by the hypercomplex method

Assume Z/P is finite separable, and let Ω be a splitting field, finite or infinite. Then

$$Z_\Omega = e_1 Z_\Omega + \cdots + e_n Z_\Omega$$

decomposes into a direct sum of fields corresponding to the distinct embeddings of Z into Ω . The idempotents e_i determine the embeddings.

If T is an intermediate field between P and Z , and $s = [Z : T]$, then every embedding of T into Ω has exactly s extensions to Z . This is proved exactly as in the non-commutative extension-principle proof: the decomposition of T_Ω refines to that of Z_Ω , and the number of idempotents above each one is s .

The class division of the embeddings of Z induced by T has T as its maximal intermediate field. When Z/P is Galois, composition with the inverse of a fixed embedding identifies the embedding set with the automorphism group, and one obtains the usual Galois correspondence.

21.8.3 3. Idempotents of Z_Ω

Let S run through the embeddings which carry Z into its conjugate copies inside the splitting field. The idempotents e^S of Z_Ω are conjugate:

$$e^S = eS,$$

with $e = e^1$. Applying S to the defining relations of e carries them to the defining relations of e^S . Since the direct-sum decomposition is unique, the idempotents are uniquely determined and hence conjugate.

If Z/P is Galois, this immediately gives the subgroup part of the main theorem of Galois theory. For if H is a subgroup, the sum

$$\sum_{S \in H} e^S$$

is fixed by all substitutions in H and by no larger subgroup. Hence the invariant field attached to H is recovered.

21.8.4 4. Complementary bases

Let $a_1, \dots, a_n$ be a basis of Z/P . Write the idempotent in the form

$$e = a_1\beta_1 + \dots + a_n\beta_n.$$

Then, for every embedding S , the element

$$e^S = a_1\beta_1^S + \dots + a_n\beta_n^S$$

expresses the image of the basis complementary to $a_1, \dots, a_n$ in the embedding defined by e^S . The relations

$$e^S e^R = 0 \quad (S \neq R), \quad (e^S)^2 = e^S$$

translate precisely into the defining equations for complementary bases. Thus the contragredience of complementary bases follows from the invariance of the idempotents, independently of the chosen basis.

This recovers Dedekind's relation between complementary bases and idempotents in the language of hypercomplex decompositions.

21.9 §9. Splitting and detachment fields of arbitrary systems

21.9.1 1. Definitions and reduction to the simple case

Let S be hypercomplex over P . A commutative extension field A/P is a splitting field of S if, in S_A , every composition series by one-sided ideals has absolutely simple composition factors. Equivalently, all irreducible representations of S over A are absolutely irreducible. An extension field T/P is a detachment field if S_T splits off at least one absolutely simple composition factor, equivalently if at least one irreducible representation over T is already absolutely irreducible.

These definitions include the earlier ones: for commutative simple systems they reduce to the definitions of §8, and for central simple algebras they reduce to the definitions of §7. In the latter case detachment and splitting fields coincide, because after a commutative extension of the center the algebra remains completely reducible, and all simple components are operator-isomorphic. Splitting off one absolutely simple component forces complete splitting.

The splitting fields of a general system are obtained by taking the compositum of splitting fields of the irreducible representations over P . A detachment field of the system is a detachment field of at least one irreducible representation. Because irreducible representations correspond one-to-one to the simple components of the radical quotient, it is enough to consider simple systems.

21.9.2 2. Simple systems with separable center

Let A be a simple system over P with center Z , and assume Z/P is separable. Let Ω be a splitting field of Z . The direct-sum decomposition of Z_Ω induces a two-sided decomposition of A_Ω :

$$A_\Omega = e_1 A_\Omega + \dots + e_n A_\Omega.$$

The components are conjugate and are isomorphic to A in the appropriate sense; each is a normal simple algebra over its own center $e_i Z_\Omega$.

Therefore A_Q has no radical for every extension field Q of P ; it is completely reducible. The components remain simple after arbitrary coefficient extension, because the center has already separated. The detachment and splitting fields are then characterized by combining §7 with §8.

If A is a division algebra with separable center Z , the irreducible embeddings of detachment fields are precisely all maximal commutative subfields of A which contain a copy of Z . Splitting fields are precisely the composita of a detachment field with a splitting field of the center, in particular with the Galois closure of Z .

A minimal detachment field can be a splitting field even if the center is not Galois. This is one of the differences from the commutative case: for example, a quaternion algebra with center isomorphic to a quadratic field may have the Galois closure of the center already as a minimal detachment and splitting field.

21.9.3 3. Inseparable center

If the center is inseparable over P , then Z_Ω has a radical, and consequently A_Ω has a radical. Let $\mathfrak{C}$ be the radical of Z_Ω , and let $\mathfrak{c}$ be the ideal it generates in A_Ω . The quotient $Z_\Omega/\mathfrak{C}$ decomposes into conjugate components, exactly as in the separable case, and this induces a direct-sum decomposition of

$$A_\Omega/\mathfrak{c}.$$

The components are again isomorphic to the original simple system in the corresponding sense and have centers equal to the corresponding components of $Z_\Omega/\mathfrak{C}$.

Rank counting shows that the composition factors belonging to a composition series of $\mathfrak{c}$ are operator-isomorphic to those of $A_\Omega/\mathfrak{c}$, not merely homomorphic. Hence the representation-theoretic conclusions remain valid after replacing direct-sum components by composition factors.

21.9.4 4. Final representation-theoretic formulation

For irreducible representations the result may be summarized as follows.

Let a representation of a hypercomplex system be irreducible over P . It becomes absolutely completely reducible after extension of the coefficient field if and only if the corresponding center is separable over P . In every case, splitting fields and detachment fields of the representation can be characterized by embeddings in the associated simple system as above.

In particular, the lowest possible degree of a detachment field is the product of the degree of the center over P and the index of the corresponding algebra class over that center. The degree of every detachment field is a multiple of this number.

The representation splits, in the sense of composition series, into as many distinct conjugate absolutely irreducible representation classes as the degree of the largest separable subfield contained in the center. Each class occurs with multiplicity equal to the product of the index and the degree of the center over that separable subfield.

When the ground field is perfect, so that all finite extensions are separable, this recovers the familiar results of I. Schur, with the additional embedding characterization already found in Brauer's work.